

Chapter 15 :

Power Amplifiers

15.1 Introduction

In general, a practical amplifying system consists of several stages that amplify weak signal (received at the input stage) until sufficient power is available to drive the output devices. In audio amplifier system, the output device is mainly a loudspeaker, while in a control system it may be a motor or some other output loads. As shown in Fig.15-1, the first stage of an amplifier system is a voltage amplifier that amplifies the voltage levels of very weak signals.

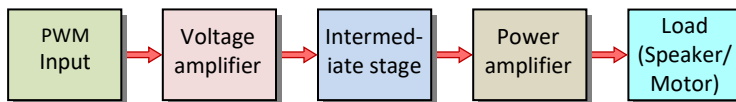


Fig.15-1: Block diagram of an amplifier system

As they amplify very weak signal their output voltage swing is very small (1 V maximum) they are called **small signal amplifiers**. Power level of output signals of these amplifiers is in the order of few milliwatts. As placed at the first-stage, they are also called the **preamplifiers**. We have already discussed small signal amplifiers in previous chapters (Chapter 9, and 10). To analyze these amplifiers small-signal equivalent models are used. The last stage of the amplifying system is **power amplifier** and also called **large signal amplifiers**. As they drive the ultimate loads, they are also called simply **drivers**. As the name implies, a **power amplifier** is one that amplifies the power (or current) level of signals and delivers that large amount of power to a load. Generally, power amplifiers increase only the current level of the signals which are already amplified by voltage amplifiers. So the signals operated by power amplifiers have large voltage swings and the device operates over the entire range of its output characteristics. For this reason, they are called **large signal amplifiers**. To perform this function, a power amplifier must itself be capable of handling large amount of power. That is, it must be designed so that the heat, generated when it is operated at high current and voltage levels, must be dissipated to surroundings. For this purpose, power amplifiers employ special type of transistors called **power transistors** with larger body and a metal case, where external heat-sinks may be attached.

15.2 Transistor Power Dissipation and Thermal Consideration

In electronic devices, electrical energy is converted to heat energy at a rate given by $P = VI$ Watts, and temperature rises when this heat energy is not removed at a comparable rate. In power amplifiers, the transistors work with a large amount of voltage and current. Generally, the B-E junction of a transistor is forward biased and the B-C junction is reversed biased. Therefore, the voltage dropped across the B-E junction is very small, while the voltage dropped across the B-C junction is very large. In fact, almost the entire voltage dropped across a transistor (V_{CE}) can be considered to be equal to the voltage drop across B-C junction. Since, the collector and emitter current is almost equal, most of the heat generated in a transistor is produced at the collector-base junction. The total power dissipated at the junction (or transistor) is,

$$P_D = V_{CB}I_C + V_{BE}I_E \approx (V_{CB} + V_{BE})I_C = V_{CE}I_C \quad (15-1)$$

The maximum capacity of P_D for a particular transistor is given in datasheets generally at 25°C ambient temperature. For a fixed value of P_D the graph of Equ.(15-2) is a hyperbola when plotted on $I_C - V_{CE}$ coordinate system. For larger values of P_D , the corresponding graphs shift outward from the axes, as illustrated in Fig.15-2. In this figure, each hyperbola represents all possible combinations of V_{CE} and I_C that gives a product equal to P_D . The load line of a transistor is shown in the same figure by the straight line. If the transistor operates over the entire load line, the operating point of the transistor will move along the load line when input signal is applied. For the save operation (proper power capacity) the load line should be always to the left of the transistor power parabola. As shown in the figure, the load line is safe for $P_D = 2\text{ W}$, but is not safe for $P_D = 1\text{ W}$. If a device operates in the unsafe region, its collector junction temperature will increase. If this temperature exceeds the maximum operating temperature, catastrophic irreversible failure of the device may occur. Depending on the transistor type, the junction temperature in the range of 100° to 110° (for germanium transistors) or 150° to 200° (for silicon transistors) will destroy the transistors.

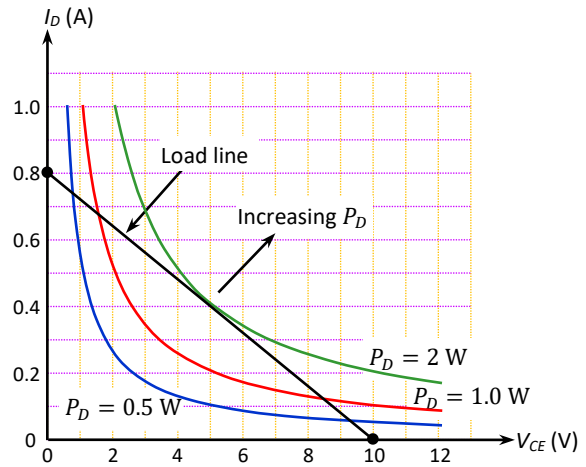


Fig.15-2: Load line and hyperbolas of constant power dissipation

15.2.1 Heat Transfer in Semiconductor Devices

Semiconductor devices are cooled by all three heat-transfer mechanisms [Fig.15-3]. Already, it is discussed that, heat is mainly produced in the junctions of a device. This heat passes from the junctions, through the semiconductor material, through the case and heat sink (if any) (all are *conduction*), and into the surrounding air (*radiation* mechanism). It is radiated from the surface of the case and heat sink. Heating of the surrounding air creates air flow around the device, so *convection* cooling occurs. Various measures are taken to enhance heat transfer by each

mechanism. To improve conduction, good physical contact is made between a junction and a device's case, or enclosure. Good conduction materials are used for the body or case. In fact, in many power transistors the collector is in direct contact with the case, so the two are electrically the same as well as being at approximately the same temperature. To improve radiation of heat from the case to the surrounding air, power devices are often equipped with *heat sink*. Heat sinks are made of good conducting material, generally aluminum, with a large number of *fins* to increase effective surface area where radiation takes place [see Fig.15-3].

Finally, convection cooling may be enhanced by using fans that blow cool air past the surface of heat sink (or into a casing) and/or through the hot air near the surface of heat sink (or from the casing). For different shapes of transistor case, heat sinks of different sizes and shapes are available in the market. Some common heat sinks are shown in Fig.15-4.

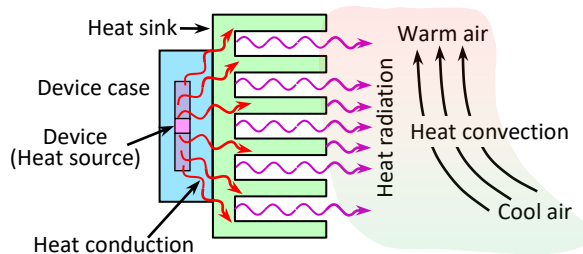


Fig.15-3: Heat transfer from a device to ambient air

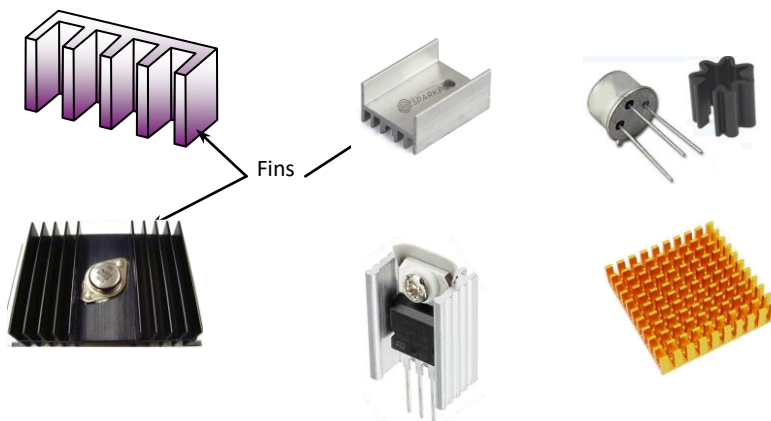


Fig.15-4: Heat sink of different shapes and sizes

15.2.2 Thermal Resistance

The heat flow mechanism by conduction method is very much similar to the conduction of electrical charges, that is, current. We know that current is the rate of flow of *charge* and is proportional to the difference in voltage across a resistance. Similarly, power is the rate of flow of *heat* energy and is proportional to the difference in temperature across the region through which heat is conducted (thermal resistance). We can consider any impedance to heat flow as *thermal resistance* which is denoted by θ . Similar to the Ohm's law, the rate of flow of heat (i.e. power) is directly proportional to temperature difference and inversely proportional to thermal resistance, that is,

$$P = \frac{\Delta T}{\theta} = \frac{T_2 - T_1}{\theta} \text{ J/s, or W} \quad (15-2)$$

Insulating materials, like wool, have large thermal resistance and metals have small thermal resistance. Solving the above equation for θ , we get the following relation that allows us to determine the value and unit of thermal resistance:

$$\theta = \frac{\Delta T}{P} = \frac{T_2 - T_1}{P} \text{ degrees/W} \quad (15-3)$$

The values of the thermal resistance of various types of heat sinks, washers, and metals can be determined from the thermal resistivity ρ ($^{\circ}\text{C} \cdot \text{in}/\text{W}$ or $^{\circ}\text{C} \cdot \text{m}/\text{W}$) as,

$$\theta = \frac{\rho t}{A} = \frac{T_2 - T_1}{P} ^{\circ}\text{C}/\text{W} \quad (15-4)$$

where, ρ = thermal resistivity of the material

t = thickness of that materials

A = cross-sectional area of the material

In a power transistor heat transfers through different materials and across different boundaries, each of which presents different values of thermal resistances. These are similar to connection of series resistance in electrical systems. Thus, the total thermal resistance will be just the sum of individual thermal resistances. For example, the thermal circuit of a power transistor with heat sink can be represented as shown in Fig.15-5.

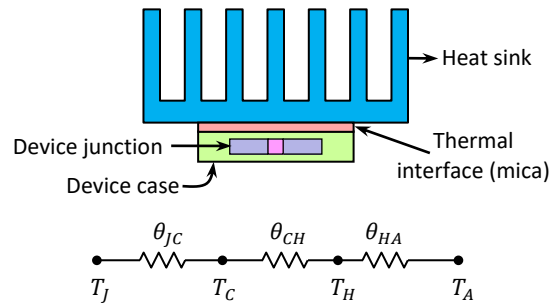


Fig.15-5: Temperature and resistance of a typical heat sink system

$$\theta_{JA} = \theta_{JC} + \theta_{CH} + \theta_{HA} \quad (15-5)$$

where, θ_{JA} = Thermal resistance from junction to air (total thermal resistance)

θ_{JC} = Thermal resistance from junction to case (or body of the transistor)

θ_{CH} = Thermal resistance from case to heat sink (including interface materials)

θ_{HA} = Thermal resistance from heat sink to air (ambient temperature)

T_J = Device (junction) or chip temperature

T_C = Device case (or body) temperature

T_H = Heat sink temperature

Therefore, the rise in temperature of the collector junction (i.e., the transistor) is given by,

$$T_2 - T_1 = \Delta T = \theta_{JA} P_D \quad (15-6)$$

Here, θ_{JA} is given by Equ.(15-5), and P_D is given by Equ.(15-1). Putting the value P_D into the above equation, we get,

$$\Delta T = \theta_{JA} V_{CE} I_C \quad (15-7)$$

Example 14-1

A power transistor used in an amplifier dissipates $V_{CE}I_C = 5 \text{ W}$. The thermal resistance from junction to case is 10°C/W , from case to heat sink is 5°C/W , and from heat sink to air is 5°C/W . The ambient temperature is $T_A = 25^\circ\text{C}$. Calculate the temperature of the case and heat sink.

Solution:

We can assume that all of the 5 W power will be dissipated in the base-collector junction. The total thermal resistance from this junction to air will be,

$$\text{Using Equ. (15-5)} \quad \theta_{JA} = \theta_{JC} + \theta_{CH} + \theta_{HA}$$

Here,

$\theta_{JC} = 10^\circ\text{C/W}$, $\theta_{CH} = 5^\circ\text{C/W}$ and $\theta_{HA} = 5^\circ\text{C/W}$. Putting these values into the above equation,

$$\theta_{JA} = 10^\circ\text{C/W} + 5^\circ\text{C/W} + 5^\circ\text{C/W} = 20^\circ\text{C/W}$$

We will calculate the junction temperature using Equ. (15-6) assuming $T_2 = T_J$, and $T_1 = T_A$ as,

$$\text{Equ. (15-6),} \quad T_J - T_A = \theta_{JA}P_D = 20^\circ\text{C/W} \times 5 \text{ W}$$

$$\therefore T_A = 25^\circ\text{C}, \quad T_J - 25^\circ\text{C} = \theta_{JA}P_D = 100^\circ\text{C/W}$$

$$\therefore \quad T_J = 100^\circ\text{C} + 25^\circ\text{C} = 125^\circ\text{C} \text{ [Ans.]}$$

Similarly, we can calculate the heat sink temperature using Equ. (15-6) assuming $T_2 = T_J$, and $T_1 = T_H$ as,

$$\text{Using Equ. (15-6),} \quad T_J - T_H = \theta_{JH}P_D = (\theta_{JC} + \theta_{CH})P_D$$

$$\text{or,} \quad T_J - T_H = (10^\circ\text{C/W} + 5^\circ\text{C/W})5 \text{ W}$$

$$\text{or,} \quad 125^\circ\text{C} - T_H = 75^\circ\text{C}$$

$$\therefore \quad T_H = 125^\circ\text{C} - 75^\circ\text{C} = 50^\circ\text{C} \text{ [Ans.]}$$

Comments: The heat sink must have sufficient surface area and air circulation to dissipate this heat to air.

Example 4-2

The maximum permissible junction temperature of a certain power transistor is 120°C . It is desired to operate the transistor with a power dissipation of 12 W in an ambient temperature of 40°C . The thermal resistance are θ_{JC} (Junction to case) = 1.5°C/W , and θ_{CA} (Case to ambient) = 10°C/W . Determine whether a heat sink is required for this application or not. If a heat sink is required, determine the maximum thermal resistance it can have. Assume that a mica washer having thermal resistance $\theta_W = 0.5^\circ\text{C/W}$ must be used between the case of the transistor and the heat sink.

Solution:

$$\text{Using Equ. (15-5),} \quad \theta_{JA} = \theta_{JC} + \theta_{CA} = 1.5^\circ\text{C/W} + 10^\circ\text{C/W} = 11.5^\circ\text{C/W}$$

$$\text{Using Equ. (15-6),} \quad T_J - T_A = \theta_{JA}P_D = 11.5^\circ\text{C/W} \times 12 \text{ W} = 138^\circ\text{C}$$

$$\text{or,} \quad T_J = 157.5^\circ\text{C} + T_A = 138^\circ\text{C} + 40^\circ\text{C} = 178^\circ\text{C}$$

Since the junction temperature (178°C) exceeds the maximum permissible temperature of the junction (120°C), a heat sink must be used to reduce the junction temperature.

Assuming $T_2 = 120^{\circ}\text{C}$ (the maximum permissible temperature) and $T_1 = 40^{\circ}\text{C}$ (ambient temperature) in Equ.(15-6),

$$\text{Equ.(15-6),} \quad T_2 - T_1 = 120^{\circ}\text{C} - 40^{\circ}\text{C} = \theta_{JA} P_D = \theta_{JA} \times 12 \text{ W}$$

$$\text{or,} \quad \theta_{JA} = \frac{80^{\circ}\text{C}}{12 \text{ W}} \approx 6.67^{\circ}\text{C/W [Ans.]}$$

If a heat sink is used the case-to-ambient thermal resistance (θ_{CA}) will not preset in the system [in Equ.(15-5)], rather we have to consider the thermal resistance of the mica washer. Thus,

$$\text{Equ.(15-5),} \quad \theta_{JA} = \theta_{JC} + \theta_W + \theta_H$$

$$\text{or,} \quad 6.67^{\circ}\text{C/W} = 1.5^{\circ}\text{C/W} + 0.5^{\circ}\text{C/W} + \theta_H$$

$$\therefore \quad \theta_H = 4.67^{\circ}\text{C/W [Ans.]}$$

Comment: Heatsink reduces the total thermal resistance by increasing the surface area from where heat can easily be radiated to air.

15.2.3 Derating Factor

As already discussed, the maximum power dissipation of a power transistor is specified by the manufacturer at a certain temperature either case temperature or ambient temperature. However, the maximum permissible power rating decreases as a function of temperature, when temperature increases beyond the given value. The decrease in allowable power dissipation at elevated temperatures is called *derating factor* and is measured in $\text{W}/^{\circ}\text{C}$. This parameter is used to find the decrease in dissipation rating beyond the certain temperature. Typical derating curves as a function of ambient or case temperature of power transistor are shown in Fig.15-. From these graphs, it is clear that as ambient or case temperature increases the power rating of the device decreases. For small and medium scale power transistors, data sheets usually give derating factors of ambient temperature but for large power transistors data sheets provide derating factors of case temperature.

A relationship between the decrease in power rating with the increase in ambient temperature and the derating factor can be established as,

$$\Delta P = D_A (T_A - 25^{\circ}\text{C}) \quad (15-8)$$

where,

ΔP = decrease in power rating of the transistor

D_A = derating factor as a function of ambient temperature

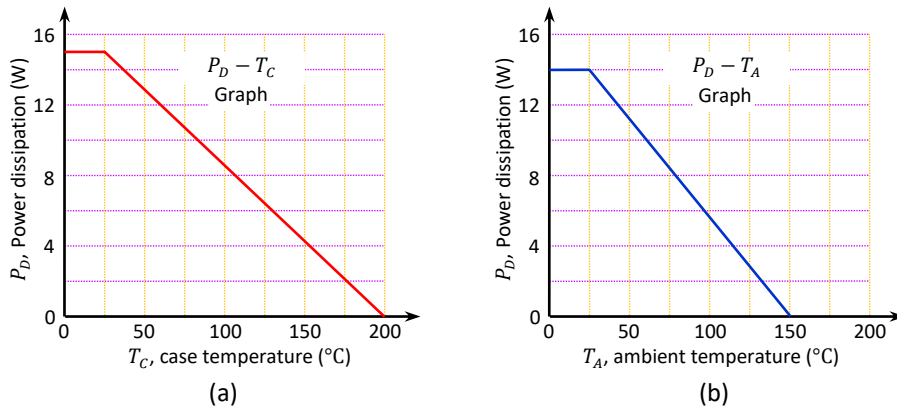


Fig.15-6: Derating graphs: (a) As a function of T_C , and (b) As a function of T_A

Similarly, for the transistor case temperature, the relationship is,

$$\Delta P = D_C(T_A - 25^{\circ}\text{C}) \quad (15-9)$$

where,

ΔP = decrease in power rating of the transistor

D_C = derating factor as a function of case temperature

Example 14-3

A power transistor has a maximum rated dissipation of $P_D = 5 \text{ W}$ at 50°C case temperature. The thermal resistance from case to ambient is $5^{\circ}\text{C}/\text{W}$.

- Can the device be operated at 5 W of dissipation without any heat sink when the ambient temperature is 40°C ?
- If not, what is the maximum permissible dissipation, with no heat sink, at the ambient temperature is 40°C .
- What derating factor should be applied to this device?

Solution:

- We will determine the case temperature when the ambient temperature is 40°C . Using Equ.(15-6) and considering $T_2 =$ case temperature T_C , and $T_1 =$ ambient temperature T_A ,

$$\text{Equ.(15-6),} \quad T_2 - T_1 = \Delta T = \theta_{CA} P_D$$

$$\text{or,} \quad T_C - T_A = 5^{\circ}\text{C}/\text{W} \times 5 \text{ W}$$

$$\text{or,} \quad T_C - 40^{\circ}\text{C} = 25^{\circ}\text{C}$$

$$\therefore \quad T_C = 65^{\circ}\text{C}$$

Since the case temperature exceeds 50°C , the full 5 W of power dissipation is not possible when the ambient temperature is 40°C .

- The derated dissipation at $T_A = 40^{\circ}\text{C}$ will be,

$$P_{D(max)} = \frac{T_C - T_A}{\theta_{CA}} = \frac{50^\circ\text{C} - 40^\circ\text{C}}{5^\circ\text{C}/\text{W}}$$

or, $P_{D(max)} = 2 \text{ W}$

- iii) We first find the maximum ambient temperature at which the device can dissipate 5 W when the case temperature is 50°C,

$$P_{D(max)} = \frac{T_C - T_A}{\theta_{CA}}$$

or, $5 \text{ W} = \frac{50^\circ\text{C} - T_A}{5^\circ\text{C}/\text{W}}$

or, $25^\circ\text{C} = 50^\circ\text{C} - T_A$

$\therefore T_A = 50^\circ\text{C} - 25^\circ\text{C} = 25^\circ\text{C}$

Using Equ.(15-9), $\Delta P = D_C(T_A - 25^\circ\text{C})$

or, $5 \text{ W} - 2 \text{ W} = D_C(40^\circ\text{C} - 25^\circ\text{C})$

$\therefore D_C = \frac{3 \text{ W}}{15^\circ\text{C}} = 0.2 \text{ W}/^\circ\text{C} [\text{Ans.}]$

Comments: Derating factor $D_C = 0.2 \text{ W}/^\circ\text{C}$ means that the power dissipation capacity of the transistor will decrease by 0.2 W from 5 W for every 1°C temperature rise in ambient temperature from 25°C.

15.3 Classification of Power Amplifiers

The main operating characteristics of an amplifier are linearity, signal gain, efficiency and power output. But in real world amplifiers, there is always a trade-off between these different characteristics. Generally, power amplifiers are used at the output stages of audio amplifier systems to drive a loudspeaker load. A typical loudspeaker has an impedance of between 4 Ω to 16 Ω, thus a power amplifier must be able to supply the high peak currents required to drive the low impedance loads.

One method used to distinguish the electrical characteristics of different types of amplifiers is by *class*. The classes are related to the *time period* (or *conduction angle*) that the active device is passing current, expressed as a fraction of the period of a sinusoidal waveform applied to the input. Amplifier classes are mainly lumped into two basic groups. The first are the classically controlled conduction angle amplifiers forming the more common amplifier classes of A, B,

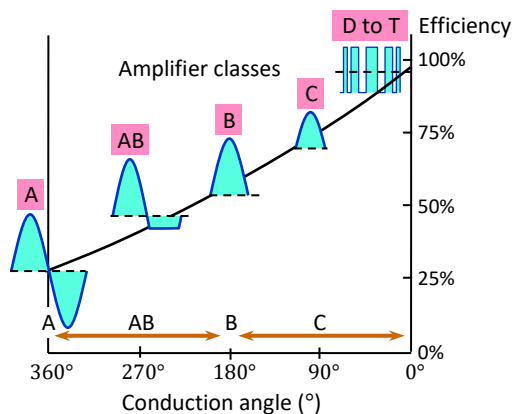


Fig.15-7: Waveforms (conduction angles) of different amplifier classes

AB and C, which are defined by the length of their conduction state over some portion of the output waveform. The second set of amplifiers are the newer *switching* amplifier classes of D, E, F, G, S, T etc, which use digital circuits and *pulse-width modulation* (PWM) to constantly switch the signal between fully-ON and fully-OFF driving the output transistors into fully saturation or fully cut-off regions.

The most commonly constructed amplifier classes which are mainly used as audio amplifiers are: class-A, B, AB and C amplifiers. As shown in Fig.15-7, the efficiency of power amplifiers increases from class-A to class-C. Descriptions of these amplifiers are given below.

Class-A Amplifiers

Class-A amplifiers are the most common type of amplifier topology as they use just one output transistor (BJT, FET, IGBT, etc). This single output transistor is biased at a quiescent current I_Q so that its Q -point lies in the middle of its load line and thus allowing it to conduct current over the full 360° of the input signal. Then, the output transistor of a class-A topology never turns OFF, which is one of its main disadvantages. Class-A amplifiers provide excellent linearity, high gain and low signal distortion levels when designed correctly. Although seldom used in high power amplifier applications due to low efficiency, class-A amplifiers are used in high-fidelity (low distortion) audio amplifier designs.

Class-B Amplifiers

Class-B amplifiers were invented as a solution to the low efficiency and heating problems associated with class-A amplifiers. The basic class-B amplifiers use two complimentary transistors either bipolar or FET and every device conducts for one half-cycle or 180° of the input signal. In class-B amplifiers, the transistors are biased at the cut-off point, that is, the value of DC quiescent current is zero, so the DC power loss is small and therefore its efficiency is much higher than that of the class-A amplifiers. However, the price paid for the improved efficiency is the reduced linearity of the devices. Class-B amplifiers suffer from cross-over distortion (discussed elaborately later).

Class-AB Amplifiers

As its name suggests, class-AB amplifier is a combination of class-A and class-B type amplifiers. The class-AB amplifier is currently one of the most common types of audio power amplifier. In fact the class-AB amplifier is a variation of a class-B amplifier. The main difference is that the transistors are biased a bit above the cut-off point, that is, the value of DC quiescent current is not zero, so the devices conduct current for greater than one half-cycle or greater than 180° but much less than 360° . This increased conduction angles help eliminating the cross-over distortion problems of the class-B amplifier.

Class-C Amplifiers

Class-C amplifiers have the greatest efficiency but the poorest linearity of all the classes. The previous classes, A, B and AB are considered linear amplifiers, as the output signals amplitude and phase are linearly related to the input signals amplitude and phase. However, in class-C amplifiers the transistors are biased below the cut-off point, and hence, the devices

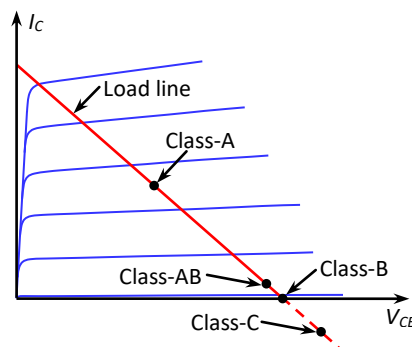


Fig.15-8: Position of operating point of different classes of amplifiers

conduct current for less than one half-cycle or less than 180° . While this form of transistor biasing gives a much improved efficiency of around 80% to the amplifier, it introduces severe distortion to the output signal. Therefore, class-C amplifiers are not suitable for use as audio amplifiers.

15.3.1 Switching Type Amplifiers

A Class-D Amplifier

A class-D amplifier or switching amplifier is an electronic amplifier in which the amplifying devices (transistors or MOSFETs) operate as electronic switches, and not as linear devices as in other amplifiers. They operate by rapidly switching back and forth between cut-off and saturation, being fed by a PWM (Pulse-Width Modulated) signal. The PWM signal is modulated by the audio input signal. The output PWM signal passes through a low-pass filter that allows only the amplified audio signal to be fed into the loudspeaker. Class-D amplifiers theoretically can reach 100% efficiency, as there is no period during a cycle where the voltage and current waveforms overlap. Class-D amplifier is discussed elaborately in Section [H_S_atten].

Class-F Amplifier

Class-F amplifiers boost both efficiency and output by using harmonic resonators in the output network to shape the output waveform into a square wave. Class-F amplifiers are capable of high efficiencies of more than 90% if infinite harmonic tuning is used. Fig.15-9 shows a class-F amplifier with two harmonic resonators and the output waveforms.

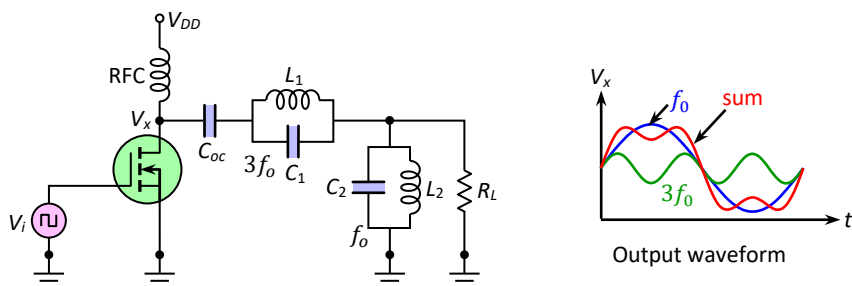


Fig.15-9: Class-F amplifier with harmonic resonators

Class-G Amplifier

Class-G offers enhancements to the basic class-AB amplifier design. Class-G uses multiple power supply rails of various voltages and automatically switches between these supply rails as the input signal changes (as shown in Fig.15-10). This constant switching reduces the average power consumption, and therefore reduces power loss caused by wasted heat.

Class-I Amplifier

The class-I amplifier has two sets of complementary output switching devices arranged in a parallel push-pull configuration with both sets of switching devices sampling

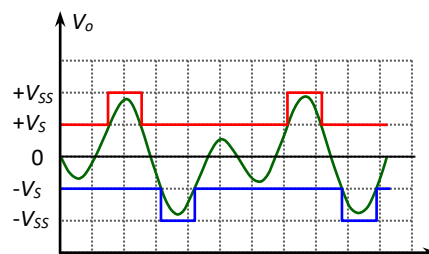


Fig.15-10: Output waveforms with rail change of class-G amplifier

the same input waveform. One device switches the positive half of the waveform, while the other switches the negative half, similar to a class-B amplifier. With no input signal applied, or when a signal reaches the zero crossing point, the switching devices are both turned ON and OFF simultaneously with a 50% PWM duty cycle cancelling out any high frequency signals.

Class-S Amplifier

A class-S power amplifier is a non-linear switching mode amplifier similar in operation to the class-D amplifier. The class-S amplifier converts analogue input signals into digital square wave pulses by a delta-sigma modulator, and amplifies them to increase the output power before finally being demodulated by a band-pass filter. As the digital signal of this switching amplifier is always either fully ON or OFF (theoretically zero power dissipation), efficiencies reaching 100% are possible.

Class-T Amplifier

The class-T amplifier is another type of digital switching amplifier design. These amplifiers are becoming more popular these days as an audio amplifier design due to the existence of digital signal processing (DSP) chips and multi-channel surround sound amplifiers. These amplifiers convert analogue signals into digital pulse-width modulated (PWM) signals for amplification which increase the amplifiers efficiency. Class-T amplifiers provide both the low distortion signal of class-AB amplifier and the power efficiency of a class-D amplifier.

15.4 Class-A Power Amplifier

Already we have discussed about class-A power amplifier. Class-A power amplifier is one in which the output current (transistor collector current) flows during the entire cycle [Fig.15-11(b)] of input signal i.e., the conduction angle of the transistor is 360° . All small-signal amplifiers, we have studied in previous chapters, have been designed so that the output voltage can vary in response to both positive and negative half-cycles of input; that is, the transistors (or FETs) are biased so that under normal operation the output never saturates or cuts off. An amplifier that has this type of property is called a class-A amplifier. Alternately, an amplifier is class-A if its output remains in the active region during a complete cycle (full period) of an input sine-wave signal. Thus, a class-A amplifier may be a small signal amplifier or a large signal amplifier. The main difference in these two designs is: in small signal amplifier the collector current is very small and the value of collector resistor is high, on the other hand, in a large signal amplifier the collector current is large and the load resistance (R_C) is very small. As R_C is very small, the voltage gain of these amplifiers will be low, but the current gain will be very high. Generally, the load (R_L) is connected to the collector, that is, the collector resistance will be the load resistance [in this description R_L and R_C have been used interchangeably].

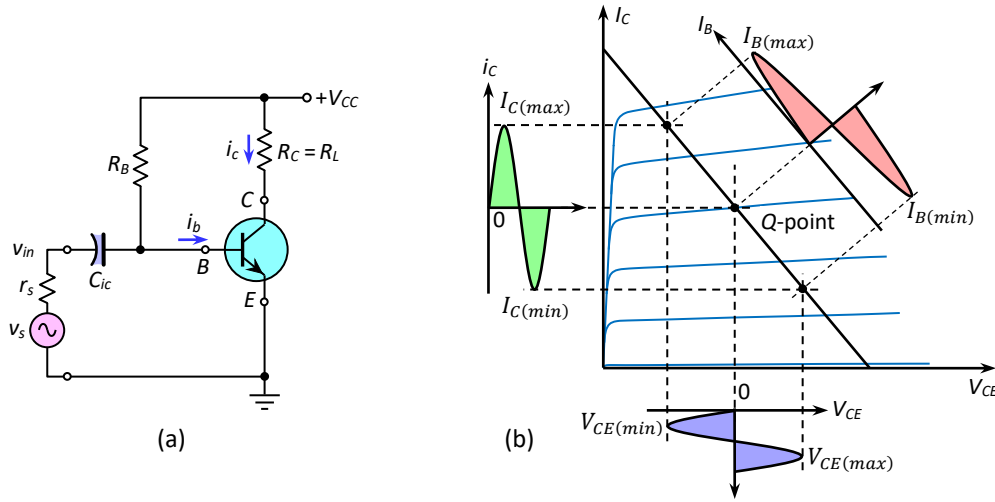


Fig.15-11: (a) Class-A power amplifier, and (b) Output characteristics of this amplifier a function of this amplifier

Fig.15-11(a) shows a CE connected transistor forming a class-A amplifier while Fig.15-11(b) shows the output characteristic with a Q -point at the centre of the load line. Here, I_{CQ} and V_{CEQ} represent quiescent (no signal) collector current, and C-E voltage, respectively. To ensure the position of the Q -point at the middle of the load line we have to set $I_{CQ} = I_{C(max)}/2 = V_{CC}/2R_C$, and $V_{CEQ} = V_{CC}/2$. When an AC input signal is applied Q -point moves along the load line from its centre position. The output current (I_C) increases to $I_{C(max)}$ and decreases to $I_{C(min)}$. Similarly, collector-emitter voltage increases to $V_{CE(max)}$, and decreases to $V_{CE(min)}$. This large collector current and C-E voltage swings will produce a large AC power in the load resistance R_C .

15.4.1 Efficiency of Class-A Amplifier

So long as the output signal variations are confined to linear region of load line, the average value of collector current will be I_{CQ} because, positive and negative input signal swings will produce equal changes in I_{CQ} . Therefore, the average DC power taken from the supply source (V_{CC}) will be,

$$P_{i(DC)} = V_{CC}I_{CQ} \quad (15-10)$$

The value of AC power supplied to the load ($R_C = R_L$) will be,

$$P_{o(AC)} = V_{rms}I_{rms} = \frac{V_p}{\sqrt{2}} \frac{I_p}{\sqrt{2}} = \frac{V_p I_p}{2} \quad (15-11)$$

Here, V_{rms} and I_{rms} are the RMS voltage and current developed in the load. From Fig.15-11(b) we can write,

$$V_{pp} = V_{CE(max)} - V_{CE(min)}$$

or,

$$V_p = \frac{V_{CE(max)} - V_{CE(min)}}{2}$$

∴

$$V_{rms} = \frac{V_p}{\sqrt{2}} = \frac{V_{CE(max)} - V_{CE(min)}}{2\sqrt{2}} \quad (15-12)$$

Here, V_{pp} is the peak-to-peak output voltage and V_p is the peak output voltage are developed across the load R_L . Similarly,

$$I_{pp} = I_{C(max)} - I_{C(min)}$$

or,

$$I_p = \frac{I_{C(max)} - I_{C(min)}}{2}$$

∴

$$I_{rms} = \frac{I_p}{\sqrt{2}} = \frac{I_{C(max)} - I_{C(min)}}{2\sqrt{2}} \quad (15-13)$$

Here, I_{pp} is the peak-to-peak output current, and I_p is the peak output current passing through the load R_L .

Using Equ.(15-11), Equ.(15-12), and Equ.(15-13) we get,

$$P_{out(AC)} = \frac{V_{CE(max)} - V_{CE(min)}}{2\sqrt{2}} \times \frac{I_{C(max)} - I_{C(min)}}{2\sqrt{2}}$$

or,

$$P_{out(AC)} = \frac{[V_{CE(max)} - V_{CE(min)}][I_{C(max)} - I_{C(min)}]}{8} \quad (15-14)$$

Like any other electronic systems, the efficiency of a power amplifier is defined as,

$$\eta = \frac{\text{AC power delivered to load}}{\text{Average power drawn from DC supply source}}$$

or,

$$\eta = \frac{P_{o(AC)}}{P_{i(DC)}} \quad (15-15)$$

Using Equ.(15-10) and Equ.(15-11),

$$\eta = \frac{V_p I_p / 2}{V_{CC} I_{CQ}} = \frac{V_p I_p}{2 V_{CC} I_{CQ}} \quad (15-16)$$

Alternately, using Equ.(15-10), and Equ.(15-14) we get,

$$\eta = \frac{[V_{CE(max)} - V_{CE(min)}][I_{C(max)} - I_{C(min)}]}{8 V_{CC} I_{CQ}} \quad (15-17)$$

15.4.2 Standby Power Loss and Efficiency

An amplifier is said to be in *standby*, when there is no input signal and no AC output developed in the load. If there is no AC input signal, the Q -point will not swing, but I_{CQ} amount of current will be drawn from the supply V_{CC} . So the power drawn from the supply will still remain same, that is,

$$P_{in(sb)} = V_{CC} I_{CQ} \quad (15-18)$$

As the Q -point is not changing, therefore, $V_{CE(max)} = V_{CE(min)} = V_{CEQ}$, and $I_{C(max)} = I_{C(min)} = I_{CQ}$. Putting these values in Equ.(15-17), we get the efficiency at standby condition as,

$$\eta_{sb} = \frac{[V_{CEQ} - V_{CEQ}][I_{CQ} - I_{CQ}]}{8 V_{CC} I_{CQ}} = 0 \quad (15-19)$$

Thus, we find that the standby power loss of Class-A amplifier is same ($V_{CC}I_{CQ}$) as that in working condition, and the standby efficiency is zero.

15.4.3 Maximum Efficiency

To get the maximum efficiency, the operating point has to be set exactly at the middle of load line, that is,

$$I_{CQ} = \frac{V_{CC}}{2R_C} \quad (15-20)$$

and,
$$V_{CEQ} = \frac{V_{CC}}{2} \quad (15-21)$$

To get the theoretical maximum efficiency, we will neglect the cut-off and saturation regions of the transistor output characteristics. Again, we will assume that the input signal is sufficient enough to make the maximum output current and voltage swings as shown in Fig.15-12.

For this type of operation, we find,

$$V_{CE(max)} = V_{CC}$$

$$V_{CE(min)} = 0$$

$$I_{C(max)} = \frac{V_{CC}}{R_L}$$

and,
$$I_{C(min)} = 0$$

Putting these values in Equ.(15-19) we get,

$$\eta_{max} = \frac{[V_{CC} - 0][V_{CC}/R_L - 0]}{8V_{CC}I_{CQ}}$$

$$\eta_{max} = \frac{V_{CC}}{8R_L I_{CQ}}$$

Putting the value of I_{CQ} from Equ.(15-20) we get,

$$\eta_{max} = \frac{V_{CC}}{8R_L (V_{CC}/2R_L)} = \frac{1}{4}$$

$$\therefore \boxed{\eta_{max} = 25\%} \quad (15-22)$$

Although Equ.(15-22) gives the maximum efficiency of class-A amplifier of 25%, but the practical class-A amplifier will have much below this value [see Example 14-4].

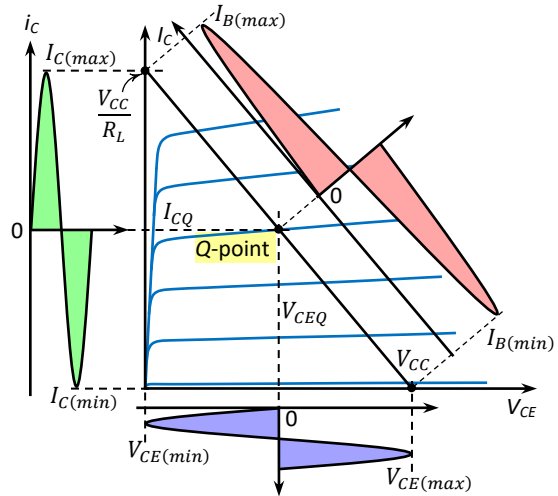


Fig.15-12: Output characteristics of class-A amplifier in maximum output power condition

Example 14-4

For the Class-A amplifier of Fig.15-14, draw the load line, calculate the input, output, and standby power loss and efficiency. Assume $V_{CE(sat)} = 1\text{ V}$, $I_{CEO} = 0.5\text{ mA}$ for the transistor.

Solution:

The minimum value of V_{CE} will be,

$$V_{CE(min)} = V_{CE(sat)} = 1\text{ V}$$

and the maximum value of V_{CE} ,

$$V_{CE(max)} = V_{CC} - I_{CEO}R_L$$

$$\therefore V_{CE(max)} = 16\text{ V} - (0.5\text{ mA})(8\ \Omega) \approx 16\text{ V}$$

The minimum value of I_C ,

$$I_{C(min)} = I_{CEO} = 0.5\text{ mA}$$

and, the maximum value of I_C ,

$$I_{C(max)} = \frac{V_{CC} - V_{CE(sat)}}{R_L} = \frac{16\text{ V} - 1\text{ V}}{8\ \Omega} = 1.875\text{ A}$$

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16\text{ V} - 0.7\text{ V}}{750\ \Omega} = 20.4\text{ mA}$$

Therefore, the collector current will be,

$$I_{CQ} = I_{BQ} \times \beta = 20.4\text{ mA} \times 50 = 1.02\text{ A}$$

Therefore, the quiescent collector-emitter voltage will be,

$$V_{CEQ} = V_{CC} - I_{CQ}R_L = 16\text{ V} - 1.02\text{ A} \times 8\ \Omega = 7.84\text{ V}$$

As the AC and DC load resistance of this circuit is same, the AC and DC load line will also be same which is shown in Fig.15-14.

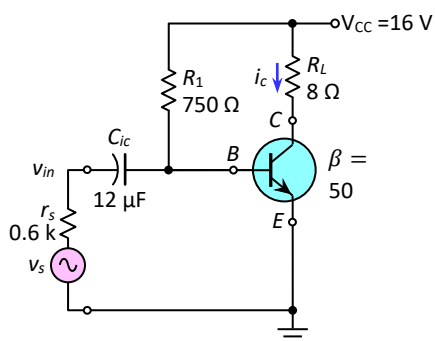


Fig.15-14: Class-A amplifier for Example 14-4

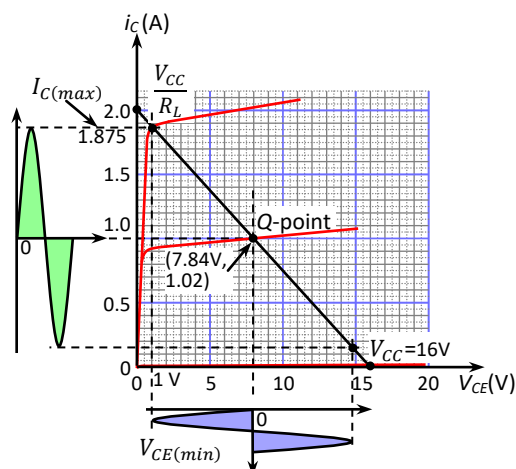


Fig.15-14: Output characteristics of Class-A amplifier for Example 14-4

From Fig.15-14, for V_{CE} the peak value of swing will be,

Increasing, $V_p = V_{CE(max)} - V_{CEQ} = 16 \text{ V} - 7.84 \text{ V} = 8.16 \text{ V}$

Decreasing $V_p = V_{CEQ} - V_{CE(min)} = 7.84 \text{ V} - 1 \text{ V} = 6.84 \text{ V}$

For I_C the peak value of swing will be,

Increasing, $I_p = I_{C(max)} - I_{CQ} = 1.875 \text{ A} - 1.02 \text{ A} = 0.855 \text{ A}$

Decreasing $I_p = I_{CQ} - I_{C(min)} = 1.02 \text{ A} - 0.5 \text{ mA} = 1.02 \text{ A}$

For proper functioning of the amplifier the current and voltage swings should be symmetrical (positive and negative half-cycles are equal). Therefore, we have to take the minimum values of V_p and I_p calculated above. Thus, the possible $V_p = 6.84 \text{ V}$, and $I_p = 0.855 \text{ A}$.

Using Equ.(15-10), $P_{i(DC)} = V_{CC}I_{CQ} = 16 \text{ V} \times 1.02 \text{ A} = 16.32 \text{ W}$ [Ans.]

Using Equ.(15-16), $\eta = \frac{V_p I_p}{2V_{CC}I_{CQ}} = \frac{(6.84 \text{ V})(0.855 \text{ A})}{2(16 \text{ V})(1.02 \text{ A})} \approx 18\%$ [Ans.]

Power drawn by the base circuit will be,

$$P_{in(B)} = I_{BQ} \times V_{CC} = 20.4 \text{ mA} \times 16 \text{ V} \approx 0.33 \text{ W}$$
 [Ans.]

Considering this biasing power loss, the efficiency of the class-A amplifier will be,

$$\eta = \frac{V_p I_p}{2V_{CC}I_{CQ} + P_{in(B)}}$$

$$\eta = \frac{(6.84 \text{ V})(0.855 \text{ A})}{2(16 \text{ V})(1.02 \text{ A}) + 0.33 \text{ W}} \approx 17.7\%$$
 [Ans.]

Comments: If the bias current is significant then we have to take it into consideration otherwise we can neglect this current.

15.5 Transformer Coupled Class-A Amplifier

Fig.15-15(a) shows the circuit diagram of a class-A transformer coupled amplifier. R_1 and R_2 provide voltage divider biasing for the transistor. R_E is the emitter resistor for bias stabilization. C_E is the bypass capacitor for R_E to prevent AC voltage from appearing across R_E which increases AC voltage gain. The input signal is applied to the input terminal (base of the transistor) via the input coupling capacitor C_{iC} . This amplifier is similar to the previously discussed class-A amplifier. The only difference is instead of connecting the load directly to the collector, transformer coupling is used. The primary coil of a transformer (T_1) is connected to

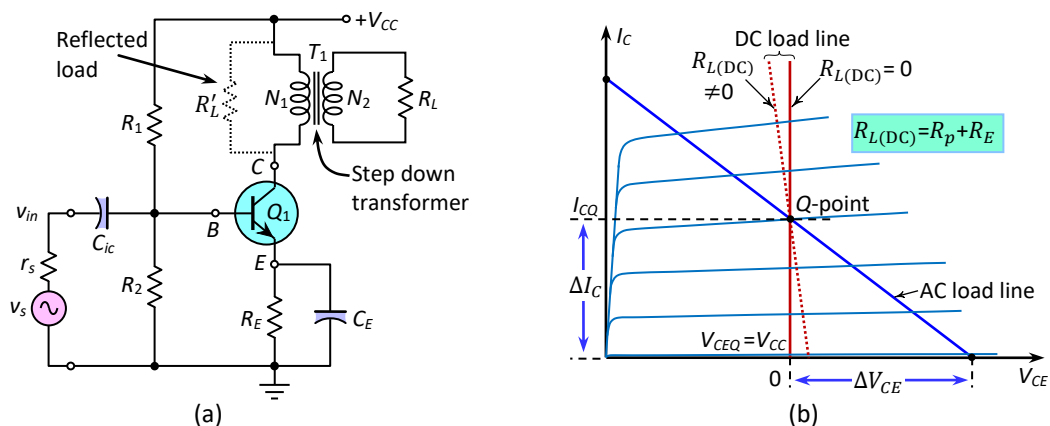


Fig.15-15: Transformer coupled Class-A power amplifier: (a) Schematic diagram, and (b) Output characteristics

the collector, and the load is connected to the secondary of the transformer. The transformer is used for impedance matching purpose. The primary coil provides high impedance to the high impedance collector circuit and the secondary coil provides low impedance to the low impedance voice coil of loudspeaker (typically 4 to 16 Ω). In addition to impedance matching, transformer coupling blocks the flow of DC current into the load. In class-A condition, transistor operates during the whole input cycle and results in distortion free amplification provided the device is linear. In order to get maximum AC output power, the maximum swing of the collector current, due to input signal, should be equal to the quiescent collector current (I_{CQ}), that is, the Q-point should be located at the middle of load line [as shown in Fig.15-15(b)]. The external load (R_L) is coupled to the collector by the transformer. If the turn ratio of the transformer is $a = N_1/N_2$, the reflected load into the collector circuit will be $R'_L = a^2 R_L$.

15.5.1 DC and AC Load Line

Fig.15-15(b) illustrates idealized characteristics of a common emitter configuration with load lines. Here, the DC and AC load lines are different as the DC load resistance is different from the AC load resistance. The total DC load resistance for the amplifier of Fig.15-15(a) is the sum of the emitter resistance (R_E), and the transformer primary winding resistance, R_p . Therefore,

$$R_{L(DC)} = R_E + R_p \quad (15-23)$$

Using this DC resistance, we can find the collector-emitter voltage as,

$$V_{CEQ} = V_{CC} - I_{CQ} R_{L(DC)} = V_{CC} - I_{CQ} (R_E + R_p) \quad (15-24)$$

Using this equation, the DC load line can be drawn which is shown in Fig.15-15(b).

To draw the AC load line, we must know the effective AC resistance at the transistor collector. To do that, let us separately consider the coupling transformer [Fig.15-16].

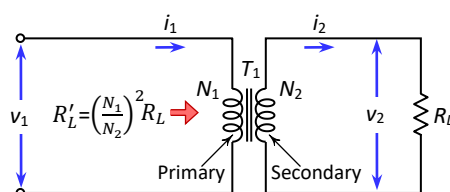


Fig.15-16: Function of the transformer: R_L connected at the secondary terminals is reflected to primary as $(N_1/N_2)^2 R_L$

Let N_1 be the number of turns in the primary winding, and N_2 be the number of turns in the secondary winding. Let the primary AC voltage and current be v_1 , and i_1 , respectively and the secondary AC voltage and current be v_2 , and i_2 , respectively. R_L is the load resistor that has to be coupled to the amplifier. This resistance can be calculated as,

$$R_L = \frac{v_2}{i_2} \quad (15-25)$$

The AC load resistance at the transformer primary can be determined as,

$$R'_L = \frac{v_1}{i_1} \quad (15-26)$$

From basic transformer theory, we can write,

$$\frac{v_1}{v_2} = \frac{N_1}{N_2} = \frac{i_2}{i_1}$$

$$\text{or,} \quad v_1 = \frac{N_1}{N_2} v_2, \quad \text{and} \quad i_1 = \frac{N_2}{N_1} i_2 \quad (15-27)$$

Putting the values of v_1 , and i_1 into Equ.(15-) we get,

$$R'_L = \frac{N_1}{N_2} v_2 \times \frac{N_1}{N_2 i_2} = \left(\frac{N_1}{N_2} \right)^2 \left(\frac{v_2}{i_2} \right) \quad (15-28)$$

Now, using Equ.(15-25), we find,

$$\boxed{R'_L = \left(\frac{N_1}{N_2} \right)^2 R_L} \quad (15-29)$$

R'_L is the load resistance seen from the primary of the transformer only for AC condition. This resistance is called the *reflected load* (R'_L). The total AC resistance at the collector of the transistor will be the sum of this reflected load and the transformer primary winding resistance, that is,

$$\boxed{R_{L(AC)} = R'_L + R_p = \left(\frac{N_1}{N_2} \right)^2 R_L + R_p} \quad (15-30)$$

In Chapter 5 (Section 5.4.1), we found that the AC load line passes through the DC operating point. Therefore, the AC load line will be found by drawing a line through the operating point with a slope of $-1/R_{L(AC)}$. This is shown in Fig.15-15(b) and illustrated further in Example 14-5.

Example 14-5

For the transformer coupled class-A amplifier of Fig.15-18(a), determine the operating point and draw the DC and AC load line. Assume, $R_p = 3 \Omega$, $N_1 = 142$ and $N_2 = 50$. The transistor has $\beta = 100$. Moreover, assume that the external capacitors are short for the range of operating frequencies.

Solution:

Q-point and DC load line:

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{3.6 \text{ k} + (100 + 1) 2} \approx 4 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 100 \times 4 \text{ mA} \approx 0.4 \text{ A}$$

$$V_{CEQ} = 16 \text{ V} - (0.4 \text{ A})(3 \Omega + 2 \Omega) = 14 \text{ V}$$

Now the DC load line is drawn by connecting two points $(V_{CC}, 0) = (16 \text{ V}, 0)$, and $(V_{CEQ}, I_{CQ}) = 14 \text{ V}, 0.4 \text{ A}$. The Q-point, $(14 \text{ V}, 0.4 \text{ A})$, will be on this load line.

AC load line

Using Equ.(15-30),
$$R_{L(AC)} = R'_L + R_p = \left(\frac{N_1}{N_2}\right)^2 R_L + R_p$$

or,
$$R_{L(AC)} = \left(\frac{142}{50}\right)^2 4 \Omega + 3 \Omega \approx 35.3 \Omega$$

For a change in I_C by $\Delta I_C = I_{CQ} = 0.4 \text{ A}$, the corresponding change in V_{CE} will be,

$$\Delta V_{CE} = \Delta I_C \times R_{L(AC)} = 0.4 \text{ A} \times 35.3 \Omega \approx 14 \text{ V}$$

If we assume that I_C has been decreased by 0.4 A from the Q-point, then V_{CE} will increase by 14 V from the Q-point. Let us denote this point as A on the x-axis. Now, we have to draw a line connecting the Q-point and point A, and extend this line up to y-axis. This will be the AC load line [shown in Fig.15-18(b)].

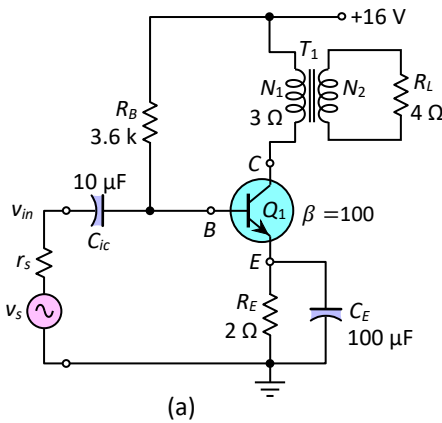


Fig.15-18 (a): Class-A amplifier for Example 14-5

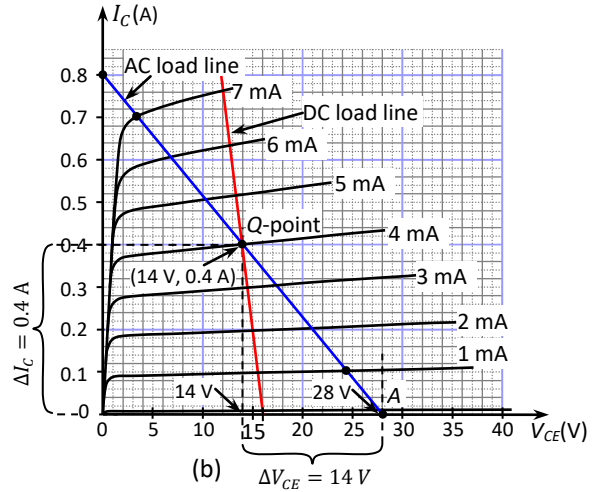


Fig.15-18 (b): DC and AC load lines for Example 14-5

Comments: The AC load line exceeds the supply voltage $V_{CC} = 16 \text{ V}$ due to the induced voltage in the transformer.

15.5.2 Operation Process

Let us assume that a sinusoidal input is applied to the amplifier. For this input signal, the base current swings around the quiescent base current (I_{BQ}). These swings in the base current are multiplied by h_{fe} of the transistor and amplified swings are produced in the collector current. Due to these changes in the collector current, voltage swings are also produced at the output. These AC current and voltage swings (AC power) are applied to the transformer's primary coil. The transformer couples this power to the secondary coil and hence to the load (R_L) with appropriate change in current and voltage levels (suitable for the load).

To make the concept clearer, the AC load line of Example 14-5 is redrawn in Fig.15-19. When the input signal causes I_B to increase from 4 mA (at I_{BQ}) to 7 mA, the output current and voltage become $I_C = 0.7$ A and $V_{CE} = 4$ V, (point B on the AC load line). The changes are: $\Delta I_C = 0.7$ A – 0.4 A = +0.3 A, and $\Delta V_{CE} = 4$ V – 14 V = –10 V. When the input signal causes I_B to decrease from I_{BQ} (= 4 mA) to 1 mA, I_C changes from 0.4 A to 0.1 A, and the V_{CE} changes from 14 V to 24 V, (point C). Now the current and voltage changes are: $\Delta I_C = -0.3$ A, and $V_{CE} = +10$ V. From this figure, it is seen that an I_B change of ± 3 mA produces a ± 0.3 A change in I_C and a ± 10 V change in V_{CE} . The V_{CE} variation appears at the primary winding of transformer (T_1) and the I_C variation flows in the primary winding. Thus, a power is developed in the primary winding, and this power is coupled to the load (R_L) by the transformer.

Note that, although $V_{CC} = 16$ V (in Fig.15-18), the transistor V_{CE} can actually go up to 28 V. This is due to the inductive effect of the transformer primary winding. Therefore, the transistor used in this type of amplifier should have a minimum breakdown voltage rating approximately equal to $2V_{CC}$.

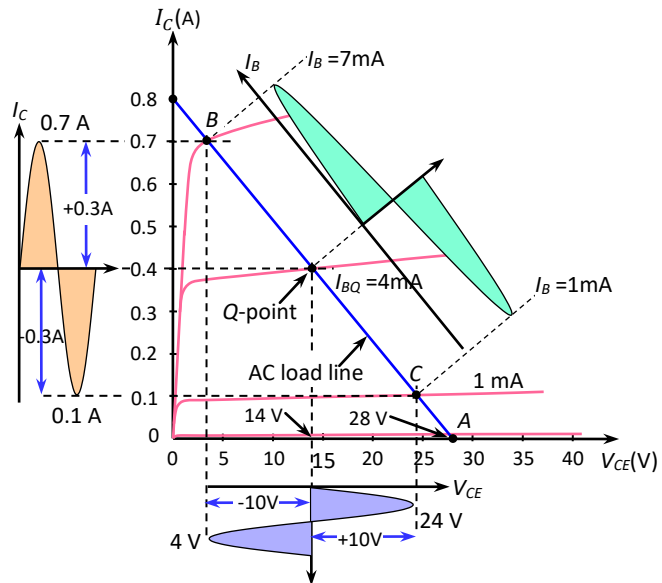


Fig.15-19: Transformer-coupled class-A amplifier with its Q-point at the centre of the load line

15.5.3 Efficiency of Transformer-Coupled Class-A Amplifier

In all amplifiers, power is drawn from the DC supply source. The amplifier converts this DC power into AC power, and supplies the AC power to the load. In this process, some of the input power is dissipated in the transistor and in other components. The different losses in a transformer-coupled class-A amplifier are:

Power loss in the primary of transformer: The primary of the transformer has some finite resistance R_p . When current passes through this resistance, there is some loss of power. As the resistance of the primary is very small, and hence, the loss in the primary is negligibly small.

Power loss in the core of transformer: The power loss may be in the form of hysteresis loss and eddy current loss. These losses are also quite small.

Power loss in the resistance R_E : Some DC power is dissipated in the emitter resistance R_E . This loss is small again.

Power loss in biasing components: Whatever the type of biasing circuit, some power will be dissipated in biasing network, because, it will draw current from the supply source. But for most of the cases this loss is also very small and can be neglected.

Power loss in the transistor itself: The main power loss occurs in the transistor itself. All other losses are small as compared to this loss. As the voltage across the C-E junction is not zero, power loss occurs when current i_C flows the transistor, and its value is $v_{CE}i_C$. This loss occurs mainly across the reverse biased B-C junction.

The efficiency of power amplifier has been defined in Section 14.4.1 [H_S_atten]. The efficiency of this amplifier will be,

$$\text{Equ. (15-15),} \quad \eta = \frac{P_{o(AC)}}{P_{i(DC)}} \quad (15-31)$$

where, $P_{o(AC)}$ is the net AC power supplied to the load and $P_{i(DC)}$ is the total DC power absorbed from the supply source.

Like the class-A amplifier, here also the average current will be I_{CQ} . Therefore, the DC power taken from the supply source (V_{CC}) will be (neglecting the loss in biasing circuit),

$$\text{Equ. (15-10),} \quad P_{i(DC)} = V_{CC}I_{CQ} \quad (15-32)$$

The AC voltage swing and current swing provide AC power to the primary of the transformer. If the peak values of these swings are I_p and V_p , the power supplied to the transformer primary (P_{Tp}) will be,

$$P_{Tp} = \frac{V_p}{\sqrt{2}} \times \frac{I_p}{\sqrt{2}} = \frac{V_p I_p}{2} \quad (15-33)$$

If the transformer used in this amplifier has an efficiency of η_t , then the output power supplied to the load (P_o) will be,

$$P_{o(AC)} = P_{Tp} \times \eta_t = \frac{V_p I_p}{2} \times \eta_t \quad (15-34)$$

Using Equ.(15-32) and Equ.(15-34) into Equ.(15-31), we get,

$$(15-31) \quad \boxed{\eta = \frac{P_{o(AC)}}{P_{i(DC)}} = \frac{V_p I_p \eta_t}{2V_{CC}I_{CQ}}} \quad (15-35)$$

Like the normal class-A amplifier, the standby power loss will be,

$$P_{i(sb)} = V_{CC}I_{CQ} \quad (15-36)$$

15.5.4 Maximum Efficiency

To achieve the maximum efficiency, we have to use highly efficient transformer, that is $\eta_t \approx 1$ and $R_p \approx 0$. We should not use the emitter resistance (R_E). With these considerations ($R_{L(DC)} = R_E + R_p = 0$), the DC load line will be a vertical line through $V_{CEQ} = V_{CC}$ as shown Fig.15-15(b). Moreover, the AC load resistance ($R_{L(AC)} = R'_L + R_p = R'_L$) should be such a value that, for an AC current swing of $I_p = I_{CQ}$, the AC voltage swing becomes $V_p = I_{CQ} \times R'_L = V_{CEQ} = V_{CC}$. Putting these values in Equ.(15-35), we get,

$$\eta_{max} = \frac{V_p I_p \eta_t}{2V_{CC} I_{CQ}} = \frac{V_{CC} I_{CQ} \times 1}{2V_{CC} I_{CQ}}$$

$$\boxed{\eta_{max} = \frac{1}{2} = 50\%}$$
(15-37)

This is the theoretical maximum efficiency of a transformer coupled class-A amplifier. Here, the saturation voltage of the transistor has not been considered. In a practical circuit, the efficiency will be far below 50%. Because, the typical value of transformer efficiency is $\eta_t = 80\%$. Not only this, power will also be lost in the emitter resistor (R_E), and in biasing circuit. The practical maximum efficiency of a transformer coupled class-A amplifier is usually around 25% a little bit higher than direct coupled class-A amplifier of Fig.15-11(a).

Example 14-6

For the transformer coupled class-A amplifier of Example 14-5, calculate the maximum efficiency, standby power consumption, and rating of the transistor. Assume $\eta_t = 90\%$, and $V_{CE(sat)} = 2.5\text{ V}$.

Solution:

From Example 14-5, we found the following values:

$$I_{CQ} = \beta I_{BQ} = 100 \times 4\text{ mA} \approx 0.4\text{ A}$$

$$V_{CEQ} = 16\text{ V} - (0.4\text{ A})(3\ \Omega + 2\ \Omega) = 14\text{ V}$$

$$R_{L(AC)} = \left(\frac{142}{50}\right)^2 \times 4\ \Omega + 3\ \Omega \approx 35.3\ \Omega$$

The value of AC load resistance ($R_{L(AC)}$) is such a value that, for an AC current swing of $I_p = \Delta I_C = I_{CQ} = 0.4\text{ A}$, the AC voltage swing should be $V_p = \Delta V_{CE} = I_{CQ} \times R_{L(AC)} = 14\text{ V} = V_{CEQ}$. **That is, the AC load is perfectly matched for maximum efficiency.** However, the maximum swing is limited by the saturation voltage of the transistor.

To avoid nonlinearity, V_{CE} should not fall below $V_{CE(sat)} = 2.5\text{ V}$. Therefore, the maximum possible swing in V_{CE} will be,

$$V_p = \Delta V_{CE} = V_{CEQ} - V_{CE(sat)} = 14\text{ V} - 2.5\text{ V} = 11.5\text{ V}$$

Similarly, the maximum possible output current swing will be,

$$I_p = \Delta I_C = \frac{V_p}{R_{L(AC)}} = \frac{11.5\text{ V}}{35.3\ \Omega} \approx 0.33\text{ A}$$

Using Equ.(15-35),

$$\eta_{\max} = \frac{V_p I_p \eta_t}{2V_{CC} I_{CQ}} = \frac{11.5 \text{ V} \times 0.33 \text{ A} \times 0.9}{2 \times 16 \text{ V} \times 0.4 \text{ A}} \approx 27 \% \text{ [Ans.]}$$

The standby power loss of this circuit can be calculated using Equ.(15-36) as,

$$P_{i(\text{sb})} = V_{CC} I_{CQ} = 16 \text{ V} \times 0.4 \text{ A} = 6.4 \text{ W}$$

From the graph of Fig.15-18(b), we find that the maximum collector current is $I_{C(\max)} = 0.8 \text{ A}$, at the intercept of the AC load line on the I_C -axis. Actually we will not operate the transistor that far into saturation, since we do not allow V_{CE} to fall below 2.5 V. However, we may use a transistor with $I_{C(\max)}$ rating of 0.8 A (or 1 A) that will ensure a safety margin.

The maximum V_{CE} rating of the transistor should be just double the supply voltage, that is $16 \text{ V} \times 2 = 32 \text{ V}$.

Answer: Efficiency = 27%, standby power loss = 6.4 W, and the rating of the transistor is $V_{CE(\max)} = 32 \text{ V}$ and $I_{C(\max)} = 1 \text{ A}$.

Comments: The standby power loss will be same as the normal class-A power amplifier.

15.6 Class-B Amplifiers

A transistor operation is said to be *class-B* when its output current flows during only one half-cycle of a sine-wave input. In other words, the transistor is in its active region, responding to input signal, only during a positive half-cycle or only during a negative half-cycle of the input. This operation is illustrated in Fig.15-20.

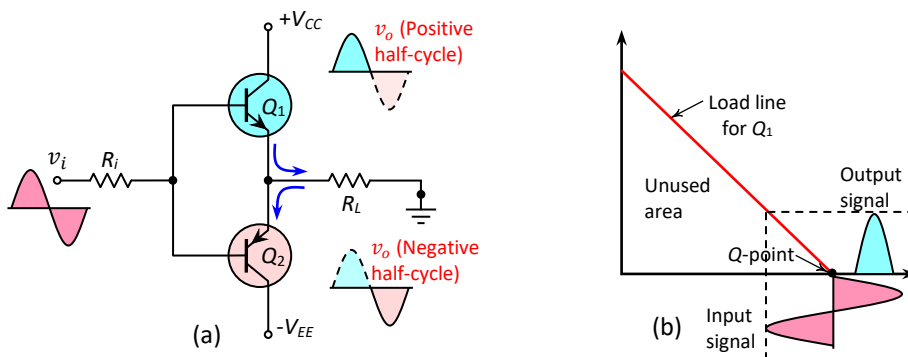


Fig.15-20: (a) Class-B power amplifier, and (b) Output characteristics of transistor Q_1

It is clear that class-B operation produces an output waveform that is severely clipped (half-wave rectified) and therefore highly distorted [Fig.15-20(b)]. The output waveform produced by a single transistor is not suitable for audio applications. However, in practical amplifiers, two transistors are operated in class-B conduction: one to amplify positive signal and the other to amplify negative signal variations. The amplifier output is obtained by combining the waveforms produced by each class-B operated transistors. An amplifier utilizing transistors that are biased at the cut-off point is called a *class-B amplifier*. As the transistors are biased at the cut-off point, no current flows

through them if there is no input signal, that is, the standby power loss is zero. For this reason the efficiency of class-B amplifiers is quite larger than that of class-A amplifiers.

The arrangement of Fig.15-20(a) is also called a push-pull amplifier. The name is derived from the fact that one transistor (Q_1) is primarily responsible for driving current through the load in one direction (pushing), while the other transistor (Q_2) drives current through the load in the opposite direction (pulling).

15.7 Push-Pull Amplifiers with Output Transformer

The push-pull amplifier of Fig.15-20(a) needs dual power supply, that is, $+V_{CC}$, and $-V_{EE}$. Using a transformer in class-B amplifier, the need for dual power supply can be eliminated. Fig.15-21 shows a push-pull amplifier arrangement with transformer (T_1). Here, only a single supply voltage ($+V_{CC}$) is used, moreover, both the transistors are NPN type. Note that the primary winding of the transformer is connected to the transistor collectors, and that its center tap is connected to the DC supply, $+V_{CC}$. The center tap is simply an electrical connection made at the center of the winding, so there are equal number of turns between each end of the winding and the center tap. The figure does not show the push-pull driver circuitry, which must produce out-of-phase signals (v_{i1} , and v_{i2}) on the bases of Q_1 and Q_2 . We will discuss that circuitry later with class-AB amplifier.

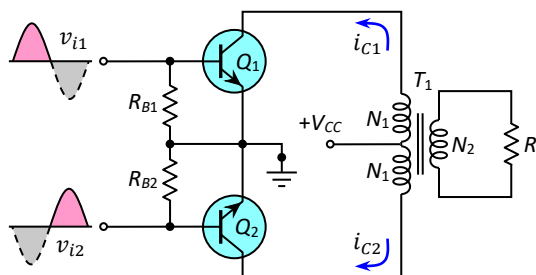


Fig.15-21: Push-pull amplifier with transformer

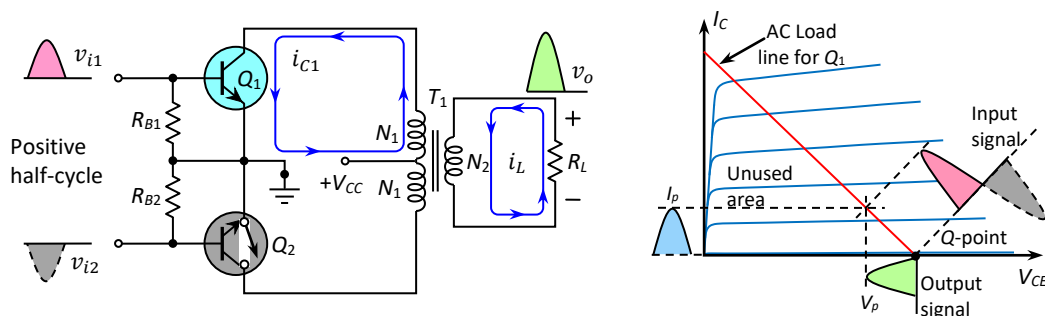


Fig.15-22: Push-pull amplifier with transformer

15.7.1 Operation of Class-B Amplifier

Fig.15-22, and Fig.15-23 illustrate how current flows through the amplifier during a positive half-cycle of input and during a negative half-cycle of input. In 15-22 (a), the input to Q_1 is the positive half-cycle of the signal, and since the input to Q_2 is out-of-phase, Q_2 is driven by a negative half-cycle. Notice that the transistor bases are grounded via resistors R_1 and R_2 and hence both transistors are biased at cut-off. Consequently, the positive base voltage on Q_1 causes it to turn on

and conducts current in the counterclockwise direction as shown in Fig.15-22(a). The negative base voltage on Q_2 keeps this transistor at cut-off (open circuit).

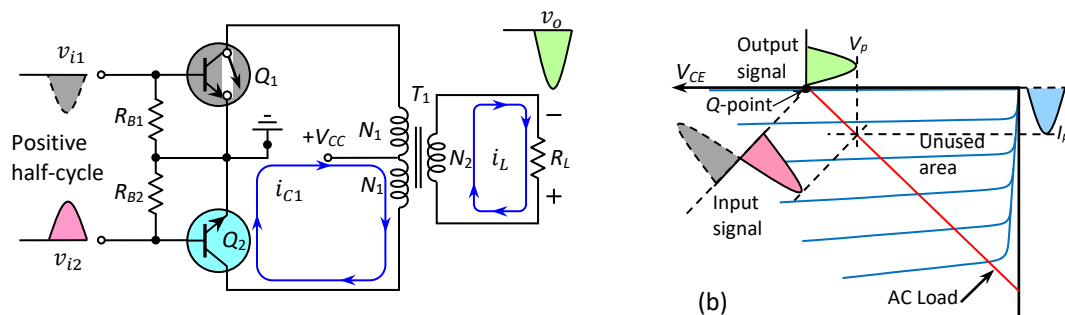


Fig.15-23: Push-pull amplifier with transformer

Current flowing in the upper half of the transformer's primary induces current in the secondary winding, and current flows through the load. In 15-23(a), the input signal on the base of Q_1 has gone negative, so its inverse on the base of Q_2 is positive. Therefore, Q_2 conducts current in the clockwise direction as shown in Fig.15-23(a), and Q_1 is cut-off. Current induced in the secondary winding is in opposite direction than that shown in Figure 15.22(a). The upshot is that current flows through the load in one direction when the input signal is positive and in the opposite direction when the input signal is negative, just as it should be in an AC amplifier. The output characteristics of the transistors with the load line and operating points are shown in Fig.15-22(b) and Fig.15-23(b). Note that the characteristics of transistor Q_2 have been drawn upside down, to represent that the movement of Q -point of this transistor is just opposite to that of Q_1 . The combined output characteristics of the two transistors of this amplifier are shown in Fig.15-24. In response to the input signal, the Q -point will swing over the entire load line from point A to B . The swing of Q -point will produce swings in output voltage and output current.

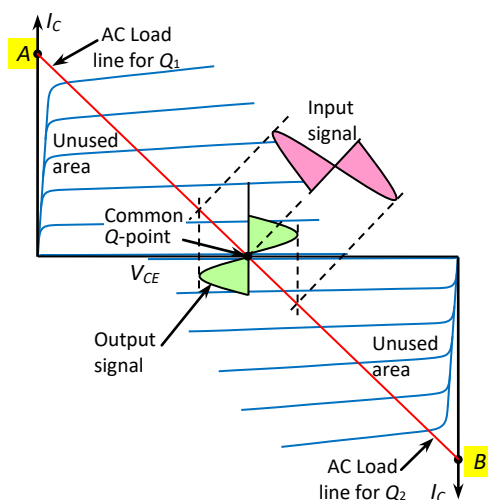


Fig.15-24: Combined AC load line of a class-B amplifier drawn on the composite BJT characteristics

15.7.2 Class-B Efficiency

The principal advantage of using a class-B power amplifier is that it is possible to achieve efficiency greater than that of class-A amplifier. The improvement in efficiency stems from the fact that no power is dissipated in a transistor when it is cut off. We will derive an expression for the maximum efficiency of a class-B push-pull amplifier assuming ideal conditions: perfectly matched transistors

and zero resistance in the transformer windings. The current flowing through each transistor is a half-wave rectified waveform, as shown in Fig.15-22 and Fig.15-23. The swing in the collector current and the swing in V_{CE} will be applied to the primary of the transformer. Let I_{Tp} represent the peak value of current, and V_{Tp} be the peak value of voltage in the primary of the transformer. The power supplied to the transformer primary will be,

$$P_{Tp} = \frac{V_{Tp}}{\sqrt{2}} \times \frac{I_{Tp}}{\sqrt{2}} = \frac{V_{Tp}I_{Tp}}{2} \quad (15-38)$$

If the transformer used in this amplifier has an efficiency of η_t , then the output power supplied to the load will be,

$$P_{o(AC)} = P_{Tp} \times \eta_t = \frac{V_{Tp}I_{Tp}}{2} \times \eta_t \quad (15-39)$$

With all these considerations, the power supply current will be a full-wave rectified waveform. Again let us assume that the peak current, supplied to the primary of the transformer, be I_{Tp} . Therefore, the average power delivered to the amplifier by the supply source will be the average value of fullwave rectified waveforems, that is,

$$P_{i(DC)} = \frac{2V_{CC}I_{Tp}}{\pi} \quad (15-40)$$

Thus, the efficiency of this class-B amplifier will be,

$$\eta = \frac{P_{o(AC)}}{P_{i(DC)}} = \frac{V_{Tp}I_{Tp}\eta_t}{2} / \frac{2V_{CC}I_{Tp}}{\pi} \quad (15-41)$$

$$\boxed{\eta = \frac{\pi V_{Tp}\eta_t}{4V_{CC}}} \quad (15-42)$$

15.7.3 Maximum Efficiency

The efficiency of this amplifier will be the maximum, if there is no loss in the transformer ($\eta_t = 1$) and the swing of the operating voltage and current should be the maximum. The maximum possible output voltage swing will be equal to supply voltage, if we neglect the cutoff and saturation voltages. That is, $V_{Tp} = V_{CC}$. Putting this value in Equ.(15-42), we get,

$$\boxed{\eta_{max} = \frac{\pi V_{CC} \times 1}{4V_{CC}} = \frac{\pi}{4} = 78.5\%} \quad (15-43)$$

Unlike the class-A amplifier, class-B amplifier will not draw any current while in standby condition, as the transistors are biased at the cut-off points. Therefore, the power consumption at standby will be,

$$P_{i(sb)} = V_{CC} \times 0 \text{ A} = 0 \quad (15-44)$$

15.7.4 Cross-Over Distortion

Although class-B amplifier provides high efficiency, they suffer from cross-over distortion. Cross-over distortion is a type of distortion which is caused by switching of current from one transistor to another. The term cross-over signifies the *crossing over* of the signal between transistors Q_1 and Q_2 and vice versa. In the amplifier circuit of Fig.15-21, both the transistors are biased at the cut-off point (zero base voltage at no signal condition) and therefore, transistors are not conducting. Consider a positive-going swing: as long as the input is less than the required forward V_{BE} drop (≈ 0.65 V) of transistor Q_1 , it will remain off or conduct very little amount of current- same as a diode operation. In fact, due to the nonlinear behavior of the input characteristics of the transistor, the base current does not increase in proportion to the input signal. As this nonlinearity is prominent at the low voltage region (0 to 0.6 V), at this region, the input current becomes distorted. Being an amplified version of base current, the output current also becomes distorted as shown in Fig.15-25. The same thing happens for the transistor Q_2 , for the negative half-cycles of input. Thus, between about ± 0.65 V of input, the output voltage is not a true replica or amplified version of the input, and we can see that as a *kink* in the output waveform near 0 V (or where one transistor stops conducting and the other starts). This kink is the most pronounced form of cross-over distortion, and it becomes more evident and intrusive when the output voltage swing is small. Thus the cross-over distortion is the main limitation of the class-B amplifiers.

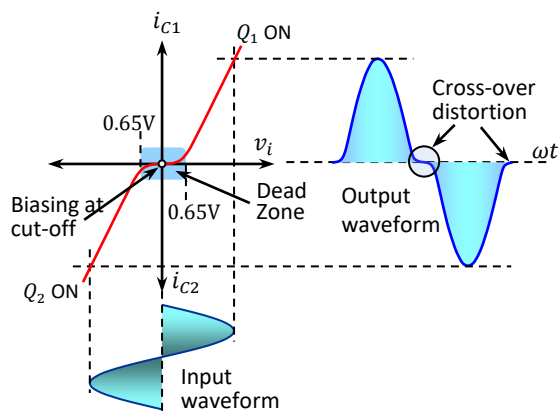


Fig.15-25: Cross-over distortion in Class-B amplifier

15.8 Class-AB Amplifier

Cross-over distortion in a push-pull amplifier can be eliminated by biasing each transistor slightly into conduction, that is, setting the operating points of the transistors just slightly above the cut-off point. Therefore, the transistors conduct current for a little bit more than 180° . When a small forward biasing voltage is applied across each base-emitter junction, and a small base current flows under no-signal conditions, it is not necessary for the input signal to overcome the built-in junction potential to start conduction of the transistors. Thus, the cross-over distortion is eliminated. The amplifier where the transistors are biased in such a way, is called a **class-AB amplifier**. Fig.15-26 shows a class-AB amplifier. This circuit is arranged with the input driving transformer as well as the output transformer. The centre-tapped transformer on the input side produces opposite polarity signals (phase-split input signals) to the two transistor inputs, and the centre-tapped output transformer combines the two halves of the AC waveforms together. In this circuit, a simple voltage-divider biasing network, consisting of R_{B1} and R_{B2} , is used to produce the biasing voltage, V_B , at the bases of the transistors. Note that instead of

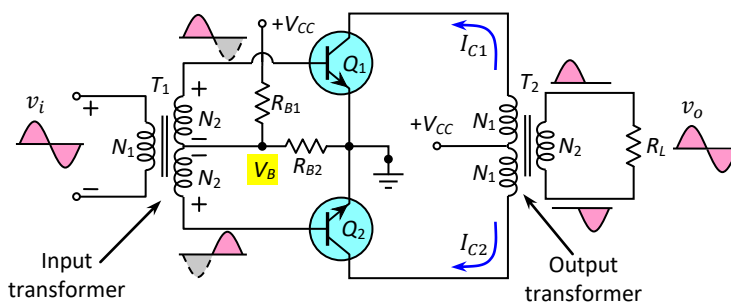


Fig.15-26: Class-AB push-pull amplifier with input and output transformers

applying this biasing voltage (V_B) directly to the base of the transistors, it is applied to the centre-tape terminal of the input transformer (T_1). If the biasing voltage were directly connected to the bases of the transistors, the transistor bases would have been shorted [H_E_atten]. As the DC resistance of the transformer windings is very small, the biasing voltage will be applied to the bases of the transistors, without any significant drop. The biasing resistors are so selected that the value of $V_B = R_{B2}V_{CC}/(R_{B1} + R_{B2})$ is typically about 0.7 V (for silicon transistors), or so that the collector current under no-signal conditions is about 1% of its peak value.

When a transistor is biased slightly into conduction, output current will flow during more than one-half cycle of a sine-wave input, as illustrated in Fig.15-27. As can be seen in the figure, conduction occurs for more than one half-cycle (or more than 180°) but less than a full cycle (or 360°) of input. This operation is neither class-A nor class-B, but is called class-AB operation.

While class-AB operation reduces cross-over distortion in a push-pull amplifier, it has the disadvantage of reducing amplifier efficiency. The fact that bias current is always present means that there is continuous power dissipation in both transistors, including the time intervals during which one of the transistors would be cut-off. The extent, to which efficiency is reduced, depends directly on how heavily the transistors are biased. The maximum achievable efficiency of class-AB amplifier is somewhere between that in class-A operation (50%) and in class-B operation (78%). In Fig.15-26, notice that the quiescent collector currents I_{C1} , and I_{C2} flow in opposite directions through the primary of the transformer. Thus, the magnetic flux created in the transformer by the DC currents oppose each other, and hence the net flux is 0. This is an advantage of class-AB amplifier, in comparison with the class-A transformer-coupled amplifier, because there is less possibility for the transformer to be saturated.

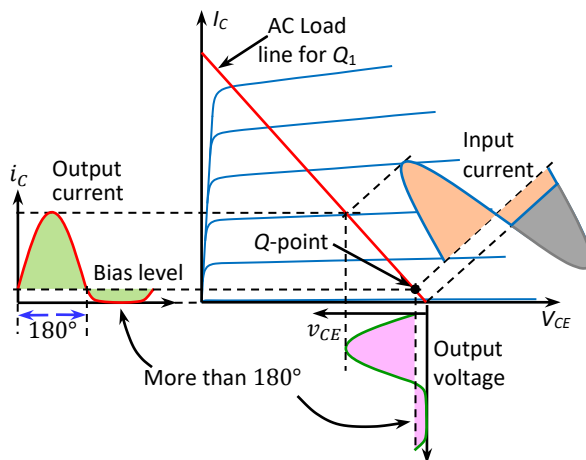


Fig.15-27: Class-AB operation; output current flows more than one-half cycle, but less than full cycle of input signal

The operation process of this amplifier is exactly same as the operation of transformer-coupled push-pull amplifier. Fig.15-27 shows the output characteristics for transistor Q_1 . The output characteristics of transistor Q_2 will be similar but it will conduct for the negative half-cycles of the input.

This type of amplifier is widely used in small electronic systems, such as radio receivers, small TV, etc. As complete amplifier with the common-emitter driver stage is shown in Fig.15-28.

Example 14-7

Fig.15-28 shows a class-AB push-pull amplifier with driver stage. Calculate the maximum possible AC power delivered to the load. Assume, $V_{CC} = 24\text{ V}$, $R_L = 8\ \Omega$, $\beta_1 = 150$, $\beta_2 = \beta_3 = 50$, $h_{ie2} = h_{ie3} = 200\ \Omega$.

Specification of the transformers: $N_1/N_2 = 315/100$, $R_{p1} = 5\ \Omega$ and $R_{s1} = 3\ \Omega$.

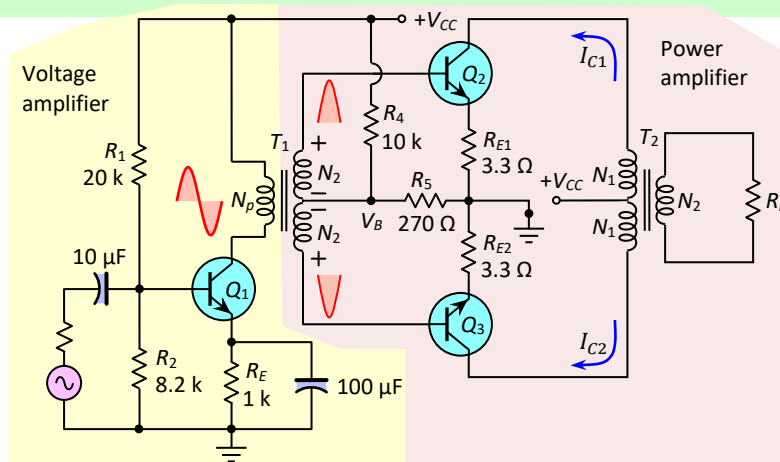


Fig.15-28: Class-AB push-pull amplifier with driver circuit for Example 14-7

Q-point of driver transistor, Q_1 :

$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{8.2\text{ k} \times 24\text{ V}}{20\text{ k} + 8.2\text{ k}} \approx 7\text{ V}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{20\text{ k} \times 8.2\text{ k}}{20\text{ k} + 8.2\text{ k}} \approx 5.8\text{ k}$$

The value of quiescent base current of Q_1 will be,

$$I_{B1} = \frac{E_{Th} - V_{BE1}}{R_{Th} + (\beta_1 + 1)R_3} = \frac{7\text{ V} - 0.7\text{ V}}{5.8\text{ k} + (150 + 1)1\text{ k}} \approx 40\ \mu\text{A}$$

The quiescent collector current of Q_1 will be,

$$I_{CQ1} = I_{B1} \times \beta_1 = 150 \times 40\ \mu\text{A} \approx 6\text{ mA}$$

The quiescent C-E voltage of Q_1 will be,

$$V_{CEQ1} = V_{CC} - I_{C1}(R_{p1} + R_3) = 24 \text{ V} - 6 \text{ mA}(5 \Omega + 1 \text{ k}) \approx 18 \text{ V}$$

The base voltage of Q_2 and Q_3 will be,

$$V_{B1} = V_{B2} = V_B = \frac{24 \text{ V} \times 0.27 \text{ k}}{10 \text{ k} + 0.27 \text{ k}} \approx 631 \text{ mV}$$

Total effective resistance to the secondary of T_1 transformer (R_{Ts1}) will be the sum of secondary DC resistance, R_{s1} , emitter resistance (R_{E1}) referred to base, and h_{ie} of Q_2 , that is,

$$R_{Ts1} = R_{s1} + h_{ie2} + \beta R_{E1}$$

or,

$$R_{Ts1} = 3 \Omega + 200 \Omega + 50 \times 3.3 \Omega = 368 \Omega$$

The total AC resistance at the primary of T_1 that will work as the collector resistance of Q_1 can be calculated as,

$$R_{C(AC)1} = \left(\frac{N_1}{N_2}\right)^2 R_{Ts1} + R_{p1}$$

$$R_{C(AC)1} = \left(\frac{315}{100}\right)^2 368 \Omega + 3 \Omega \approx 3.65 \text{ k}$$

For a change of collector current of Q_1 from $I_{C1} = 6 \text{ mA}$ to zero, that is $\Delta I_{C1} = 6 \text{ mA}$, the corresponding change in V_{CE1} will be, $\Delta V_{CE1} = 6 \text{ mA} \times 3.65 \text{ k} \approx 22 \text{ V}$. Using these values the AC load line is drawn in Fig.15-29.

From this AC load line, we find that the maximum possible swing in V_{CE1} will be $\Delta V_{CE1} = V_{CEQ1} - V_{CE(sat)1} = 18 \text{ V} - 1.5 \text{ V} = 16.5 \text{ V}$, and the corresponding change (increase) in I_{C1} will be, $\Delta I_{C1} = 10.9 \text{ mA} - 6 \text{ mA} = 4.9 \text{ mA}$.

This current and voltage swings will be applied to the primary of T_1 . If we neglect the loss in the transformer, the swings in the secondary or bases of Q_2 , and Q_3 will be,

$$\Delta I_{s1} = \Delta I_{B2} = \Delta I_{B3} = \frac{N_1}{N_2} \Delta I_{C1} = \frac{315}{100} 4.9 \text{ mA} \approx 15.435 \text{ mA}$$

and,

$$\Delta V_{s1} = \Delta V_{B2} = \Delta V_{B3} = \frac{N_2}{N_1} \Delta V_{CE1} = \frac{100}{315} \times 16.5 \text{ V} \approx 5.2 \text{ V}$$

The current flowing through collector of Q_2 and Q_3 will be,

$$\Delta I_{C2} = \Delta I_{C3} = 15.435 \text{ mA} \times \beta = 15.435 \text{ mA} \times 50 \approx 772 \text{ mA}$$

$$R'_L = \left(\frac{315}{100}\right)^2 \times 8 \Omega \approx 79 \Omega$$

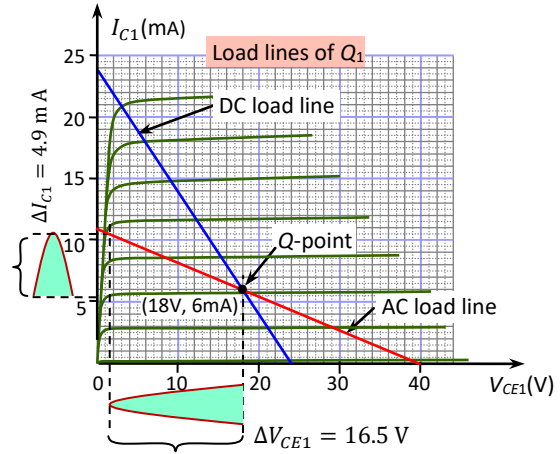


Fig.15-29: Load lines of transistor Q_1 of Example 14-7

The voltage swing across Q_2 will be, $772 \text{ mA} \times 79 \Omega = 60 \text{ V}$! That is, the maximum swings of the driving amplifier will not be used by the power stage.

The possible maximum swing of V_{CE2} will be approximately 2 V less than V_{CC} , that is $\Delta V_{CE2} = 22 \text{ V}$. Thus the current swing will be,

$$\Delta I_{C2} = \Delta V_{CE2} / (R'_L + R_{E2} + R_{p2})$$

$$\therefore \Delta I_{C2} = 22 \text{ V} / (79 \Omega + 3.3 \Omega + 5 \Omega) \approx 252 \text{ mA}$$

Therefore, the load current will be,

$$\Delta I_L = \left(\frac{315}{100} \right)^2 \times 252 \text{ mA} \approx 794 \text{ mA}$$

Neglecting the voltage drop across R_{E2} and the primary DC resistance, all of the ΔI_{C2} will be transferred to the secondary of T_2 ,

$$\Delta I_L = \left(\frac{N_1}{N_2} \right)^2 \times \Delta I_{C2} = \left(\frac{315}{100} \right)^2 \times 252 \text{ mA} \approx 794 \text{ mA}$$

Therefore, the AC power delivered to the load will be

$$P_{o(AC)} = \left(\frac{V_p}{\sqrt{2}} \right)^2 R_L = \left(\frac{\Delta I_L}{\sqrt{2}} \right)^2 R_L = \left(\frac{0.794}{\sqrt{2}} \right)^2 \times 8 \Omega \approx 2.5 \text{ W}$$

Comments: Although the driver amplifier has the capacity to produce a large voltage and current swings, the power amplifier cannot use the full capacity of the driver amplifier.

15.8.1 Design of Class-AB Amplifier

Design process of transformer coupled class-B and class-AB amplifiers are almost same. As class-B is seldom used, only the design process of class-AB amplifier is discussed here. The design of a transformer-coupled class-AB amplifier is mainly to determine the component values, transistor and transformer specifications for a specified load resistance and output power. The amplitude of the input signal voltage may also be stated, as well as the upper and lower cut-off frequencies for the amplifier. If a supply voltage is given, the design must determine the specifications for the required transformer. If the design requirement is for an available transformer, then the supply voltage is to be calculated to suit the transformer. The maximum levels of V_{CE} , I_C , and P_D must be calculated for each transistor. The power delivered to the primary of T_2 can be determined as [here subscript 2 indicates output transformer],

$$P_{p2} = \frac{P_{s2}}{\eta_t} = \frac{P_L}{\eta_t} = \frac{P_o}{\eta_t} \quad (15-45)$$

The power delivered to the primary of T_2 (the output transformer) can also be expressed as,

$$\therefore P_{p2} = \frac{V_{rms}^2}{R'_L} \quad (15-46)$$

where V_{rms} is the RMS value of the primary voltage, and R'_L is the AC resistance offered by the transformer primary. Using peak voltages, the equation becomes,

$$P_{p2} = \frac{(V_p/\sqrt{2})^2}{R'_L} \quad (15-47)$$

This gives the equation for R'_L as,

$$R'_L = \frac{V_p^2}{2P_{p2}} \quad (15-48)$$

The value of R'_L can be calculated in terms of the transformer turn ratio and the load resistance as,

$$R'_L = \left(\frac{N_{p2}}{N_{s2}}\right)^2 R_L \quad (15-49)$$

The primary winding of the output transformer has a centre tap, the upper half and the lower half of the coil contains N_p number of turns. Sometimes transformers are specified with the number of turns in half of the primary coil. In that case,

$$R'_L = \left(\frac{2N_{p2}}{N_{s2}}\right)^2 R_L \quad (15-50)$$

$$\therefore R'_L = 4 \left(\frac{N_{p2}}{N_{s2}}\right)^2 R_L = 4R'_L \quad (15-51)$$

The output transformer can now be specified in terms of the output power $P_{s2} = P_{o(AC)}$, the load resistance R_L , and the load referred to primary R'_L .

The power delivered to the transformer primary can also be determined from the peak values of current and voltage as,

$$P_{p2} = \frac{V_p}{\sqrt{2}} \times \frac{I_p}{\sqrt{2}} = \frac{V_p I_p}{2} \quad (15-52)$$

$$\therefore I_p = \frac{2P_{p2}}{V_p} \quad (15-53)$$

The above expression gives the maximum current requirement for the transistor, that is, the transistors must have $I_{C(max)}$ rating greater than I_p given by Equ.(15-53).

To find the maximum voltage rating of the transistors, refer to Fig.15-30. Note that when Q_2 is OFF and Q_3 is ON the entire supply voltage ($+V_{CC}$) is dropped across the lower half of the transformer. In addition to that a voltage with a peak value of approximately $+V_{CC}$ is induced in the upper half of the transformer primary connected to Q_2 . The induced voltage occurs because the current flowing through the lower half of the primary will produce magnetic flux in the core and the upper half of the coil is wound on the same core. As the voltage developed across the coils of the primary winding due to the same magnetic flux, their polarity is same (positive on the upper terminal). Now apply KVL in the entire loop we get,

$$+V_{CE2} - V_{CC} - V_{CC} - V_{CE3} = 0 \quad (15-54)$$

As Q_3 is ON, so, $V_{CE3} = 0$. Therefore,

$$V_{CE2} = 2V_{CC} \quad (15-55)$$

This expression gives the maximum voltage rating of the transistor which is

$$V_{CE(max)} = 2V_{CC} \quad (15-56)$$

The power dissipated in the two output transistors (Q_2 and Q_3) is the difference between the DC power drawn by the amplifier from the supply voltage and the AC power delivered to the transformer primary. That is,

$$2P_T = P_{i(DC)} - P_{p2} = P_{i(DC)} - P_o/\eta_t \quad (15-57)$$

The two transistors conduct current for half of each cycle of input signal. So the total power loss will be equally divided between them. Therefore, the power dissipated by a single transistor will be,

$$P_T = 0.5(P_{i(DC)} - P_{p2}) = 0.5(P_{i(DC)} - P_o/\eta_t) \quad (15-58)$$

For class-B operation, the average power delivered to the amplifier by the DC supply source is,

$$P_{i(DC)} = \frac{2V_{CC}I_p}{\pi} + I_{CQ} \times V_{CC} \quad (15-59)$$

However, for class-AB operation, the DC input power will be higher than the power calculated using Equ.(15-59) because, the transistors draw current when there is no input signal.

The transistors are specified in terms of the device power dissipation (P_T), the peak current (I_p), and the maximum collector-emitter voltage ($V_{CE(max)}$). The transistors must also be operated below the maximum power dissipation curve, (see Section 8-7). To do that we may need to use heat sinks with the transistors.

Example 14-8

Design a class-AB push-pull amplifier to deliver 6 W power to a load of 4Ω with a supply voltage of 24 V. Assume, $h_{fe} = 50$, $h_{ie} = 600 \Omega$, transformer efficiency $\eta_t = 92\%$. To operate the transistors in class-AB mode set $I_{CQ} = 5\%$ of peak output current.

Solution:

$$\text{Using Equ.(15-45)} \quad P_{p2} = \frac{P_{s2}}{\eta_t} = \frac{P_L}{\eta_t} = \frac{P_o}{\eta_t} = \frac{6 \text{ W}}{0.92} \approx 6.5 \text{ W}$$

With 5% drop across emitter resistance, (when peak current flows), the maximum swing in the primary of T_2 will be, $V_{p2} = V_{CC} - V_{CC} \times 5\% = 24 \text{ V} - 24 \text{ V} \times 5\% = 22.8 \text{ V}$

$$\text{Using Equ.(15-48),} \quad R'_L = \frac{V_p^2}{2P_{p2}} = \frac{(22.8 \text{ V})^2}{2 \times 6.5 \text{ W}} \approx 40 \Omega$$

$$\text{Using Equ.(15-51),} \quad R''_L = 4R'_L = 4 \times 40 \Omega = 160 \Omega$$

Therefore, the transformer specifications will be,

$P_o = 6 \text{ W}$, $R''_L = 160 \Omega$, and the load resistance is $R_L = 4 \Omega$.

The peak current will be,

Using Equ.(15-53),
$$I_p = \frac{2P_{p2}}{V_p} = \frac{2 \times 6.5 \text{ W}}{22.8 \text{ V}} \approx 570 \text{ mA}$$

The value of the emitter resistances will be,

$$R_{E1} = R_{E1} = \frac{5\% V_{CC}}{I_p} = \frac{5\% \times 24 \text{ V}}{570 \text{ mA}} \approx 2 \Omega$$

Let the DC quiescent current of transistors Q_1 and Q_2 be, 5% of I_p , that is $I_{CQ1} = I_{CQ2} = 570 \text{ mA} \times .05 = 28.5 \text{ mA}$. Therefore, the base current will be,

$$I_{BQ1} = I_{BQ2} = \frac{I_{CQ1}}{h_{fe1}} = \frac{I_{CQ2}}{h_{fe2}} = \frac{28.5 \text{ mA}}{50} \approx 0.57 \text{ mA}$$

Thus, the required base voltage (for Q_1 and Q_2) can be calculated as,

$$I_{BQ} = \frac{V_B - V_{BE}}{h_{ie} + h_{fe} R_E}$$

or,
$$0.57 \times 10^{-3} \text{ A} (600 \Omega + 50 \times 2 \Omega) = V_B - 0.7 \text{ V}$$

$$\therefore V_B = 1.1 \text{ V}$$

Let the current flowing through R_5 be $I_5 = 10 \times I_{BQ} = 5.7 \text{ mA}$. Therefore,

$$R_5 = \frac{V_B}{I_5} = \frac{1.1 \text{ V}}{5.7 \text{ mA}} = 192 \Omega, [\text{use standard } 200 \Omega]$$

Current flowing through R_4 will be, $I_4 = I_5 + I_{BQ} = 5.7 \text{ mA} + 0.57 \text{ mA} = 6.27 \text{ mA}$

Therefore,

$$R_4 = \frac{V_{CC} - V_B}{I_4} = \frac{24 \text{ V} - 1.1 \text{ V}}{6.27 \text{ mA}}$$

$$\therefore R_4 \approx 3.7 \text{ k}, [\text{use standard } 3.6 \text{ k}]$$

Using Equ.(15-56), the maximum C-E voltage rating of the transistor will be,

$$V_{CE(max)} = 2V_{CC}$$

or,
$$V_{CE(max)} = 2 \times 24 \text{ V} = 48 \text{ V}$$

Neglecting the power loss in the biasing network, the total power drawn from the supply voltage will be

Using Equ.(15-59),
$$P_{i(DC)} = \frac{2V_{CC}I_p}{\pi} + I_{CQ} \times V_{CC}$$

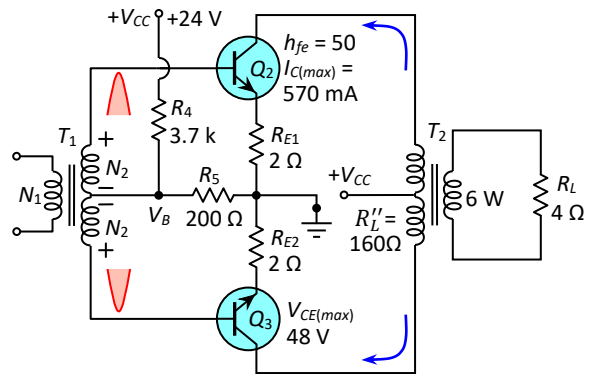


Fig.15-30: Class-AB push-pull amplifier designed for Example 14-8

$$\text{or,} \quad P_{i(DC)} = \frac{2 \times 24 \text{ V} \times 570 \text{ mA}}{\pi} + 5.7 \text{ mA} \times 24 \text{ V}$$

$$\text{or,} \quad P_{i(DC)} = \frac{2 \times 24 \text{ V} \times 570 \text{ mA}}{\pi} + 5.7 \text{ mA} \times 24 \text{ V} \approx 8.85 \text{ W}$$

$$\text{Using Equ.(15-58),} \quad P_T = 0.5(P_{i(DC)} - P_{p2})$$

$$\text{or,} \quad P_T = 0.5(8.85 \text{ W} - 6.5 \text{ W})$$

$$\therefore \quad P_T \approx 1.16 \text{ W}$$

Therefore, the transistor specifications will be, $P_T = 1.16 \text{ W}$, $V_{CE(max)} = 48 \text{ V}$, and $I_{C(max)} = 570 \text{ mA}$. The designed values are shown in Fig.15-30.

Comments: In fact the ratings of the transistors should be higher than those found in calculations.

15.9 Transformerless Push-Pull Amplifiers

The principal disadvantage of the push-pull amplifier circuits, discussed so far, is the cost and bulk of their output transformers. High-power amplifiers in particular are encumbered by the need for very large transformers capable of conducting large currents without saturating the core. A solution to this problem is to use transformerless amplifiers. Transformerless amplifiers are discussed in the following subheadings.

15.9.1 Complementary Push-Pull Amplifiers

Fig.15-31 shows a popular design of power amplifier using two emitter followers. Although one transistor is NPN and the other is PNP, the devices are selected to have identical characteristics (similar parameters), so the transistors are called **complementary transistors**, and the amplifier is also called **complementary emitter follower**. This design eliminates the need for an output transformer in push-pull operation, and also eliminates the need for a driver transformer or any other drive circuitry used for phase splitting purpose. The transistors of the circuit are biased in class-B mode using a split power supply.

Fig.15-32(a) shows that current flows in a counterclockwise path through the load when the input signal on the base of NPN transistor (Q_1) is

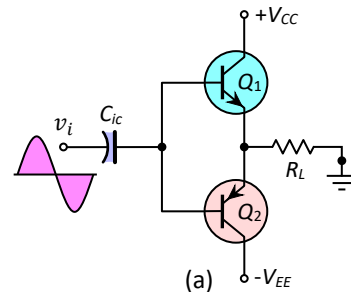


Fig.15-31: Complementary push-pull amplifier or complementary emitter follower

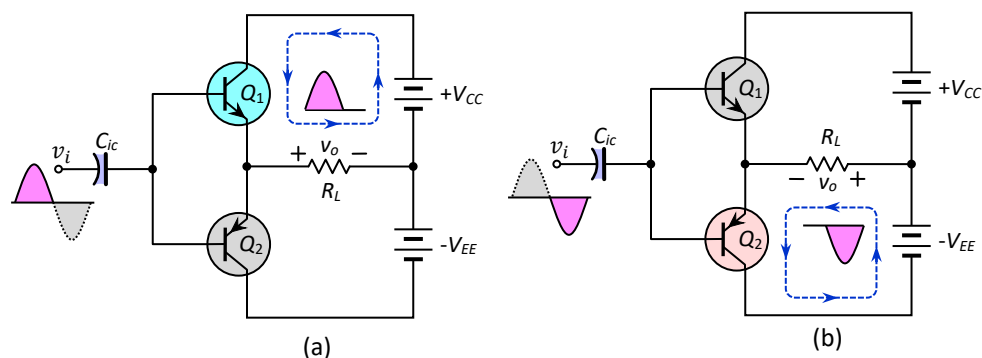


Fig.15-32: Complementary push-pull amplifier or emitter follower

positive. The positive input simultaneously appears on the base of PNP transistor (Q_2) and keeps it cut-off. When the input is negative, Q_1 is cut-off and Q_2 conducts current through the load in the opposite direction, as shown in Fig.15-32(b). Note that the load is connected to the emitter terminal of each transistor which ensure the emitter-follower configuration. For that reason, this circuit is also called the complementary emitter follower. The advantageous consequence is that emitter follower provides very high current gain, and very low output impedance. Therefore, this amplifier can drive low-impedance loads without sacrificing any power. Also, the large negative feedback that is inherent in emitter follower amplifier reduces the problem of output distortion. However, as is the case in all emitter followers, voltage gains greater than unity cannot be realized. The maximum positive voltage swing is $+V_{CC}$, and the maximum negative swing is V_{EE} . Normally, $+V_{CC} = |V_{EE}|$, so the maximum peak-to-peak swing is $2V_{CC}$ volts. Since the voltage gain is near unity, the input must also swing through $2V_{CC}$ volts to achieve maximum output swing. As shown in Fig.15-32(a), when Q_2 is cut-off, the collector-to-emitter voltage drop of this transistor will be the sum of v_o and V_{EE} . As the maximum value of v_o is V_{CC} , the maximum collector-to-emitter voltage will be $2V_{CC}$. For transistor Q_1 the maximum collector-to-emitter voltage will also be $2V_{CC}$.

15.10 Complementary Emitter Follower in Class-AB Operation

Fig.15-33(a) shows a complementary emitter follower amplifier where the transistors are operated in class-AB mode to eliminate the cross-over distortion and incorporates some additional features. The complementary amplifier in Fig.15-33(a) needs a split power supply, but the circuit in Fig.15-33(b) eliminates the need for split power supply using a large capacitor, C_{oc} at the output. This capacitor is called **reservoir capacitor**. In both circuits the output v_o (at point A in the figures) is biased at half the supply voltage, that is, in the first circuit point A is at zero potential and in the second circuit point A is at $V_{CC}/2$ voltage. The resistor-diode network connected across the transistor bases is used to bias each transistor slightly above the cut-off point. Cross-over distortion can be reduced or eliminated by adjusting the resistor in series with the diodes to bias the transistors into AB operation. Assuming that all components are perfectly matched, the supply voltage will divide equally across each half of the amplifier (further illustrated in Example 14-9). Resistors R_E provide bias stability to prevent thermal runaway (see Section 5-6), but should be made as small as possible, since they adversely affect efficiency. Since, each half of the amplifier has half of the supply voltage across it, the forward-biased diode drops (V_{D1} and V_{D2}) plus half of

small voltage dropped across R_B will appear at the base with respect to point A with the proper polarity to bias each transistor towards conduction (a little bit above the cut-off point). The diodes are selected so that their characteristics track the base-emitter junctions under temperature changes and thus ensure bias stability. The diodes and output transistors can be thermally-coupled by mounting D_1 and Q_2 on a single heat sink, and D_2 and Q_3 on another single heat sink. In that case, V_D follows the V_{BE} level changes with temperature, thus stabilizing the transistor bias conditions over a wide temperature range.

In AC operation, when input signal v_i is positive and Q_1 is conducting, current is drawn from the power supply, V_{CC} , and flows through Q_1 to the load, and the coupling capacitor, C_{oc} of Fig.15-33(b) will be charged. When Q_1 is cut-off by a negative input voltage, no current can flow from V_{CC} . At that time, Q_2 conducts current from ground to $-V_{EE}$ through the load for Fig.15-33(a) and for Fig.15-33(b) the capacitor C_{oc} discharges through that transistor. Thus in both circuits current flows through the load, and through the collector terminal of the PNP transistor (Q_2). The $R_L C_{oc}$ time constant must be quite larger than the period of the lowest frequency of operating signals. The lower cutoff frequency due to C_{oc} , will given by

$$f_{L(C_{oc})} = \frac{1}{2\pi(R_L + R_E)C_{oc}} \quad (15-60)$$

In calculating $f_{L(C_{oc})}$, the output impedance of the emitter follower transistors have been neglected as they are very small.

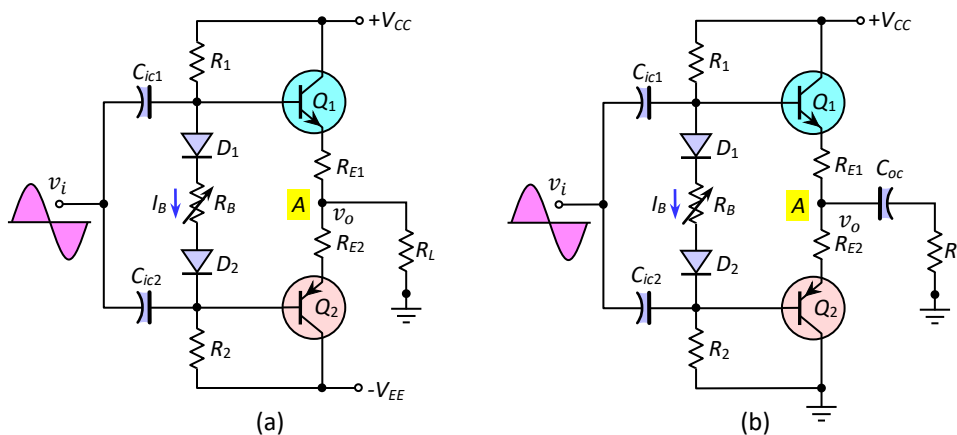


Fig.15-33: Complementary push-pull amplifier or emitter follower

Example 14-9

- For the complementary emitter-follower of Fig.15-34, determine the operating point of transistors Q_1 and Q_2 .
- Do the same again considering $V_{CC} = +24 \text{ V}$, and $V_{EE} = 0 \text{ V}$ (that is for single ended power supply).

Solution:

i) The amplifier consists of two complementary emitter follower circuits. Let us divide R_B into two equal resistors ($10\ \Omega$ each) as shown in Fig.15-34(b). If we neglect the base current of the transistors, the supply voltage will be equally divided around the point B . So, the current flowing through the biasing network can be calculated as,

$$I_1 = \frac{V_{CC} + |V_{EE}| - 2V_D}{R_1 + R_2 + R_B} = \frac{12\text{ V} + 12\text{ V} - 2 \times 0.7\text{ V}}{2.2\text{ k} + 2.2\text{ k} + 0.02\text{ k}} \approx 5.1\text{ mA}$$

So, the voltage drop across D_1 plus R_{B1} will be,

$$V_{D1} + V_{RB1} = V_{D1} + I_1 R_{B1} = 0.7\text{ V} + 5.1\text{ mA} \times 10\ \Omega = 0.75\text{ V}$$

This voltage will be applied to $B-E$ junction and R_3 . The value of $B-E$ junction will be,

$$V_{BE1} = V_{BE2} = 0.7\text{ V}$$

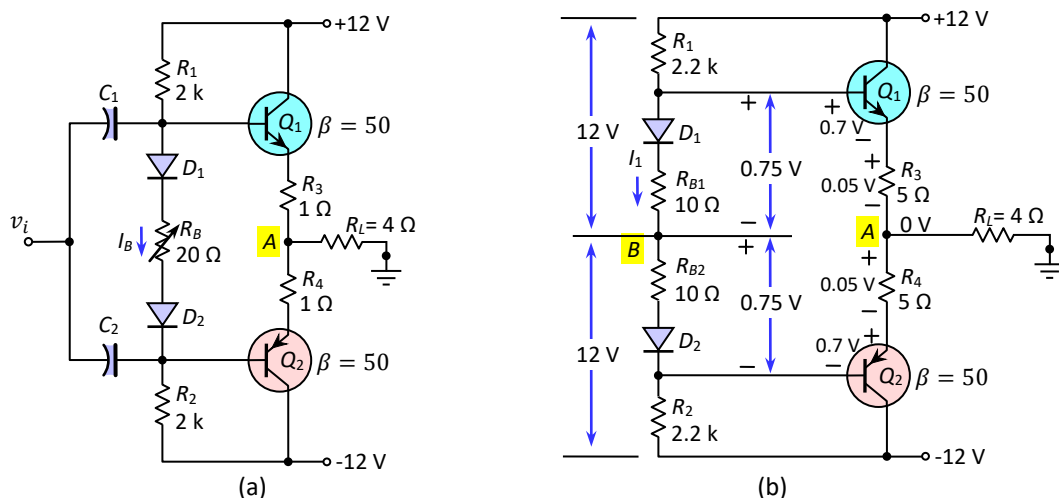


Fig.15-34: Complementary push-pull amplifier for Example 14-9

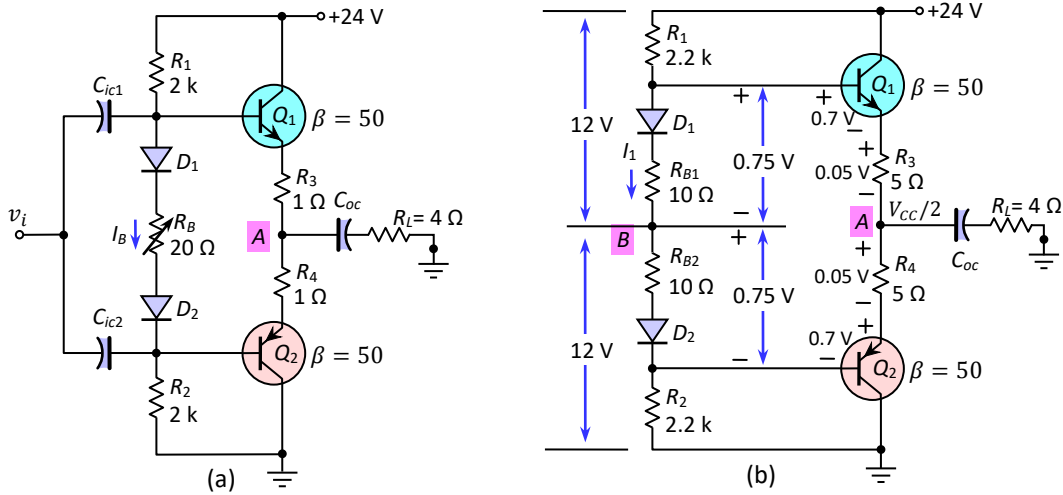


Fig.15-35: Complementary push-pull amplifier for Example 14-9

Therefore,

$$V_{R3} = V_{R4} = 0.75 \text{ V} - 0.7 \text{ V} = 0.05 \text{ V}$$

So, the current flowing through, R_3 which is also the emitter current of Q_1 will be,

$$I_3 = I_{E1} = I_{E2} = \frac{0.05 \text{ V}}{5 \Omega} = 10.2 \text{ mA} \approx I_{C1} = I_{C2}$$

As $I_{E1} = I_{E2}$, no current will flow through the load R_L . Therefore, the potential at point A will be zero.

The values of collector-emitter voltage will be,

$$V_{CE1} = V_{CE2} = 12 \text{ V} - 0.05 \text{ V} \approx 12 \text{ V}$$

ii) For this problem the circuit is redrawn in Fig.15-35(a). For this circuit all voltages and current will be same as the first part of Fig.15-34. Only the difference is the point B and A will be at $V_{CC}/2$ voltage. The voltage drops are shown in Fig.15-35(b).

Comments: For dual power supply as well as for single power supply, the point A to which the load is connected will be always at half of the supply voltage.

15.10.1 Complementary Emitter Follower with Driver Amplifier

The basic circuit of a Class-AB amplifier using a complementary emitter follower output stage and a capacitor-coupled load is shown Fig.15-36. The circuit is termed a **complementary symmetry amplifier**.

Transistor Q_1 and resistors R_1 , R_2 , R_C , and R_{E1} comprise a common emitter amplifier that produces all of the voltage gain. The output of Q_1 is developed across R_C and is applied to the bases of Q_2 and Q_3 . This amplifier is called the **driver** stage as it drives the output power amplifier. The output stage is same as described in section 14.9. Capacitor C_{oc} AC couples R_L , and DC isolates R_L to keep it from affecting the circuit bias conditions. The total voltage drop (V_B) across diodes D_1 and D_2 and resistor R_B forward biases the base-emitter junctions of Q_2 and Q_3 to eliminate cross-over distortion. Emitter resistors R_{E2} and R_{E3} help limit the quiescent current through Q_2 and Q_3 and provide bias stability discussed as before. Adjustment of the bias voltage (V_B) is provided by variable resistor R_B . The diodes have voltage drops (V_D) that approximately match the output transistor's V_{BE} levels. In Fig.15-33, the same input signal has been applied to the bases of two output transistors, but here, the bases of output transistors are connected to different voltage levels. However as the diode voltage (V_D) does not change significantly when the diode current changes, so, when the AC voltage swings at the collector of Q_1 almost same AC voltage will be applied to the base of Q_2 and Q_3 . In fact, the diodes will provide required DC biasing for the output transistors, as well as bypasses the AC voltage across them (see Example 14-10). The diodes and output transistors can be thermally-coupled as described before. The junction of R_{E2} and R_{E3} (point A) must be biased to $V_{CC}/2$, so that the output voltage to C_{oc} can swing by equal amounts in positive- and negative-going directions.

The operation process of this amplifier is same as discussed for other push-pull amplifiers. As illustrated in Fig.15-37(a), for the positive half-cycles Q_3 is OFF, and Q_2 is ON. So Q_2 conducts current through the load R_L . At this time the output capacitor (C_{oc}) is charged by a voltage approximately equal to V_{CC} . Similarly as shown in Fig.15-37(b), for the negative half-cycles Q_2 is OFF, and Q_3 is ON. So Q_3 conducts current through the load R_L in opposite direction. At this time the output capacitor (C_{oc}) works as the supply voltage source.

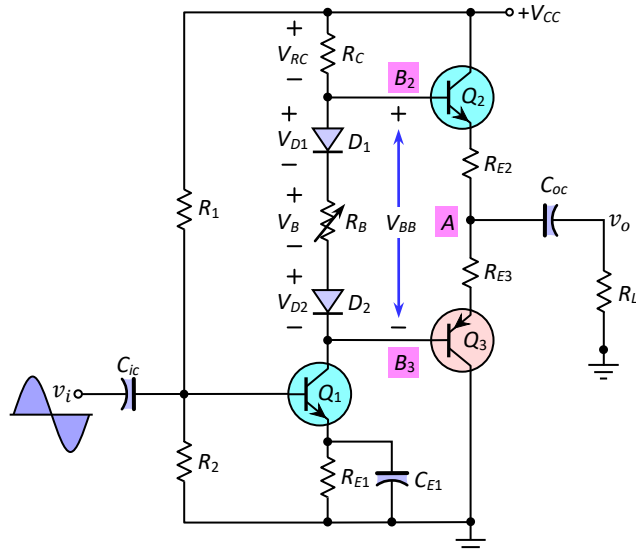


Fig.15-36: Complementary emitter follower with driver amplifier

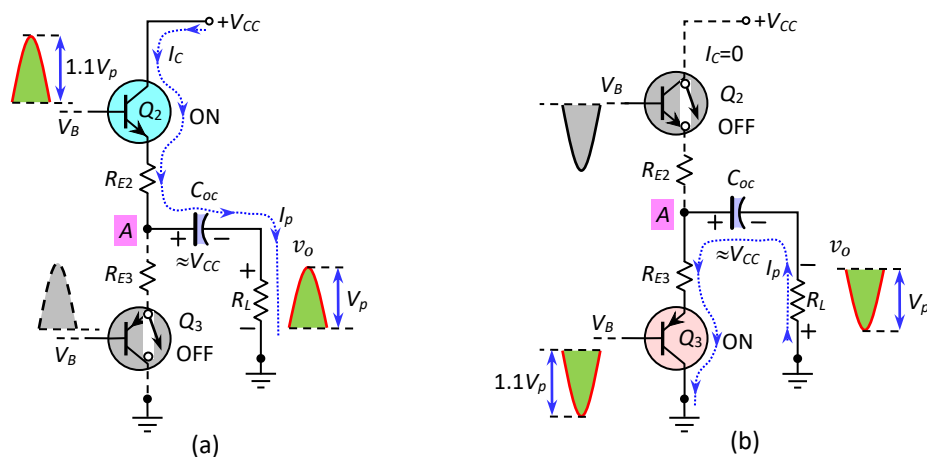


Fig.15-37: Push-pull operation of the amplifier of Fig.15-36

Example 14-10

The driver amplifier of the push-pull amplifier in Fig.15-30 is redrawn in Fig.15-37 with component values. Assume that the base currents of the output transistors (Q_2 and Q_3) are negligible. Determine the base voltage swings developed at the output transistors for a sinusoidal input signal with amplitude of ± 20 mV.

Solution:

Thevenin's equivalent of the biasing network:

$$E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{3 \text{ k} \times 24 \text{ V}}{16 \text{ k} + 3 \text{ k}} \approx 3.8 \text{ V}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{16 \text{ k} \times 3 \text{ k}}{16 \text{ k} + 3 \text{ k}} \approx 2.5 \text{ k}$$

Base current of Q_1 will be,

$$I_{B1} = \frac{E_{Th} - V_{BE1}}{R_{Th} + (\beta_1 + 1)R_3} = \frac{3.8 \text{ V} - 0.7 \text{ V}}{2.5 \text{ k} + (100 + 1) \times 0.62 \text{ k}} \approx 47.6 \text{ } \mu\text{A}$$

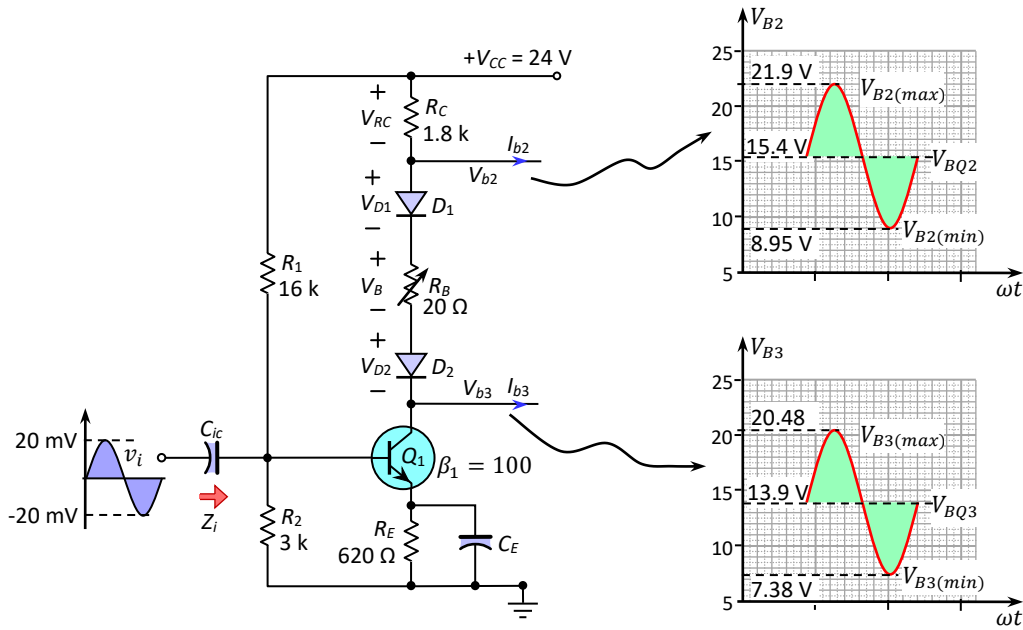


Fig.15-38: Driver amplifier circuit for Example 14-10

Quiescent current of Q_1 will be,

$$I_{CQ1} = I_{B1} \times \beta_1 = 100 \times 47.6 \mu\text{A} \approx 4.76 \text{ mA}$$

$$\therefore V_{CEQ1} = V_{CC} - I_{C1}(R_E + R_C + R_B) - V_{D1} - V_{D2}$$

$$\text{or, } V_{CEQ1} = 24 \text{ V} - 4.76 \text{ mA}(1.8 \text{ k} + 0.02 + 0.62) - 2 \times 0.7 \text{ V} \approx 11 \text{ V}$$

The value of emitter resistance will be,

$$r_{e1} = \frac{26 \text{ mV}}{I_{E1}} \approx \frac{26 \text{ mV}}{I_{CQ1}} = \frac{26 \text{ mV}}{4.76 \text{ mA}} \approx 5.5 \Omega$$

$$\therefore h_{ie1} \approx r_{e1} \times \beta_1 = 5.5 \Omega \times 100 = 0.55 \text{ k}$$

The input resistance will be,

$$Z_i = R_B \parallel h_{ie} \approx h_{ie} = 0.55 \text{ k}$$

At zero signal condition, the base voltage of Q_2 , and Q_3 will be,

$$V_{BQ2} = V_{CC} - I_{CQ1} \times R_C = 24 \text{ V} - 4.76 \text{ mA} \times 1.8 \text{ k} \approx 15.4 \text{ V}$$

$$\text{and, } V_{B3} = V_{B1} - V_{D1} - V_{D2} - I_{CQ1} \times R_B$$

$$\therefore V_{B3} = 15.4 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} - 4.76 \text{ mA} \times 20 \Omega \approx 13.9 \text{ V}$$

Due to the input voltage the swing in the base current will be,

$$\Delta I_B = \frac{v_i}{Z_i} = \frac{\pm 20 \text{ mV}}{0.55 \text{ k}} \approx \pm 36.36 \mu\text{A}$$

Therefore, the swing in the collector current will be,

$$\Delta I_C = \pm 36.36 \mu\text{A} \times \beta_1 = \pm 36.36 \mu\text{A} \times 100 = \pm 3.6 \text{ mA}$$

The minimum collector current will be,

$$I_{C(\min)} = 4.76 \text{ mA} - 3.6 \text{ mA} = 1.16 \text{ mA}$$

The maximum collector current will be,

$$I_{C(\max)} = 4.76 \text{ mA} + 3.6 \text{ mA} = 8.36 \text{ mA}$$

For $I_{C(\min)}$ the base voltages will be,

$$V_{B2(\max)} = V_{CC} - I_{C(\min)} \times R_C = 24 \text{ V} - 1.16 \text{ mA} \times 1.8 \text{ k} \approx 21.9 \text{ V}$$

and,

$$V_{B3(\max)} = V_{b1} - V_{D1} - V_{D2} - I_{C(\min)} \times R_B$$

∴

$$V_{B3(\max)} = 21.9 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} - 1.16 \text{ mA} \times 20 \Omega \approx 20.48 \text{ V}$$

For $I_{C(\max)}$ the base voltages will be,

$$V_{B2(\min)} = V_{CC} - I_{C(\max)} \times R_C = 24 \text{ V} - 8.36 \text{ mA} \times 1.8 \text{ k} \approx 8.95 \text{ V}$$

and,

$$V_{B3(\min)} = V_{b1} - V_{D1} - V_{D2} - I_{C(\max)} \times R_B$$

∴

$$V_{B3(\min)} = 8.95 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} - 8.36 \text{ mA} \times 20 \Omega \approx 7.38 \text{ V}$$

These base voltages will swing sinusoidal around their quiescent value as shown in Fig.15-37. The peak-to-peak base voltage in Q_2 will be,

$$V_{B2} = V_{B2(\max)} - V_{B2(\min)} = 21.9 \text{ V} - 8.95 \text{ V} \approx 13 \text{ V}$$

The peak-to-peak base voltage in Q_3 will be,

$$V_{B3} = V_{B3(\max)} - V_{B3(\min)} = 20.48 \text{ V} - 7.38 \text{ V} \approx 13 \text{ V}$$

Comments: The peak-to-peak value at the bases of two output transistors is same.

15.10.2 Complementary Emitter Follower Amplifier Design

Like most of the power amplifiers, the design process of the power amplifier circuit shown in Fig.15-36 is largely a matter of selecting appropriate resistor voltage drops and current levels, and then applying Ohm's law to calculate the resistor values. The main design requirement is the output power (P_o) for a given value of load resistance (R_L). The peak output voltage (V_p) and peak output current (I_p) can be determined as,

$$P_o = \frac{V_{rms}^2}{R_L} = \frac{(V_p/\sqrt{2})^2}{R_L} = \frac{V_p^2}{2R_L}$$

∴

$$\boxed{V_p = \sqrt{2R_L P_o}} \quad (15-61)$$

Also,

$$P_o = \frac{V_p}{\sqrt{2}} \times \frac{I_p}{\sqrt{2}} = \frac{V_p I_p}{2}$$

$$\therefore \boxed{I_p = \frac{2P_o}{V_p}} \quad (15-62)$$

As large value of R_{E2} , and R_{E3} will cause power loss, the voltage drops across R_{E2} , and R_{E3} may be considered typically 5% to 10% of the peak output voltage when the peak output current flows through them. As the same peak current flows through the load and emitter resistances as illustrated in Fig.15-37(a) and (b), 5% to 10% of the peak output voltage will drop if R_{E2} , and R_{E3} are set to 5% to 10% of load resistance (R_L), that is,

$$\boxed{R_{E2} = R_{E3} = 0.05 R_L \text{ to } 0.1 R_L} \quad (15-63)$$

Already we know that the function of emitter resistors is to stabilize the DC biasing point. Here, the function of R_{E2} and R_{E3} is to stabilize the quiescent currents of Q_2 and Q_3 at the level to eliminate the cross-over distortion in the output waveforms. The amplifier circuits shown in Fig.15-36 employs no overall negative feedback (like the feedback we employed in op-amps). So, for this circuit the values of R_{E2} and R_{E3} should be as large as possible. **However, the amplifier of Fig.15- (with differential stage) employs AC and DC negative feedback. As this negative feedback will reduce the cross-over distortion, smaller emitter resistance can be used in that circuit.**

From Fig.15-39(a), the voltage swing at the base of Q_2 (ΔV_{B2}) should be,

$$\boxed{\Delta V_{B2} = V_p + V_{RE2}} \quad (15-64)$$

Similarly, the voltage swing, ΔV_{B3} , at the base of Q_3 will be,

$$\boxed{\Delta V_{B3} = V_p + V_{RE3}} \quad (15-65)$$

For symmetrical swings,

$$\Delta V_{B2} = \Delta V_{B3} \quad (15-66)$$

Considering 10% drop across the emitter resistance, ΔV_{B2} , and ΔV_{B3} can also be calculated as,

$$\boxed{\Delta V_{B2} = \Delta V_{B3} = 1.1 V_p} \quad (15-67)$$

The quiescent collector-emitter voltage, V_{CEQ1} , of the driving transistor Q_1 must be large enough to allow the base currents of Q_2 and Q_3 to swing negatively by $V_p + V_{RE2}$ or $V_p + V_{RE3}$ without pushing the transistor into saturation. We may allow collector-emitter voltage to fall to minimum 1 V (or to $V_{BE(sat)}$ if known). In that case,

$$\boxed{V_{CEQ1} = \Delta V_{B2} + 1 \text{ V} = \Delta V_{B3} + 1 \text{ V}} \quad (15-68)$$

The minimum voltage at the collector of Q_1 , during the negative peak will be ,

$$V_{C1(min)} = V_E + 1 \text{ V} \quad (15-69)$$

Since, generally $V_E = 3 \text{ V}$, we may set $V_{C1(min)} = 4 \text{ V}$ as shown in Fig.15-39(b).

Similarly, the minimum voltage across the collector resistor, R_C , during the positive peak of the output signal, should be

$$V_{RC(min)} = V_{C1(min)} = 4 \text{ V} \quad (15-70)$$

The DC voltage drop across the collector resistor, R_C , should be large enough to allow the base voltage of Q_2 and Q_3 to swing by ΔV_{B2} voltage positively without pushing the transistor Q_1 into cut-off. Like the collector-emitter voltage, we may allow V_{RC} to fall to 1 V. Hence,

$$V_{RC} = \Delta V_{B2} + 1 = \Delta V_{B3} + 1 \quad (15-71)$$

The voltage drop, V_{BB} should be large enough to bias the transistors Q_2 and Q_3 as class-AB operation eliminating cross-over distortion. Therefore, the minimum value of V_{BB} should be,

$$V_{BB} = V_{BE2} + I_{CQ}(R_{E2} + R_{E3}) + V_{BE3} \quad (15-72)$$

Here, I_{CQ} is the quiescent collector current of transistors Q_2 and Q_3 and its value can be set to 5% to 10% of peak output current I_p . It is expected that, $V_{BE2} = V_{D1}$ and $V_{BE3} = V_{D2}$. Therefore, $I_{CQ}(R_{E2} + R_{E3})$ should be supplied by adjusting the variable resistance R_B , that is,

$$I_{CQ}(R_{E2} + R_{E3}) = I_{CQ1}R_B$$

$$\therefore \quad R_B = \frac{I_{CQ}(R_{E2} + R_{E3})}{I_{CQ1}} \quad (15-73)$$

Peak base current of Q_2 and Q_3 will be,

$$I_{Bp2} = I_{Bp3} = \frac{I_p}{h_{fe2}} = \frac{I_p}{h_{fe3}} \quad (15-74)$$

The quiescent collector current of transistor Q_1 should be large enough to provide this current. A thumb rule is to consider,

$$I_{CQ1} = 10 \times I_{Bp2} \text{ or } = \frac{I_p}{10} \quad (15-75)$$

The emitter voltage of Q_1 should be 3 V to 5 V as described for single-stage amplifier in Chapter 10. The value of R_1 , R_2 , and C_{E1} have to be calculated as discussed in the same chapter.

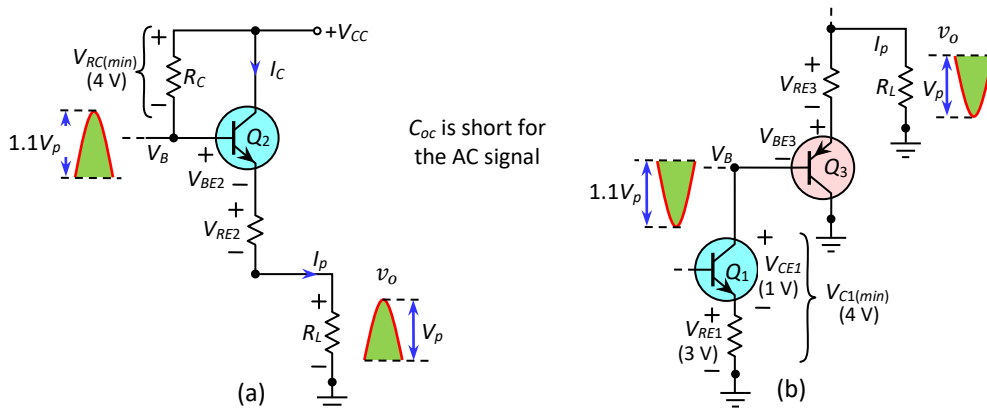


Fig.15-39: Effect of positive and negative half-cycles on the complementary emitter follower

As shown in Fig.15-39(a), the supply voltage required to produce positive output peak voltage is,

$$V_+ = V_p + V_{RE2} + V_{BE2} + V_{RC(min)}$$

Similarly, from Fig.15-39(b), the supply voltage required to produce the negative output peak voltage is,

$$V_- = V_p + V_{RE3} + V_{BE3} + V_{C1(min)}$$

and for symmetrical voltage swing,

$$V_+ = V_-$$

Therefore, the total supply voltage for this amplifier should be,

$$V_{CC} = 2(V_p + V_{RE3} + V_{BE3} + V_{C1(min)}) \quad (15-76)$$

However, if the supply voltage is already fixed (or given) then for symmetrical swings of positive and negative half-cycles, we may distribute this voltage as,

$$V_{C1(DC)} = V_{RC(DC)} = \frac{V_{CC} - V_{BB}}{2} \quad (15-77)$$

The output transistors should be specified in terms of their maximum C-E voltage, collector current, and power dissipation. The maximum V_{CE} for Q_2 , and Q_3 (in Fig.15-39) is the full supply voltage (V_{CC}) and their maximum current rating should be the peak load current plus the selected quiescent current (I_{Q2} or I_{Q3}) which is normally $1.1I_p$. Transistor power dissipation is calculated as,

$$P_T = \frac{P_{i(DC)} - P_o}{2} \quad (15-78)$$

Recall that the transistors must be operated within the safe operating area of their characteristics.

With a capacitor-coupled load, current is drawn from the power supply during the positive half-cycle of the output, but not during the negative half cycle. The capacitor acts as an energy reservoir to supply load current during the negative half cycle. Consequently, the current drawn from the supply voltage source has a half-wave rectified waveform (see Fig.15-40).

In addition to this AC current, the transistors will draw the quiescent DC current. So the total current drawn from the supply source will be,

$$I_{i(DC)} = I_{avg} + I_Q = \frac{I_p}{\pi} + 0.1I_p$$

$$\text{or,} \quad I_{i(DC)} = 0.32I_p + 0.1I_p$$

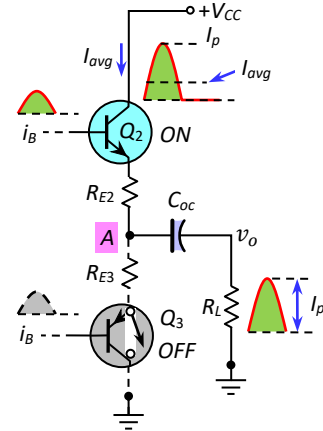


Fig.15-40: Average current drawn for the supply voltage

$$\therefore \quad I_{i(DC)} = 0.42I_p \quad (15-79)$$

The DC power supplied to the output stage will be,

$$P_{i(DC)} = V_{CC} \times I_{i(DC)} = 0.42 V_{CC} I_p \quad (15-80)$$

Using Equ.(15-77) and (15-79) the power rating of the transistors Q_2 and Q_3 can be determined.

Most audio power amplifiers typically have $R_L = 4 \Omega$ to 16Ω . So, the output coupling capacitor (C_{oc}) may have the largest capacitance. In that case, the lower cut-off frequency or 3 dB frequency is set by this capacitor not by the emitter bypass capacitor of Q_1 . Therefore, the value of this capacitor should be,

$$C_{oc} = \frac{1}{2\pi f_L R_L} \quad (15-81)$$

The values of other capacitor should be calculated considering their impedance one tenth of the impedance in series with the capacitor. As described in Chapter 9, the series resistance seen by C_{E1} is h_{ib1} , so,

$$C_{E1} = \frac{10}{2\pi f_L h_{ib1}} \quad (15-82)$$

and,

$$C_{ic} = \frac{10}{2\pi f_L Z_{i1}} \quad (15-83)$$

The rest of the driver amplifier can be designed using the same process as described in Chapter 10.

Example 14-11

Design an audio amplifier of Fig.15-36 to deliver $P_o = 2 \text{ W}$ to a load of $R_L = 16 \Omega$. Assume, $h_{fe2} = h_{fe3} = 50$, $h_{fe1} = 100$, $h_{ib1} = 12 \Omega$, $h_{ie1} = 1.6 \text{ k}$ and the lower cut-off frequency is $f_L = 100 \text{ Hz}$.

Solution:

Using Equ.(15-61), $V_p = \sqrt{2R_L P_o} = \sqrt{2 \times 16 \Omega \times 2 \text{ W}} = 8 \text{ V}$

Using Equ.(15-62), $I_p = V_p / R_L = 8 \text{ V} / 16 \Omega = 0.5 \text{ A}$

Using Equ.(15-63), $R_{E2} = R_{E3} = 0.1 R_L = 16 \Omega \times 0.1 = 1.6 \Omega$

Select $I_{CQ2} = I_{CQ3} = I_{CQ} = 0.1 I_p = 0.5 \text{ A} \times 0.1 = 50 \text{ mA}$.

The biasing voltage of Q_2 and Q_3 will be,

Using Equ.(15-72), $V_{BB} = V_{BE2} + I_{CQ2}(R_{E2} + R_{E3}) + V_{BE3}$

or, $V_{BB} = 0.7 \text{ V} + 50 \text{ mA}(1.6 \Omega + 1.6 \Omega) + 0.7 \text{ V} = 1.56 \text{ V}$

Therefore, the maximum base current of Q_2 and Q_3 will be,

$$I_{B2(max)} = I_p / h_{fe2} = 0.5 \text{ A} / 50 = 10 \text{ mA}$$

Let $I_{RC(min)} = I_{B2(max)} + 2 \text{ mA} = 10 \text{ mA} + 2 \text{ mA} = 12 \text{ mA} = I_{CQ1}$

Select $V_{E1} = 3 \text{ V}$ and $V_{CE1(min)} = 1 \text{ V}$. Therefore, $V_{C1(min)} = 3 \text{ V} + 1 \text{ V} = 4 \text{ V}$.

Similarly, select $V_{RC(min)} = 4 \text{ V}$.

$$R_C = V_{RC(min)}/I_{RC(min)} = 4 \text{ V} / 12 \text{ mA} \approx 333 \Omega \text{ [Use standard } 330 \Omega \text{]}$$

Using Equ.(15-76), $V_{CC} = 2[V_p + V_{RE2} + V_{BE2} + V_{RC(min)}]$

or, $V_{CC} = 2[8 \text{ V} + 0.8 \text{ V} + 0.7 \text{ V} + 4 \text{ V}] = 27 \text{ V}$

Let us select $V_{CC} = 30 \text{ V}$.

Using Equ.(15-77), $V_{RC(DC)} = (V_{CC} - V_{BB})/2 = (30 \text{ V} - 1.56 \text{ V})/2 = 14.22 \text{ V}$

or, $I_{C1(DC)} = V_{RC(DC)}/R_{C1} = (14.22 \text{ V})/330 \Omega = 43 \text{ mA}$

Using Equ.(15-73), $R_B = \frac{I_{CQ}(R_{E2} + R_{E3})}{I_{CQ1}} = \frac{50 \text{ mA}(1.6 \Omega + 1.6 \Omega)}{12 \text{ mA}}$

$\therefore R_B \approx 13 \Omega$ [use standard 100Ω pot]

$$R_{E1} = \frac{V_E}{I_{C1}} = \frac{3 \text{ V}}{12 \text{ mA}} = 250 \Omega \text{ [use standard } 240 \Omega \text{]}$$

$$V_{B1} = V_E + V_{BE1} = 3 \text{ V} + 0.7 \text{ V} = 3.7 \text{ V}$$

$$I_{B1} = \frac{I_{CQ1}}{h_{fe1}} = \frac{12 \text{ mA}}{100} = 120 \mu\text{A}$$

Let $I_1 = I_{B1} \times 10 = 120 \mu\text{A} \times 10 = 1.2 \text{ mA}$.

$$R_2 = \frac{V_{B1}}{I_1} = \frac{3.7 \text{ V}}{1.2 \text{ mA}} \approx 3 \text{ k}$$

$$R_2 = \frac{V_{CC} - V_{B1}}{I_1} = \frac{30 \text{ V} - 3.7 \text{ V}}{1.2 \text{ mA}} \approx 22 \text{ k}$$

Output transistor specifications:

$$V_{CE(max)} = V_{CC} = 30 \text{ V}$$

$$I_{C(max)} = 1.1I_p = 1.1 \times 0.5 \text{ A} = 0.55 \text{ A}$$

$$P_{i(DC)} = 0.42V_{CC}I_p = 0.42 \times 30 \text{ V} \times 0.5 \text{ A} = 6.3 \text{ W}$$

Using Equ.(15-78), $P_T = \frac{P_{i(DC)} - P_o}{2} = \frac{6.3 \text{ W} - 2 \text{ W}}{2} \approx 2.15 \text{ W}$

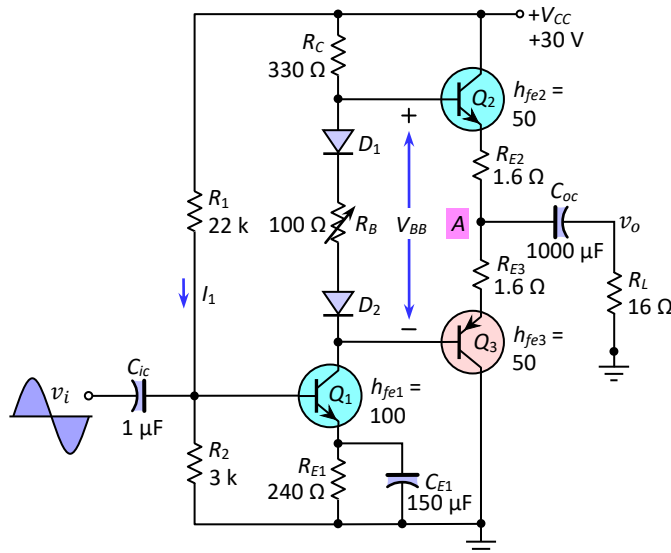


Fig.15-41: Complementary emitter follower with driver amplifier

Calculation of Capacitors:

As $h_{ib1} < R_L$, so the lower cut-off frequency will be set by C_{E1} .

$$C_{E1} = \frac{1}{2\pi(100\text{ Hz}) \times 12\text{ }\Omega} \approx 133\text{ }\mu\text{F}, [\text{use } 150\text{ }\mu\text{F capacitor}]$$

$$x_c = h_{ib}$$

$$C_{oc} = \frac{10}{2\pi(100\text{ Hz}) \times 16\text{ }\Omega} \approx 1000\text{ }\mu\text{F}$$

Let $Z_{i1} \approx h_{ie1}$

$$C_{ic} = \frac{10}{2\pi f_L \times Z_{i1}} = \frac{10}{2\pi(100\text{ Hz}) \times 1.6\text{ k}} \approx 1\text{ }\mu\text{F}$$

Comments: The designed amplifier is shown in Fig.15-41.

15.10.3 Complementary Emitter Follower with Split Supply

The class-AB power amplifier of Fig.15-36, needs a reservoir capacitor (C_{oc}). Normally, this capacitor is very large, and hence, needs more space and cost. Instead of using this coupling capacitor, the load can be direct coupled to the power amplifier, if a split power supply is used. Although, split power supply may add

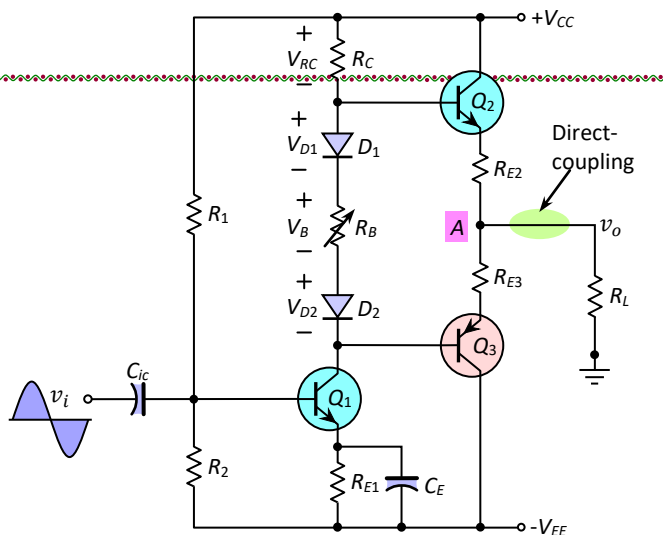


Fig.15-42: Complementary emitter follower with split power supply

extra cost, but in some systems, split power is already available for some other purpose. Fig.15-42 shows the class-AB power amplifier with direct coupled load and a split power supply ($+V_{CC}$, $-V_{EE}$). Note that in the amplifier of Fig.18-18, point A must be at $V_{CC}/2$, but, here the potential level at this point should be zero. This is necessary to avoid current flow through the load and hence to avoid power loss at zero signal condition. The operation process of this amplifier is exactly same as that of capacitor-coupled amplifier of Fig.18-18. The design process of this amplifier is also same. Here the total voltage will $V_{CC} + |V_{EE}|$. So the maximum value V_{CE} of the operation transistors should be greater than $V_{CC} + |V_{EE}|$.

The value of the supply voltage can be calculated using Equ.(15-76) as,

$$V_{CC} = V_p + V_{RE3} + V_{BE3} + V_{C1(min)} \quad (15-84)$$

$$V_{EE} = -[V_p + V_{RE3} + V_{BE3} + V_{C1(min)}] \quad (15-85)$$

Example 14-12

Design an audio amplifier of Fig.15-42 to deliver $P_o = 2 \text{ W}$ to a load of $R_L = 16 \Omega$. Assume, $h_{fe2} = h_{fe3} = 50$, $h_{fe1} = 100$, $h_{ib1} = 12 \Omega$, $R_L = 16 \Omega$, $h_{ie1} = 1.6 \text{ k}$ and the lower cut-off frequency is $f_L = 100 \text{ Hz}$.

Solution:

The design requirements are same as in Example 14-11. Here the main difference is the supply voltage. All calculation processes will be exactly similar to that of Example 14-11 except the supply voltage.

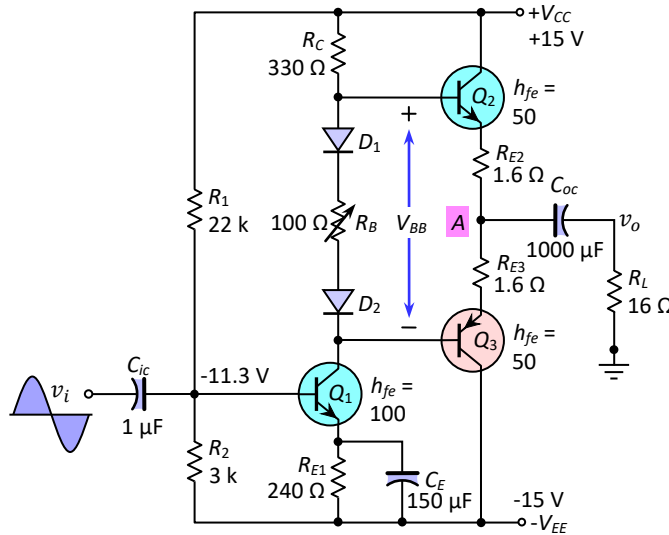


Fig.15-43: Complementary emitter follower with driver amplifier

Using Equ.(15-84),

$$V_{CC} = [V_p + V_{RE2} + V_{BE2} + V_{RC(min)}]$$

$$\therefore V_{CC} = [8 \text{ V} + 0.8 \text{ V} + 0.7 \text{ V} + 4 \text{ V}] = 13.5 \text{ V}$$

Let us select $V_{CC} = 15 \text{ V}$. Similarly $-V_{EE} = -15 \text{ V}$.

From Example 14-11, $I_1 = I_2 = I_{B1} \times 10 = 120 \mu\text{A} \times 10 = 1.2 \text{ mA}$.

$$\therefore R_1 = \frac{V_{R2}}{I_2} = \frac{-11.3 \text{ V} + 15 \text{ V}}{1.2 \text{ mA}} \approx 3 \text{ k}$$

$$\text{and, } R_2 = \frac{V_{CC} - V_{B1}}{I_1} = \frac{15 \text{ V} + 11.3 \text{ V}}{1.2 \text{ mA}} \approx 22 \text{ k}$$

Comments: All the design parameters are same as Example 14-11. Here the split power supply has been used, but the total voltage is same (30 V). Therefore, the design components remain similar.

15.11 Class-AB Push-Pull Amplifier with Darlington Pair

High-power transistors usually have low current gain (small value of β). For example, a power transistor TIP3055 has maximum current capacity of 15 A, but the β may be as low as 20. When this type of transistors are used at the output stage (Q_2 and Q_3) of power amplifier (Fig.15-36), relatively large base current must flow into Q_2 and Q_3 to supply the high load current. For example, to supply 50 W to a load of 4Ω , the peak value of the output current should be 5 A. Therefore, the peak value of the base currents of Q_2 and Q_3 should be $I_B = 5 \text{ A}/20 = 250 \text{ mA}$. This means that the quiescent current through driving transistor must be large (up to $10I_B$), and consequently the value of the collector resistor R_C will be small. The small value of R_C results in very low voltage gain of the driver amplifier. To overcome this problem, Darlington-connected output transistors may be used, as illustrated in Fig.15-44.

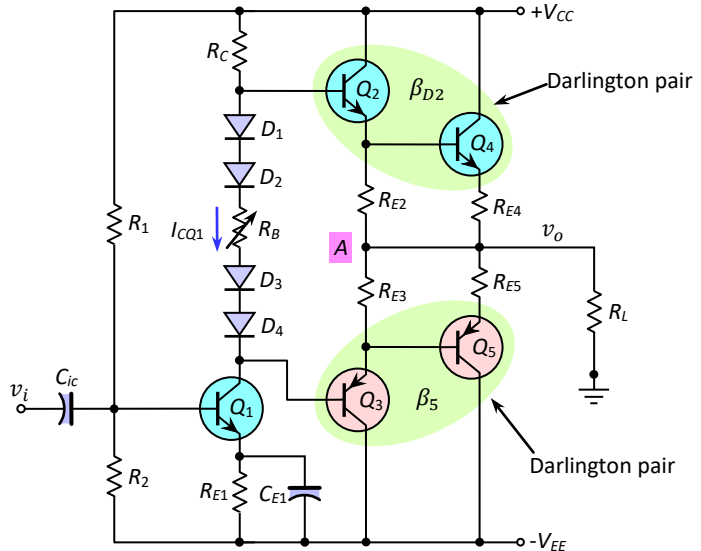


Fig.15-44: Complementary emitter follower with Darlington pairs

Transistors Q_2 and Q_3 are low-power devices that supply base current to output transistors (Q_4 and Q_5), respectively. Being low-power transistors, Q_2 and Q_3 will have comparatively larger value of β . Therefore, as derived in Chapter 4, the effective current gain of the NPN Darlington pair should be,

$$\boxed{\beta_{D2} = \beta_2\beta_4 + \beta_2 + \beta_4} \quad (15-86)$$

and for the PNP Darlington pair it should be,

$$\beta_{D3} = \beta_2\beta_4 + \beta_2 + \beta_4 \quad (15-87)$$

For perfectly matched transistors, $\beta_{D2} = \beta_{D3}$. Note that four biasing diodes have been used in Fig.15-44 to bias the four transistor base-emitter junctions. When peak load current, I_p flows through these Darlington pairs, the peak base current flowing through Q_4 and Q_5 will be,

$$I_{Bp4} = \frac{I_p}{\beta_{D2}} = I_{Bp4} = \frac{I_p}{\beta_{D3}} \quad (15-88)$$

As β_{D2} and β_{D3} will be very large, the base currents will be very small compared to those without Darlington pair. This reduced base current allows I_{CQ1} to be smaller, giving a larger resistance for R_C , which results in a larger voltage gain. Resistors R_{E2} and R_{E3} in Fig.15-44 are included to bias Q_2 and Q_3 OFF when Q_4 and Q_5 are in cut-off. The largest possible resistance values should normally be selected for R_{E2} and R_{E3} . When Q_2 and Q_3 are OFF the collector-base leakage current I_{CBO} flows in R_{E2} and R_{E3} as shown in Fig.15-. The voltage drop across these resistors should be much smaller than the normal transistor base-emitter voltage. We can select $I_{CBO}R_{E2} = 0.01 \text{ V to } 0.05 \text{ V}$ to get satisfactory resistor values. R_{E2} and R_{E3} are not required when power Darlington pairs are used, as illustrated in Fig.15-46.

The amplifier can be operated under push-pull class-AB condition only if a quiescent current is allowed to flow through the driver transistors Q_2 and Q_3 . In that case the voltage drops across R_{E4} and R_{E5} will bias Q_4 and Q_5 in class-AB mode.

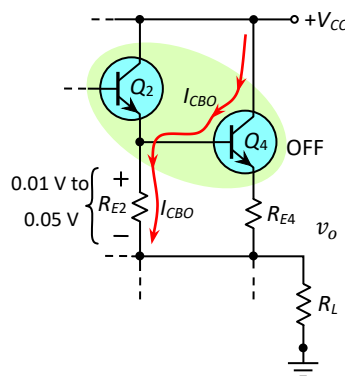


Fig.15-45: Leakage current flowing through R_{E2} when Q_4 is OFF

Example 14-13

Design the complementary emitter follower using Darlington pair to deliver 6 W to a load of $R_L = 8 \Omega$. Assume $h_{fe2} = h_{fe3} = 100$, $h_{fe4} = h_{fe5} = 25$, and $I_{CBO4} = I_{CBO5} = 0.1 \text{ mA}$.

Solution:

Using Equ.(15-61), $V_p = \sqrt{2R_L P_o} = \sqrt{2 \times 8 \Omega \times 6 \text{ W}} \approx 9.8 \text{ V}$

Using Equ.(15-62), $I_p = V_p / R_L = 9.8 \text{ V} / 8 \Omega \approx 1.2 \text{ A}$

Using Equ.(15-63), $R_{E4} = R_{E5} = 0.1 R_L = 8 \Omega \times 0.1 \approx 1 \Omega$

Select $I_{CQ4} = I_{CQ5} = I_{CQ} = 0.1 I_p = 1.2 \text{ A} \times 0.1 = 0.12 \text{ A}$, and $V_{RE2} = 0.02 \text{ V}$. Therefore,

$$R_{E2} = R_{E3} = \frac{V_{RE2}}{I_{CBO2}} = \frac{0.02 \text{ V}}{0.1 \text{ mA}} = 200 \Omega$$

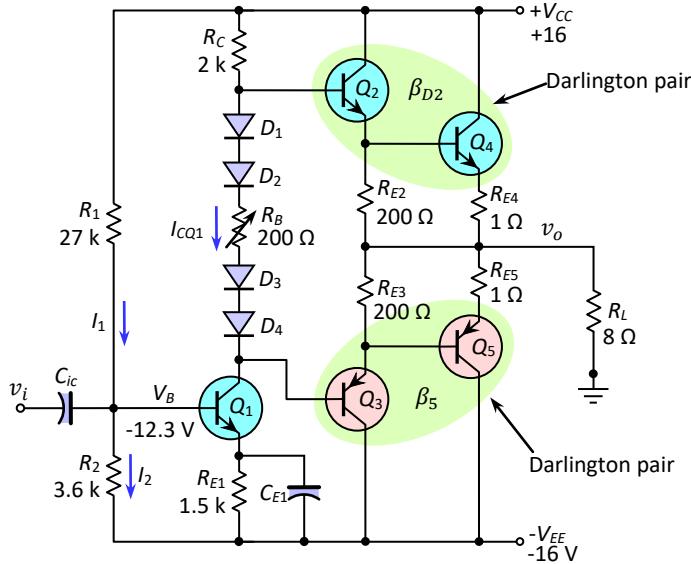


Fig.15-46: Complementary emitter follower with Darlington pairs

The current gain for the Darlington pairs will be,

Using Equ.(15-84), $\beta_{D2} = \beta_{D3} \approx \beta_2 \times \beta_4 = \beta_3 \times \beta_5 = 100 \times 25 = 2500$

The maximum base current of the Darlington pairs will be,

$$I_{B2} = I_{B3} = \frac{I_p + I_{CQ4}}{\beta_{D2}} = \frac{1.2 \text{ A} + 0.12 \text{ A}}{2500} = 0.53 \text{ mA}$$

Let $I_{CQ1} = 2 \text{ mA}$.

Let $V_{E1} = 3 \text{ V}$ and $V_{CE1(\min)} = 1 \text{ V}$. Therefore, $V_{C1(\min)} = 3 \text{ V} + 1 \text{ V} = 4 \text{ V}$. For symmetrical voltage distribution let $V_{RC(\min)} = 4 \text{ V}$.

The biasing voltage will be,

Using Equ.(15-72), $V_{BB} = V_{BED2} + I_{CQ4}(R_{E4} + R_{E5}) + V_{BED3}$

or, $V_{BB} = 1.4 \text{ V} + 0.12 \text{ A}(1 \Omega + 1 \Omega) + 1.4 \text{ V} = 3.04 \text{ V}$

Using Equ.(15-84), $V_{CC} = [V_p + V_{RE2} + V_{BE2} + V_{RC(\min)}]$

or, $V_{CC} = 9.8 \text{ V} + 0.98 \text{ V} + 1.4 \text{ V} + 4 \text{ V} = 16.18 \text{ V}$

Let us select $V_{CC} = 16 \text{ V}$, and $-V_{EE} = -16 \text{ V}$.

The biasing resistance can be calculated as,

$$R_B = (V_{BB} - 4V_D)/I_{C1(DC)} = (3.04 \text{ V} - 2.8 \text{ V})/2 \text{ mA}$$

$\therefore R_B \approx 120 \Omega$ [Use a 200Ω pot]

Collector and emitter resistance of Q_1 will be,

$$R_C = \frac{V_{RC(min)}}{I_{CQ1}} = \frac{4 \text{ V}}{2 \text{ mA}} = 2 \text{ k}$$

and,

$$R_{E1} = \frac{V_E}{I_{CQ1}} = \frac{3 \text{ V}}{2 \text{ mA}} = 1.5 \text{ k}$$

The base voltage of Q_1 will be, $V_B = -16 \text{ V} + 3 \text{ V} + 0.7 \text{ V} = -12.3 \text{ V}$. Let $I_1 = 1 \text{ mA} \approx I_2$. Therefore,

$$R_2 = \frac{V_B - V_{EE}}{I_2} = \frac{3.7 \text{ V}}{1 \text{ mA}} = 3.7 \text{ k} \text{ [use standard 3.6 k]}$$

$$R_1 = \frac{V_{CC} - V_B}{I_1} = \frac{16 \text{ V} - (-12.3 \text{ V})}{1 \text{ mA}} \approx 28.3 \text{ k} \text{ [use standard 27 k]}$$

Comments: The designed amplifier is shown in Fig.15-46. The values of the capacitors have not been calculated as the necessary parameters are not given.

Instead of using two transistors as Darlington pair, Darlington single transistors can be used. These transistors contain Darlington connected two transistors in a single package. The circuit diagram is shown in Fig.15-47. This circuit is similar to that of Fig.15-44, but as the Darlington transistor has $V_{BE} \approx 1.4 \text{ V}$, four diodes are used to bias two Darlington transistors. In this circuit there is no scope to use R_{E4} and R_{E5} resistors.

15.12 Class-AB Push-Pull Amplifier with Complementary Darlington Pair

In the past, NPN power transistors offered much better performance compared to PNP power transistors.

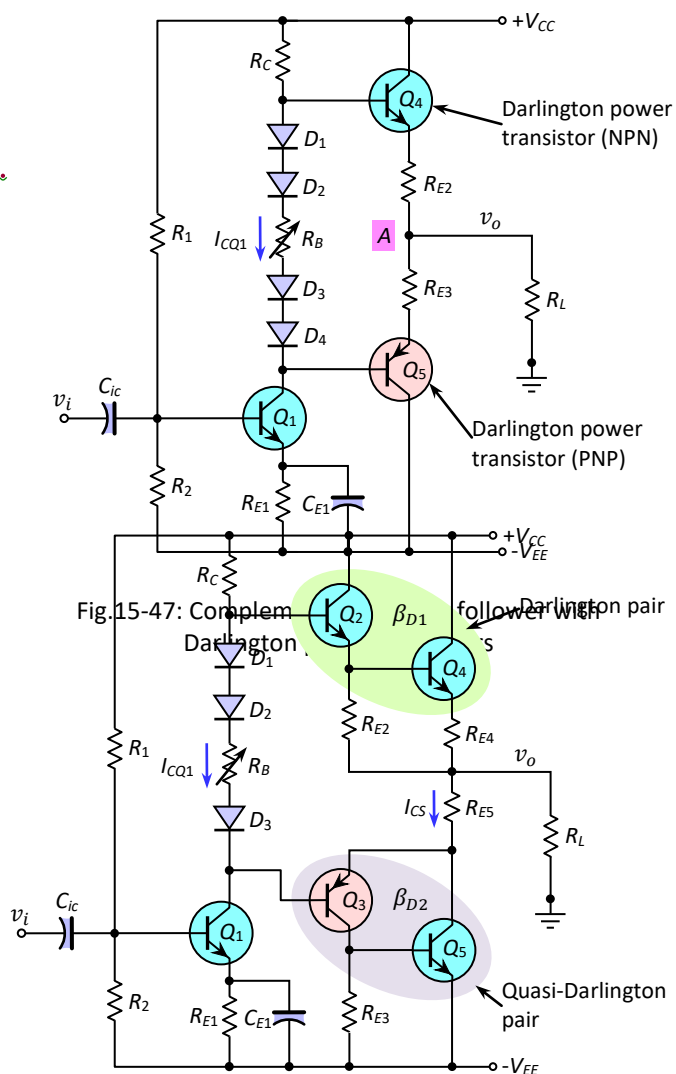


Fig.15-48: Complementary emitter follower with quasi-Darlington pairs

As a result the quasi-complementary output was an ideal option. As shown in Fig.15-48, a quasi-complementary amplifier consists of an NPN Darlington pair and a PNP complementary-Darlington pair (also called Sziklai pair). It had the advantage that both output transistors are the same (NPN) and often the very common transistors such as the 2N3055 can be used. Although the circuit has been widely used in the past, quasi-complementary symmetry amplifiers are used less these days because NPN and PNP pairs with very similar performance are available now. Consider the arrangement in Fig.15-48, the upper portion is a normal Darlington pair as described in previous section. Complementary-Darlington pair or Sziklai pair is used in the lower part consisting of Q_3 and Q_5 . Here, Q_5 is a high-power NPN transistor, and Q_3 is a low-power PNP device. The effective current gain of this Sziklai pair will be, $\beta_{D2} = \beta_3\beta_5 + \beta_3 + 1$ as described in Chapter 4. When Q_3 base current (I_{B3}) flows, the collector current (I_{CS}) of this Sziklai pair, which is the load current, should be,

$$I_{CS} = \frac{I_p}{\beta_{D2}} = I_{B3}\beta_{D2} \quad (15-89)$$

This is the same as the current gain with Darlington-connected transistors. Now note that when a negative-going input voltage is applied to the base of Q_3 a negative-going output occurs, because the emitter of Q_3 (and the collector of Q_5) follows the input voltage. Thus, the combination of transistors Q_3 and Q_5 in Fig.15-48 behaves as a high-power PNP emitter follower, just like Darlington-connected transistors Q_3 and Q_5 in Fig.15-44. Because transistors Q_4 and Q_5 in Fig.15-48 are both NPN devices, they can be the same type of transistor. This eliminates any problem with finding suitable complementary high-power transistors. Resistor R_{E3} in Fig.15-48 ensures that Q_5 is biased OFF when Q_3 goes into cut-off. Moreover, voltage drop across this resistor biases the base-emitter junction of Q_5 and that ensures the class-AB operation to eliminate cross-over distortion.

Example 14-14

Design the amplifier of Fig.15-48 to deliver 5 W power to a load of $16\ \Omega$. Assume, $h_{fe4} = h_{fe5} = 20$, $h_{fe2} = h_{fe3} = 80$, and $h_{fe1} = 120$.

Solution:

Using Equ.(15-61), $V_p = \sqrt{2P_o R_L} = \sqrt{2(5\text{ W})(16\ \Omega)} \approx 12.7\text{ V}$

Using Equ.(15-62), $I_p = \frac{V_p}{R_L} = \frac{12.7\text{ V}}{16\ \Omega} \approx 0.8\text{ A}$

Let, $V_{RE4} = V_{RE5} = \frac{V_p}{10}$ [when I_p flows]

or, $V_{RE2} = V_{RE3} = \frac{12.7\text{ V}}{10} = 1.27\text{ V}$

$\therefore R_{E2} = R_{E3} = \frac{V_{RE2}}{I_p} = \frac{1.27\text{ V}}{0.8\text{ A}} \approx 1.6\ \Omega$

Let the quiescent current of Q_4 and Q_5 be,

$$I_{CQ4} = I_{CQ5} = \frac{I_p}{10} = \frac{0.8 \text{ A}}{10} = 80 \text{ mA}$$

The base current of Q_4 and Q_5 will be,

$$I_{BQ4} = I_{BQ5} = \frac{I_p}{h_{fe4}} = \frac{80 \text{ mA}}{20} = 4 \text{ mA}$$

Let the quiescent current of Q_2 and Q_3 be,

$$I_{CQ2} = I_{CQ3} = I_{BQ4} \times 10 = 4 \text{ mA} \times 10 = 40 \text{ mA}$$

The value of R_{E4} and R_{E5} can be calculated as,

$$R_{E4} = R_{E5} = \frac{V_{BE2} + I_{CQ2} \times R_{E2}}{I_{CQ4}}$$

or,

$$R_{E4} = R_{E5} = \frac{0.7 + 80 \text{ mA} \times 1.6 \Omega}{40 \text{ mA}} \approx 20 \Omega$$

The design procedure of the driver stage is same as in Example 14-13.

Comments: Values of R_{E4} and R_{E5} can be calculated using I_{CBO} as in Example 14-13.

15.13 Complementary Emitter-Follower with Current Limiting Circuit

Due to short circuit or connected load with lower resistance than specified, excessive current may flow through the power transistor of amplifiers discussed so far (Fig.15-36 to Fig.15-48). This excessive current may even destroy the power transistors (Q_2 and Q_3 of Fig.15-33). For this reason, output current limiting circuits are often included in a power amplifier. Fig.15-49 shows the typical arrangement for a current limiting circuit.

The current limiting circuit is shown in the shaded area. Emitter resistors R_{E2} and R_{E3} are each made up of two components (R_a and R_b), as illustrated in the figure. The current limiting transistors, Q_4 and Q_5 are connected such a way that the voltage drop across R_{b2} and R_{b3} due to the overloaded current will turn Q_4 and Q_5 ON. This occurs only when I_L is greater than the maximum designed load current $I_{L(max)}$. When Q_4 turns ON, it pulls the base of Q_2 down to its emitter voltage, so that Q_2 cannot supply current more than $I_{L(max)}$. Similarly, turning Q_5 ON (during negative half-cycles) causes the base of Q_3 to be pulled up toward its emitter voltage, thus again limiting the output current to $I_{L(max)}$.

The component values shown in Fig.15-49 limit the peak output current to approximately 1 A, as

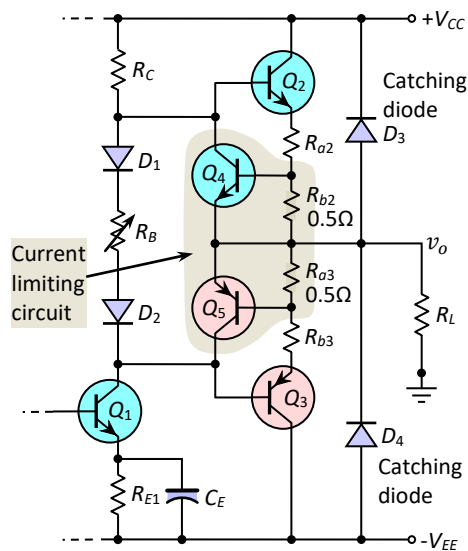


Fig.15-49: Complementary emitter follower with current limiting circuit

for this current the voltage drop across R_{b2} and R_{b3} will be 0.5 V which turns the transistor Q_4 and Q_5 ON. The diodes D_3 and D_4 included in Fig.15-49 are called *Catch diodes*. These diodes are reverse biased power diodes, to absorb transients generated by fast current-changes into an inductive load. All moving-coil loudspeakers present inductive impedance over some frequencies. Thus they protect both the power transistors and the load from exceeding the supply voltage level.

15.14 V_{BE} Multiplier

So far we have used diodes (and a potentiometer) to bias the output transistors in class-AB mode. The problem is a number of diodes have to be used, and there is no flexibility to control the biasing voltage V_{BB} . This problem can be overcome by using a V_{BE} multiplier circuit. The class-AB amplifier of Fig.15-36 has been redrawn in Fig.15-50 using a V_{BE} multiplier (voltage divider network of driving transistor is not shown). The circuit shown in the shaded area is the V_{BE} multiplier and is used as alternative to diode biasing. This circuit is also known as *adjustable diode*. V_{BE} multiplier is also used as DC level shifting.

This circuit is known as V_{BE} multiplier because it produces a bias voltage ($V_{BB} = V_{CE4}$) which is a multiple of the V_{BE4} of transistor Q_4 . As the base biasing current of Q_4 is very small, the currents flowing through the voltage divider resistors (R_3 , and R_4) can be considered equal, that is $I_3 = I_4$ (let it be denoted by I_B). Moreover, as the base current is small, I_B will also be small compared to the quiescent collector current of Q_4 . Therefore, we can assume that $I_{CQ1} = I_{CQ4}$. Therefore,

$$I_4 = \frac{V_{BE4}}{R_4} = I_3 = I_B \quad (15-90)$$

From the circuit, we can write,

$$V_{BB} = V_{CE4} = I_B(R_3 + R_4)$$

$$\text{or, } V_{BB} = V_{CE4} = \frac{V_{BE4}}{R_4}(R_3 + R_4)$$

$$\therefore \boxed{V_{BB} = V_{CE4} = V_{BE4} \times \left(1 + \frac{R_3}{R_4}\right)} \quad (15-91)$$

As the name implies, the output voltage (V_{CE4}) of the V_{BE} multiplier circuit is the product of base-emitter voltage and a constant $(1 + R_3/R_4)$. By changing the value of R_3/R_4 any multiple of V_{BE4} can be produced. In the figure, V_{BE4} is used to bias the output transistors into class-AB mode. It is seen that the base bias voltage for the output transistors can be set by suitable selection of the V_{BE} multiplier resistors. The bias voltage remains constant even though I_{C1} changes. If V_{BE} of the output transistors changes with temperature, the V_{BE4} will also change similarly and hence the

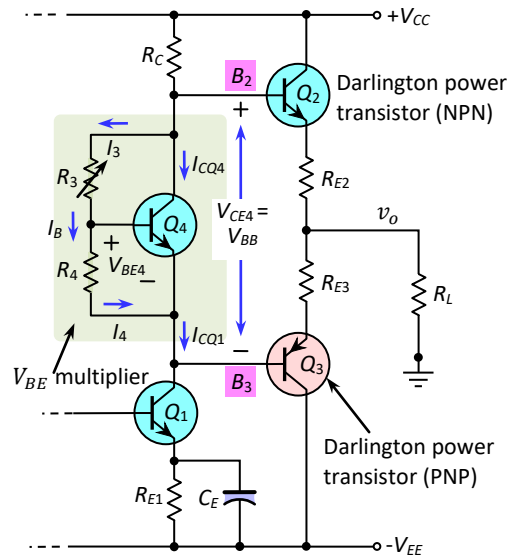


Fig.15-50: Complementary emitter follower with V_{BE} multiplier

biasing voltage V_{CE4} will also be changed. Thus the effect of temperature change will readily be compensated by the V_{BE} multiplier. Although, here, the V_{BE} multiplier has been used with the complementary emitter follower consisting of Darlington power transistors, this circuit can also be used for other class-AB power amplifiers (Fig.15-36 to Fig.15-48).

Example 14-15

Calculate the biasing voltage range (V_{BB}) of the circuit of Fig.15-50 assuming $I_{CQ1} = 1 \text{ mA}$, $R_4 = 1.2 \text{ k}$, and $R_3 = 3.3 \text{ k}$ variable.

Solution:

As R_3 is a 1.2 k variable resistor, its minimum value can be set to 0Ω . Thus the minimum biasing voltage will be,

$$\text{Using Equ. (15-91), } V_{BB(\min)} = V_{BE4} \times \left(1 + \frac{R_3}{R_4}\right) = 0.7 \text{ V} \left(1 + \frac{0 \Omega}{1.2 \text{ k}}\right) = 0.7 \text{ V}$$

The maximum value of R_3 can be set 3.3 k . Thus the maximum biasing voltage will be,

$$\text{Using Equ. (15-91), } V_{BB(\max)} = 0.7 \text{ V} \left(1 + \frac{3.3 \text{ k}}{1.2 \text{ k}}\right) = 2.6 \text{ V [Ans.]}$$

Comments: When R_3 is set to zero, the base and collector terminal of transistor Q_4 will be shorted and hence, this transistor will work as a diode. So $V_{BB} = 0.7 \text{ V}$.

15.15 Complementary emitter follower with Current Source (or Active Load)

The collector resistor (R_C) of the driving transistor (Q_1) in the amplifier circuits discussed so far (Fig.15-36 to Fig.15-48) has its resistance limited by the need to supply base current to the NPN output transistor (Q_2). The resistance value of R_C will also be limited by the collector current of Q_1 , that is, larger value of I_{C1} results in lower value of R_C for a constant power supply voltage. But if, we want to use negative feedback in power amplifier a larger voltage gain is necessary. It is shown that (Chapter 12) negative feedback reduces distortion and increases circuit bandwidth, so its use is necessary in all power amplifiers. Thus, we need higher voltage gain, but it is not possible to attain that gain by increasing R_C . A solution to this

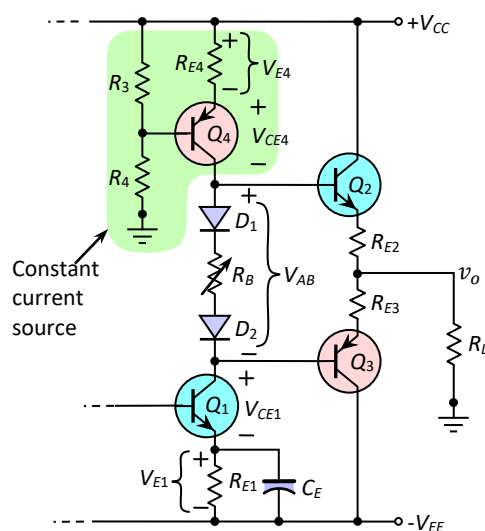


Fig.15-51: Complementary emitter follower with current source (or active load)

problem is to use constant current source in place of collector resistor R_C . They work as the load of transistor Q_1 , so they are also called *active load*.

Fig.15-51 shows a complementary power amplifier with a constant current source in place of R_C . The constant current source is consisted of transistor Q_4 together with emitter resistor R_{E4} , and bias resistors R_3 and R_4 . The minimum level of V_{CE4} and the R_{E4} voltage drop are selected to be equal to the corresponding values of Q_1 . This allows the output voltage to swing positively by the same amount as it can go negatively. Even without changing this voltage level (sum of V_{RE4} , and V_{CE4}), the current level of this active load can be selected by proper selection of its biasing resistors (R_3 and R_4). The constant current circuit offers a high AC load resistance, which is equal to $1/h_{oe4}$, for the Q_1 driving amplifier stage, to give the highest possible voltage gain.

15.16 Distortion in Large Signal Amplifier

Distortion in amplifier has been discussed in Chapter 9. Here we will elaborate the nonlinear distortion produced by the driver stage of power amplifiers. It is desired that the output of an amplifier should be a pure sine wave, if the input signal is a sine wave. But in all the cases it may not be so. The *deviation* of the output waveform of an amplifier for the input waveform is called the *distortion*. There are three types of distortion which are described in Chapter 9. In the amplifier of Fig.15-36 to Fig.15-48, the driver amplifiers have to develop an output with

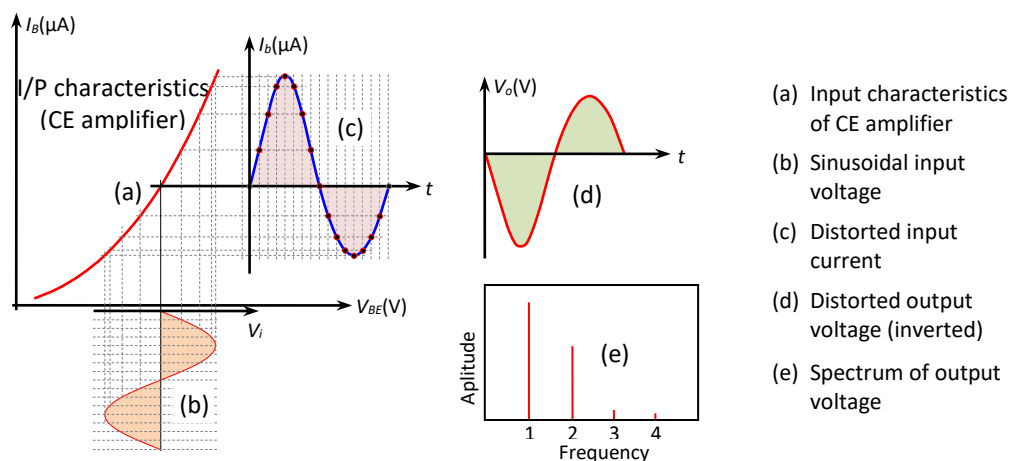


Fig.15-52: Nonlinear distortion produced in a power amplifier

quite large peak-to-peak value (large signal) and hence they will use almost their entire load line. As we know, the input characteristic of an amplifier is nonlinear, and the relation between the output current (I_C) with the input current (I_B) is also nonlinear. Due to this nonlinearity, the output signal gets distorted as illustrated in Fig.15-52. In this figure, graph (a) is a input characteristics of a typical CE amplifier and graph (a) is a sinusoidal input signal. The amplitude of the input signal is large compared to small signal amplifier. Using projection method, the input base current is determined in graph (c). The output voltage at the collector terminal of a CE amplifier is proportional to the base current with 180° phase shift. Thus the output of the CE amplifier will be as shown in graph (d). The shape of the output has been deviated a lot from the shape of input

signal. This type of distortion is called the *amplitude distortion* or *non-linear distortion*. The distorted output signal will produce many harmonic frequencies in addition to the input fundamental frequency [as shown in graph (e)]. Though the cross-over distortion can be fully eliminated by proper biasing of the transistors, but it is not possible to remove fully nonlinear distortion. Using negative feedback we can reduce this distortion. For this reason the amplifiers of Fig.15- x to Fig.15-x cannot be used to deliver very power to the load.

15.17 Load Compensation

All design, analysis, and calculations for power amplifiers, covered so far, assume a resistive load with a given resistance value (R_L). For audio amplifiers, the load is usually the coil of a loudspeaker which is illustrated in Fig.15-53. Being a coil, the impedance of the loudspeaker

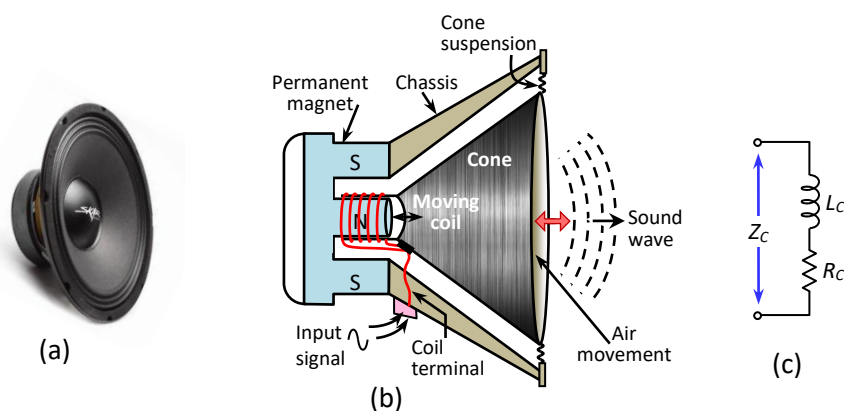


Fig.15-53: Loud speaker: (a) Photograph, (b) Cross-sectional view, and (c) Coil impedance

has a resistive component (R_C) and an inductive component (L_C). The load impedance is, $Z_L = R_C + j2\pi fL_C$, and clearly Z_L increases (from a low of R_C at DC) as the signal frequency increases. Moreover, when the coil of the loudspeaker moves in a magnetic field, a back e.m.f. is also induced in the coil. The combination of this back e.m.f. and Z_L is called dynamic impedance of loudspeaker. An $8\ \Omega$ speaker might offer an impedance of $8\ \Omega$ only at a frequency around 400 Hz. As the load is inductive, the load current lags the load voltage, and typically the phase angle could be as high as 60° . Similarly, when capacitive loads are involved the load current can lead the load voltage. The phase difference between load current and voltage can offer extra stress on the output transistors. To eliminate this extra stress to some extent, output compensating components are often included to minimize the phase difference. Fig.15-54 shows the typical arrangement of the compensating components. Inductor L_p and its parallel-connected resistor R_p

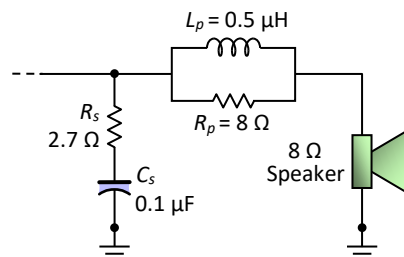


Fig.15-54: Loud speaker connection with compensating circuit

are usually recommended by device and IC manufacturers for isolating a capacitive load. Capacitor C_s and series-connected resistor R_s help to correct the lagging phase angle of an inductive load.

In most of the Hi-Fi (high-fidelity) sound systems, three to five loudspeakers (woofer, subwoofer, tweeter etc.), with different frequency response, are used. Instead of directly connected, the speakers are connected through cross-over circuits. The primary purpose of the cross-over circuit in a loudspeaker is to split an incoming audio signal into frequency bands that are passed to the speaker best suited. For

example, in a three-way system, a cross-over circuit will pass low frequencies (30 Hz to 1 kHz) to the woofer, mid-frequencies (1 kHz to 8 kHz), and high frequencies (8 kHz to 16 kHz) to the tweeter. Here, we do not need the extra compensating circuits.

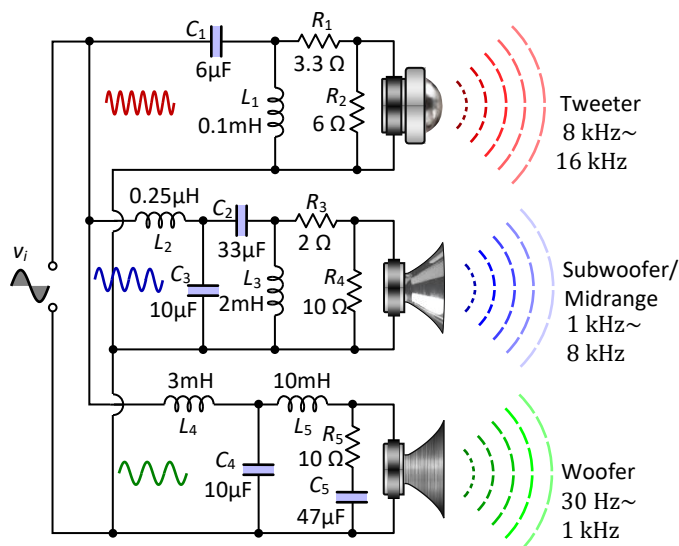


Fig.15-55: Loud speaker connection with cross-over circuit

15.18 Power Amplifier with Differential Input Stages

The direct-coupled amplifier in Fig.15-56 is a BJT power amplifier with differential input stage constituted by transistors Q_1 and Q_2 . It also has an intermediate stage (Q_3) with a constant current load (Q_4). Both pairs of output transistors (Q_5 and Q_7) and (Q_6 and Q_8) are in quasi-complementary configuration, instead of the usual complementary form. This arrangement helps to minimize the required supply voltage by removing the V_{BE} of the power transistors from the V_{CC} equation. Now, the required supply voltage will be,

$$V_{CC} = V_P + V_{R13} + V_{BE5} + V_{CE3(min)} + V_{R8} \quad (15-92)$$

and,

$$V_{EE} = -(V_P + V_{R14} + V_{BE6} + V_{CE4(min)} + V_{R10})$$

Differential input stage has very large voltage gain that facilitates negative feedback (NFB), and the whole circuit functions like an operation amplifier. Q_1 base is the noninverting input, Q_2 base is the inverting input, and the junction of R_{13} and R_{14} is the output terminal. As the capacitor, C_2 , remains open for DC, there is 100% DC NFB provided from the output via R_5 to the base of Q_2 (inverting input). As we described earlier, the DC voltage level of the output terminal is 0. Therefore, this zero voltage will be applied to the base of Q_2 . Thus for DC condition, both transistors' bases are grounded. The value of C_2 is so selected that, it behaves as a short-circuit for the operating frequency. Thus, the AC NFB to the base of Q_1 is divided by R_4 and R_6 . That is, the

feedback factor will be $\beta = R_4/(R_4 + R_5)$. As we know the gain of an amplifier with feedback is approximately $1/\beta$, the voltage gain of the driving stage will be $A_f = (R_4 + R_5)/R_4$.

The dynamic impedance of D_1 combined with R_4 functions as an AC voltage divider to attenuate the ripple at the emitters of Q_1 and Q_2 . Ripple at this point is amplified just like an input signal. Capacitors C_3 and C_5 are frequency-compensating components.

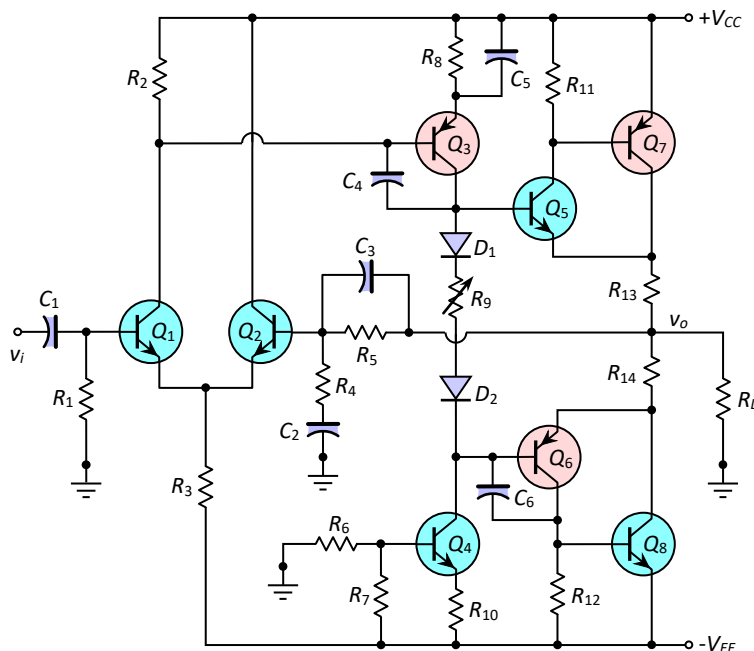


Fig.15-56: Power amplifier with differential input

Here, the voltage level of the input signal is amplified by the differential amplifier, and the CE amplifier (transistor Q_3). Transistor Q_4 acts as the active load for Q_3 . Hence, these two stage of voltage amplifier provide huge gain, but due to the negative feedback, the gain settles to $1/\beta$. After voltage amplification, the current level of the signal is amplified by the output complementary Darlington pair push-pull amplifier. As before, diodes D_1 , D_2 , and R_9 bias the output stage in class-AB mode.

15.18.1 Amplifier Design

The design procedure for the output and intermediate stages of the circuit in Fig.15-56 is similar to the procedures already discussed. Design of the BJT amplifier with differential input stages is very simple. The DC collector currents for Q_1 and Q_2 should be larger than the peak base current for Q_3 , and the voltage drop across R_2 is $(V_{E3} + V_{BE3})$. The value of R_1 should be calculated, from the required base current of Q_1 . To ensure symmetrical DC biasing for the two transistors of differential amplifier, the value of R_5 should be equal to R_1 . R_4 is calculated in terms of R_5 to give the desired closed-loop gain. The impedance of C_2 is made equal to R_4 at the desired lower cutoff frequency (f_L), so that C_2 sets lower cut-off frequency limit (f_L). Capacitors C_1 and C_5 are determined in the usual way describe in Chapter 9. In fact, the values of C_1 and C_5 should be large enough, so that

they do not affect the lower cut-off frequency. An additional capacitor (C_3) might be included to set the upper cut-off frequency (f_H) as $X_{C_6} = R_5$.

What is the function of C_5 , not mentioned. So it must be found out.

Example 14-16

The amplifier circuit in Fig.15-56 is to deliver 10 W power to a load of $R_L = 8 \Omega$. Design the output and intermediate stage of this amplifier. Assume, the lower cut-off frequency $f_L = 50 \text{ Hz}$, $h_{fe7} = h_{fe8} = 25$, $h_{fe5} = h_{fe6} = 100$, $h_{fe3} = h_{fe4} = 150$, $I_{CB07} = I_{CB08} = 0.2 \text{ mA}$.

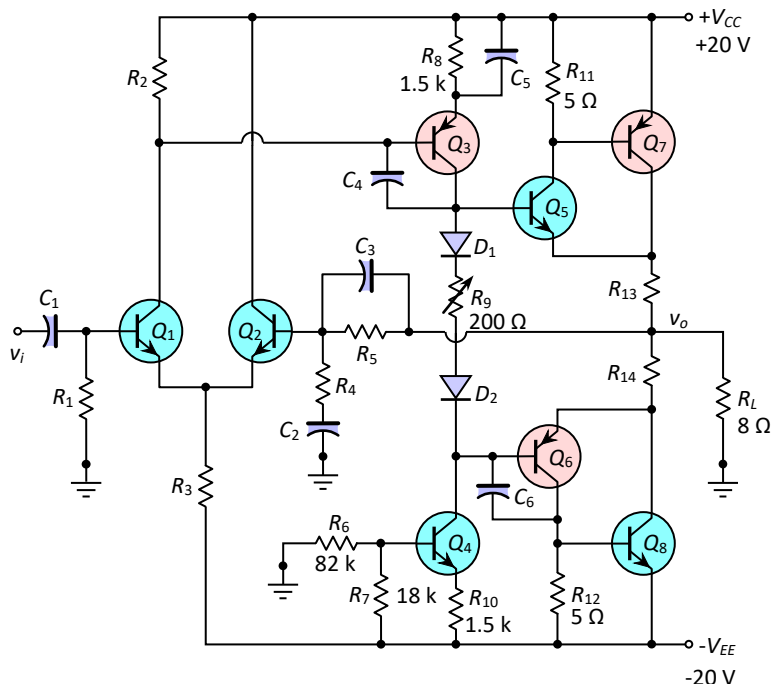


Fig.15-57: Power amplifier with differential input designed in Example 14-16

Solution:

Using Equ.(15-61),
$$V_p = \sqrt{2P_o R_L} = \sqrt{2(10 \text{ W})(8 \Omega)} \approx 12.7 \text{ V}$$

Using Equ.(15-62),
$$I_p = \frac{V_p}{R_L} = \frac{12.7 \text{ V}}{8 \Omega} \approx 1.6 \text{ A}$$

Let $V_{R13} = V_{R14} = V_p/10 = 12.7 \text{ V}/10 \approx 1.3 \text{ V}$

Let,
$$R_{13} = R_{14} = \frac{R_L}{10} = \frac{8 \Omega}{10} \approx 1 \Omega$$

Select $I_{CB07} R_{11} = I_{CB08} R_{12} = 0.04 \text{ V}$

$$R_{11} = R_{12} = \frac{0.04 \text{ V}}{0.2 \text{ mA}} = 200 \Omega$$

Let $V_{R10} = V_{R8} = 3 \text{ V}$, and $V_{CE3(\min)} = V_{CE3(\min)} = 1 \text{ V}$

Using Equ.(15-92), $V_{CC} = V_p + V_{R13} + V_{BE5} + V_{CE3(\min)} + V_8$

or, $V_{CC} = 12.7 \text{ V} + 1.27 \text{ V} + 0.7 \text{ V} + 1 \text{ V} + 3 \text{ V} = 18.67 \text{ V}$

Let $V_{CC} = 20 \text{ V}$, and $-V_{EE} = -20 \text{ V}$.

Let the quiescent current of Q_7 and Q_8 be,

$$I_{CQ7} = I_{CQ8} = \frac{I_p}{10} = \frac{1.6 \text{ A}}{10} = 160 \text{ mA}$$

The peak base current of Q_5 and Q_6 will be,

$$I_{BQ5p} = I_{BQ6p} = \frac{1.1 I_p}{h_{fe7} \times h_{fe5}} = \frac{1.1 \times 1.6 \text{ A}}{25 \times 100} = 0.7 \text{ mA}$$

Let the quiescent current of Q_3 and Q_4 be,

$$I_{CQ3} = I_{CQ4} = 2 \text{ mA}$$

The value of R_9 can be calculated as,

$$R_9 = \frac{(R_{13} + R_{14})I_{CQ7}}{I_{CQ3}} = \frac{(1 \Omega + 1 \Omega)160 \text{ mA}}{2 \text{ mA}} \approx 160 \Omega \text{ [use a } 200 \Omega \text{ pot]}$$

$$R_8 = R_{10} = \frac{V_{R8}}{I_{CQ3}} = \frac{3 \text{ V}}{2 \text{ mA}} = 1.5 \text{ k}$$

Let $I_7 = I_{CQ4}/10 = 2 \text{ mA}/10 = 0.2 \text{ mA}$

$$\therefore R_7 = \frac{V_{R8} + V_{BE4}}{I_7} = \frac{3 \text{ V} + 0.7 \text{ V}}{0.2 \text{ mA}} = 18.5 \text{ k [use standard } 18 \text{ k]}$$

$$\text{and, } R_6 = \frac{|V_{EE}| - 3.7 \text{ V}}{I_7} = \frac{20 \text{ V} - 3.7 \text{ V}}{0.2 \text{ mA}} = 81.5 \text{ k [use standard } 82 \text{ k]}$$

As half-wave rectified current is drawn from both V_{CC} and $|V_{EE}|$, the input DC power to the amplifier will be,

$$P_{i(DC)} = 0.42V_{CC}I_p + 0.42V_{EE}I_p = 2 \times 0.42 \times 20 \text{ V} \times 1.6 \text{ A} \approx 27 \text{ W}$$

$$\text{Using Equ.(15-78), } P_T = \frac{P_{i(DC)} - P_o}{2} = \frac{27 \text{ W} - 10 \text{ W}}{2} = 8.5 \text{ W}$$

Output transistor specifications:

$P_T = 8.5 \text{ W}$, $V_{CE(\min)} = 2 V_{CC} = 2 \times 20 \text{ V} = 40 \text{ V}$, and $I_{C(\max)} = 1.1 I_p = 1.1 \times 1.6 \text{ A} \approx 2 \text{ A}$

As there is no output capacitor, the lower cut-off frequency will be determined by the bypass capacitor C_5 .

$$\therefore C_5 = \frac{1}{2\pi(50 \text{ Hz}) \times 12 \Omega} \approx 265 \mu\text{F, [use } 333 \mu\text{F capacitor]}$$

Also design the differential stage.

Comments: The designed amplifier is shown in Fig.15-57. The power transistors must be mounted on a good heat sink.

15.19 Complementary MOSFET Common Source Power Amplifier

15.19.1 Advantages of MOSFETs

Power MOSFET has several advantages over power BJTs for large signal amplifier applications. One of the most important differences is that MOSFET transfer characteristics (I_D vs V_{GS}) are more linear than I_C vs V_{BE} characteristics of BJTs. This helps to minimize distortion in the output waveform. Thermal runaway does not occur with power MOSFETs, so the emitter resistors in the BJT output stage are not needed in a MOSFET amplifiers. Thus, the wastage of power in the emitter resistors is eliminated.

Power MOSFETs can be operated in parallel to reduce the total channel resistance and increase the output current level. Unlike BJTs, operated in parallel, there is no need for resistors to equalize current distribution between parallel-connected MOSFETs. For class-AB operation, the MOSFET gate-source should be biased to the threshold voltage (V_{Th}) for the device, to ensure that it is conducting at a low level when no signal is present which sets the DC operating point.

15.19.2 Power Amplifier with MOSFET Output Stage

The four output transistors in the amplifier circuit of Fig.15-56 can be replaced with two power MOSFETs operating as source followers. The circuit arrangement is shown in Fig.15-58. As we know, the input impedance of MOSFET transistor is very high, and hence the gate current required to drive it is very small (ideally zero). Thus, the output Darlington and Sziklai pair (in Fig.15-56) can be replaced by single power MOSFETs. One thing to be remembered that, the threshold gate-source voltage of FET (V_{Th}) is larger compared to V_{BE} in BJT amplifier. Therefore, the FET gate-source voltage must be included when calculating the supply voltage, and hence, a larger supply voltage would be required for a MOSFET amplifier than a BJT amplifier. In the power amplifier in Fig.15-58, the complementary MOSFETs output devices operate as complementary MOSFET common source power amplifier. As will be seen, this permits the peak output voltage to approach almost the supply voltage level. The function of the rest of the circuit is same as described before. The diodes (D_1 to D_2) and the variable resistor, R_B are used to bias the MOSFET in class-AB mode. Here, though two diodes are shown, but in practical circuit more diodes, or even a Zener diode, may be used. The differential amplifier consisting of Q_1 and Q_2 and the intermediate CE amplifier (Q_3) provides necessary voltage gain of the circuit. Transistor Q_4 works as the active load for transistor Q_3 .

As in the case of the BJT circuit (in Fig. 18-56), the differential amplifier input stage allows the use of negative feedback (via R_5) to stabilize the output voltage level, and AC negative feedback (via R_4 and R_5) to set the closed-loop voltage gain. Output transistors Q_5 and Q_6 are complementary MOSFETs and both are operated as complementary MOSFET common source amplifier.

15.19.3 Design Process of MOSFET Power Amplifier

In a class-AB MOSFET power amplifier, the FET gate-source voltage should be biased to the minimum specified threshold voltage (V_{Th}) of the transistors to eliminate the cross-over distortion. For this purpose a suitable value of R_9 has to be set as,

$$R_9 = \frac{V_{Th5} + V_{Th6} - 1.4 \text{ V}}{I_{CQ3}} \quad (15-93)$$

The output voltage and output current can be calculated as before. The required supply voltage can be calculated as,

$$V_{DD} = |V_{SS}| = V_p + I_D R_{DS(ON)} \quad (15-94)$$

where, $R_{DS(ON)}$ is the ON-state channel resistance FET (described in Section 6.5.1) [H_S_atten].

The required gate-source voltage swings (ΔV_{GS}) can be determined as,

$$\Delta V_{GS} = I_p / g_m \quad (15-95)$$

The differential amplifier and the intermediate stage will produce $\pm \Delta V_{GS}$ at the collector of Q_3 (or the gate terminals). The current capacity of the output MOSFET transistor should be $1.1 I_p$ and the maximum drain-source voltage should be $2V_{DD}$. The power dissipation in the FETs will be,

$$P_T = \frac{P_{i(DC)} - P_o}{2}$$

The design process of the differential amplifier and the intermediate stage will be same as before.

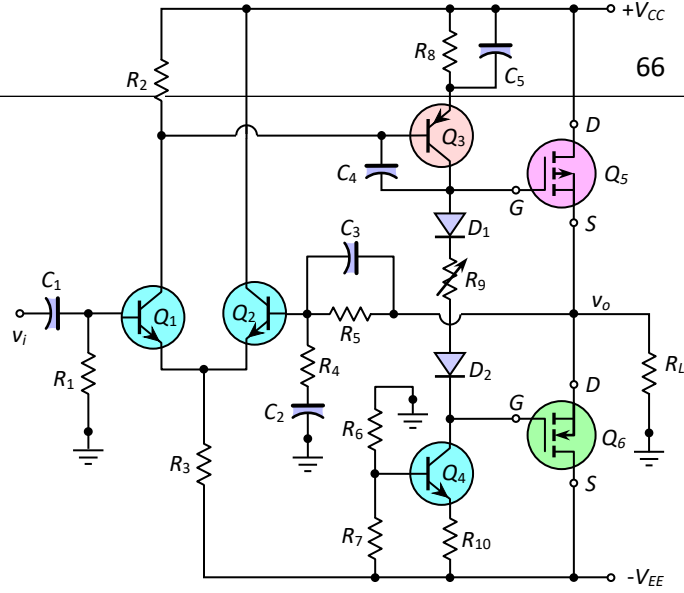


Fig.15-58: Power amplifier with MOSFET output stage

Example 14-17

Design the output power amplifier of Fig.15-58, to deliver 12 W power to a load of $R_L = 8 \Omega$. Assume that the output MOSFETs have $g_m = 2 \text{ S}$, $V_{Th} = 2 \text{ V}$, and $R_{DS(ON)} = 1 \Omega$. The lower cut-off frequency of the amplifier is $f_L = 50 \text{ Hz}$.

Solution:

Using Equ.(15-61),
$$V_p = \sqrt{2P_o R_L} = \sqrt{2(12\text{ W})(8\ \Omega)} \approx 14\text{ V}$$

Using Equ.(15-62),
$$I_p = \frac{V_p}{R_L} = \frac{14\text{ V}}{8\ \Omega} \approx 1.75\text{ A}$$

Let the quiescent drain current be $I_{DQ} = 0.1I_p = 0.1 \times 1.75\text{ A} = 0.175\text{ A}$.

The total drain current will be, $I_D = I_p + I_{DQ} = 1.75\text{ A} + 0.175\text{ A} = 1.925\text{ A}$.

Using Equ.(15-94),
$$V_{DD} = |V_{SS}| = V_p + I_D R_{D(on)} = 14\text{ V} + 1.925\text{ A} \times 1\ \Omega$$

$$\therefore V_{DD} = |V_{SS}| \approx 16\text{ V}$$

DC power drawn by the power amplifier will be,

$$P_{i(DC)} = 0.42V_{DD}I_p + 0.42V_{SS}I_p = 2 \times 0.42 \times 16\text{ V} \times 1.75\text{ A} \approx 24\text{ W}$$

Using Equ.(15-78),
$$P_T = \frac{P_{i(DC)} - P_o}{2} = \frac{24\text{ W} - 12\text{ W}}{2} = 6\text{ W}$$

Output transistor specifications:

$P_T = 6\text{ W}$, $V_{DS(min)} = 2V_{CC} = 2 \times 16\text{ V} = 32\text{ V}$, and $I_{D(max)} = 1.1I_p = 1.1 \times 1.75\text{ A} \approx 2\text{ A}$

Complete the whole design as previous Example.

Comments: The design process of the rest of the amplifier will be same as in Example 16.

15.20 BJT Power Amplifier with Op-Amp Driver

The class-AB power amplifier shown in Fig.15-59 uses an operational amplifier (A_1) at the input stage. Instead of a class-A CE amplifier, here the op-amp provides the necessary voltage gain of the amplifier. The power transistors used in this circuit (D44H8G, and D45H8G) are complementary Darlington transistors. They need $\approx 1.3\text{ V}$ across B-E junction to bias them as class-AB mode. For this biasing purpose, diodes D_1 to D_4 together with resistors R_6 and R_7 have been used. If we use normal power transistors, then we have to use only two diodes. Here, the supply voltage is $\pm 30\text{ V}$, but the op-amp needs $\pm 15\text{ V}$. So two Zener diodes (Z_1 and Z_2) and resistors R_4 and R_5 have been used to bias the op-amp.

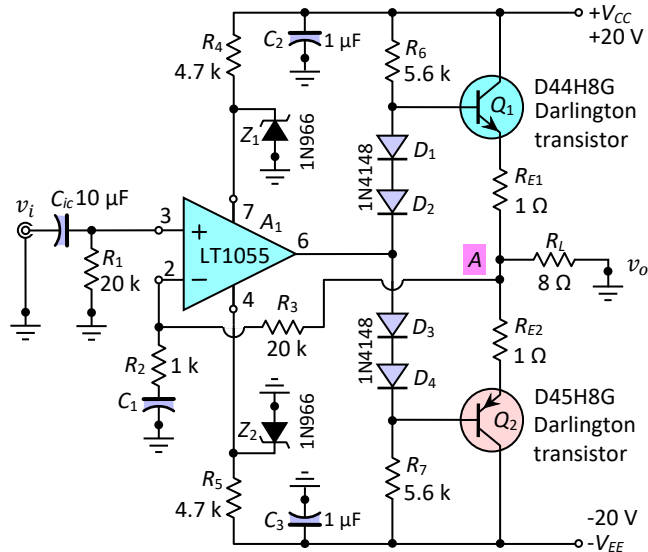


Fig.15-59: Complementary emitter follower with op-amp driver

The capacitors C_2 and C_3 are power supply decoupling capacitors. For DC input voltage, capacitor C_1 is open, and hence, there is 100% negative feedback via R_3 for DC signal. 100% negative feedback means unity voltage gain, that is, the output terminal (point A) and the input terminal (non-inverting input) will remain at the same voltage. But the non-inverting terminal is at ground voltage through resistor R_1 . So for DC condition point A (output of the amplifier) will be at zero potential. For the same reason, the output of the op-amp (pin no 6) will also be at zero voltage for DC condition. But for AC signal (for the range of operating frequency) capacitor C_1 will be short, so the feedback factor will be $\beta = R_2/(R_2 + R_3)$ which will set the voltage gain of the amplifier to $1/\beta = (1 + R_3/R_2)$.

When the op-amp output at the junction of D_2 and D_3 is increased in a positive direction, A_1 supplies current through D_3 , D_4 , and R_7 . So, the voltage drop across R_7 is increased which results in reduction in voltage drop across R_6 . Thus the base voltage of Q_1 is increased and current flows through this transistor and the load (current is pushed). Similarly, when the op-amp output goes in a negative direction, A_1 sources current through D_1 , D_2 , and R_6 . So, the voltage drop across R_6 is increased which results in reduction in voltage drop across R_7 . Thus the base voltage of Q_2 is decreased and current flows through this transistor and the load (current is pulled through load). The operation is further illustrated in Example 14-18.

Example 14-18

For the op-amp driven push-pull amplifier of Fig.15-59, $h_{fe1} = h_{fe2} = 40$, $v_{ip} = 570$ mV, voltage across each diode is 0.65 V, voltage drop across B-E junction of each power transistor is 1.2 V. Determine the all voltage levels of the power amplifier.

Solution:

The overall voltage gain of the amplifier will be $1/\beta$, that is,

$$A_{vf} = \left(1 + \frac{R_3}{R_2}\right) = \left(1 + \frac{20 \text{ k}}{1 \text{ k}}\right) = 21$$

Therefore, the output voltage of the op-amp will be,

$$v_{o(op)} = v_i \times A_{vf} = 570 \text{ mV} \times 21 \approx 12 \text{ V}$$

The voltage drop across D_1 and D_2 will be equal to that across D_3 and D_4 . This value will be,

$$V_{D12} = V_{D34} = 0.65 \times 2 = 1.3 \text{ V}$$

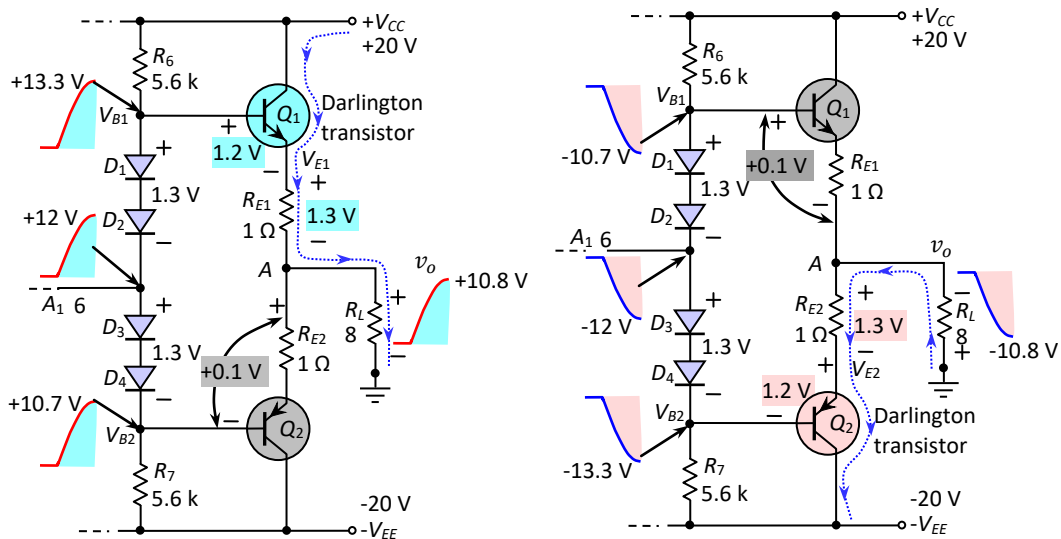


Fig.15-60: Complementary emitter follower with op-amp driver

Fig.15-60(a) depicts the situation when the op-amp's output produces the positive peak voltage, +12 V. The base voltage of the output transistor Q_1 will be,

$$V_{B1} = 12 \text{ V} + 1.3 \text{ V} = 13.3 \text{ V}$$

For this positive voltage Q_1 transistor conducts (push) current through the load.

The voltage at the emitter of Q_1 will be,

$$V_{E1} = 13.3 \text{ V} - 1.2 \text{ V} = 12.1 \text{ V}$$

This voltage will be divided between R_{E1} and the load resistor R_L . Therefore,

$$V_{RE1} = \frac{12.1 \text{ V} \times 1 \Omega}{1 \Omega + 8 \Omega} \approx 1.3 \text{ V}$$

and,

$$V_{RL} = \frac{12.1 \text{ V} \times 8 \Omega}{1 \Omega + 8 \Omega} \approx 10.8 \text{ V}$$

The base voltage of the output transistor Q_2 will be,

$$V_{B1} = 12 \text{ V} - 1.3 \text{ V} = 10.7 \text{ V}$$

Note that the voltage at the output terminal (point A) is 10.8 V, and the voltage at the base of Q_2 is 10.7 V. Therefore, only 0.1 V will be dropped across R_{E2} and the B-E junction of transistor Q_2 . Hence, this transistor will not conduct.

Fig.15-60(b) depicts the situation when the op-amp's output produces the negative peak voltage, -12 V. The base voltage of the output transistor Q_2 will be,

$$V_{B2} = -12 \text{ V} - 1.3 \text{ V} = -13.3 \text{ V}$$

For this negative voltage Q_2 transistor conducts (pull) current through the load.

The voltage at the emitter of Q_2 will be,

$$V_{E2} = -13.3 \text{ V} + 1.2 \text{ V} = -12.1 \text{ V}$$

This voltage will be divided between R_{E2} and the load resistor R_L . Therefore,

$$V_{RE2} = \frac{-12.1 \text{ V} \times 1 \Omega}{1 \Omega + 8 \Omega} \approx -1.3 \text{ V}$$

and,

$$V_{RL} = \frac{-12.1 \text{ V} \times 8 \Omega}{1 \Omega + 8 \Omega} \approx -10.8 \text{ V}$$

The base voltage of the transistor Q_1 will be,

$$V_{B1} = -12 \text{ V} + 1.3 \text{ V} = -10.7 \text{ V}$$

Here also, the voltage at the output terminal (point A) is -10.8 V , and the voltage at the base of Q_1 is -10.7 V . Therefore, only 0.1 V will be dropped across R_{E1} and the B-E junction of transistor Q_1 . Hence, this transistor will not conduct.

Comments: In Fig.15-60, the driver amplifier consisting of the op-amp is not shown.

15.21 Use of Bootstrapping Capacitors

The resistance of (with equal values) R_6 and R_7 in Fig.15-59, or the resistance R_C in Fig.15-36 is limited by the need to supply base current to the output transistors. There is a need for minimum voltage drop across R_6 and R_7 (R_C in Fig.15-36) to produce the base current of the output transistor. This voltage drops, limit the output voltage swings, because the sum of V_o , V_{RE1} , V_{BE1} , and V_{R6} cannot exceed the supply voltage V_{CC} . Thus, even for the peak output voltage, there will have a substantial voltage drop across transistor Q_1 ($V_{CE1} = V_{CC} - V_{op} - V_{RE}$). This large voltage drops results in a greater loss in the transistors which limits the amplifier

efficiency.

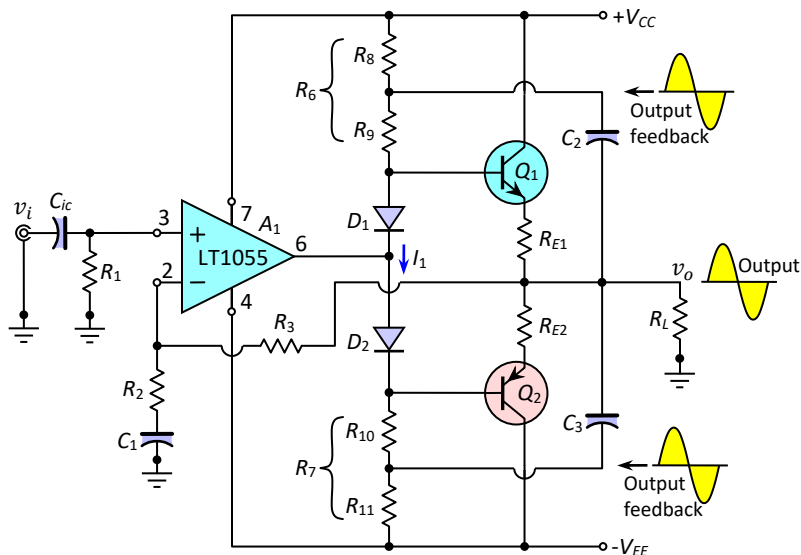


Fig.15-61: Power amplifier with bootstrap capacitors

The situation can be substantially improved by the use of two capacitors C_2 and C_3 as shown in Fig. 18-61. This process is called *bootstrapping* and capacitors are called *bootstrap capacitors*. Resistors R_6 and R_7 in Fig.15-59 are divided into two equal-value resistors ($R_6 = R_8 + R_9$, and $R_7 = R_{10} + R_{11}$), and the capacitors couple the output voltage back to the junctions of these resistors as illustrated in the figure.

At first, we will discuss the *bootstrapping* process for the positive output voltage of the op-amp. The DC bias (zero-signal condition) level is shown in Fig. 18-61(a), and the two other circuits show the situation for some positive voltage.

For zero-signal condition, the output of A_1 is at 0 V, Q_1 base is at +0.7 V, and the load voltage is 0 V. As R_8 , and R_9 are of equal values, $20\text{ V} - 0.7\text{ V}$ will be equally divided between them. So, the voltage at the junction of R_8 , and R_9 will be $(20\text{ V} - 0.7\text{ V})/2 + 0.7\text{ V} = +10.35\text{ V}$, and the voltage across C_3 is also +10.35 V.

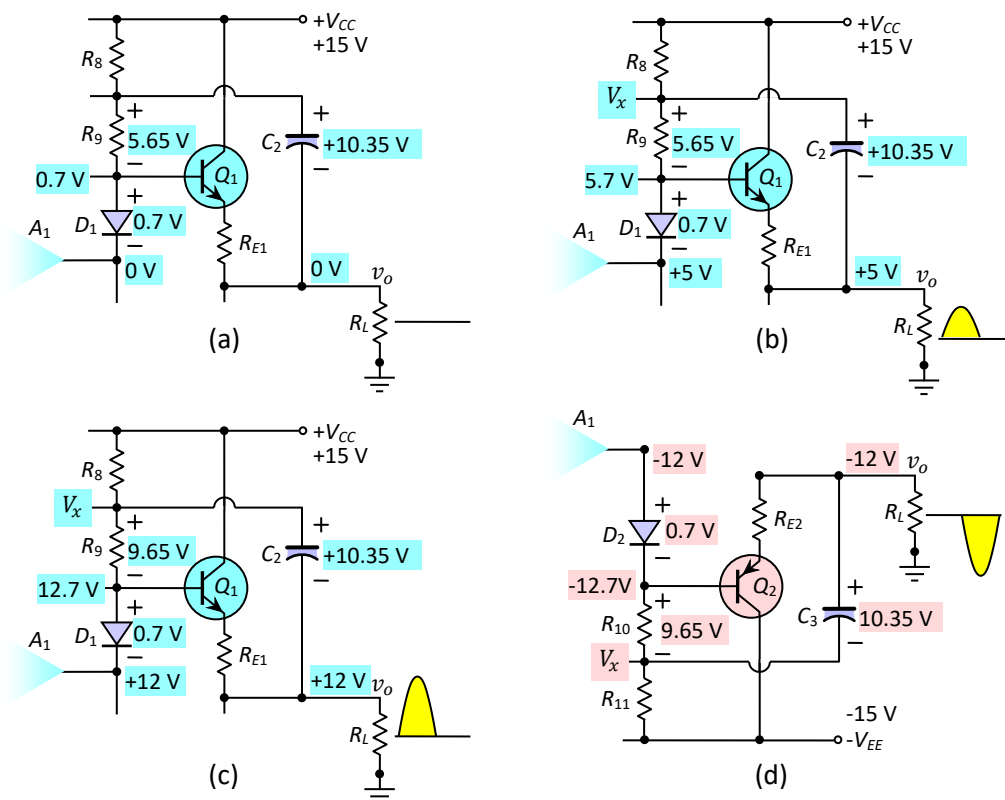


Fig.15-62: Different voltage levels at the output stage with bootstrap capacitor

New voltage levels that occur when the op-amp output increases by 5 V are shown in Fig.15-62(b). Now, Q_1 base is at +5.7 V, and the load voltage (v_o) is +5 V. Here for simplicity, voltage dropped across emitter resistors has been neglected. Because C_2 is a large capacitor, its terminal voltage remains substantially constant at +10.35 V, so the junction of R_8 , and R_9 is pushed up to,

$$V_x = v_o + V_{C2} = 5 \text{ V} + 10.35 \text{ V} = 15.35 \text{ V}$$

With 15.35 V at one end of R_9 and 5.7 V at the other end, the voltage across R_9 is 9.65 V. This is the same level of V_{R9} that occurs when the op-amp output is zero. Thus, C_2 keeps V_{R9} constant. Recall that without the bootstrapping capacitor, V_{R9} decreases when the op-amp output rises.

Now consider the voltage levels shown in Fig.15-62(c), where the op-amp output is +12 V. Q_1 base voltage is 12.7 V, $v_o = +12 \text{ V}$, and the capacitor will hold its previous voltage at $V_C = +10.35 \text{ V}$ and the voltage at the R_8 , and R_9 junction is,

$$V_x = v_o + V_C = +12 \text{ V} + 10.35 \text{ V} = 22.35 \text{ V}$$

So, the voltage across R_9 will be,

$$V_{R9} = V_x - 12.7 \text{ V} = 22.35 \text{ V} - 12.7 \text{ V} = 9.65 \text{ V}$$

Once again, V_{R9} remains constant, however, note that the bootstrapping capacitor has actually driven the R_8 , R_9 junction to a level (+22.35 V) higher than the supply voltage ($V_{CC} = 20 \text{ V}$). This

allows the output transistors to be driven into saturation, and the voltage drop across ($R_8 + R_9$) is no longer involved in the supply voltage calculation. In this case, the required supply voltage is,

$$V_{CC} = |V_{EE}| = V_p + V_{RE1} + V_{CE1(sat)} \quad (15-96)$$

Finally, let us consider the op-amp output is -12 V [to avoid lengthy description -5 V has not been consider]. Now, the base voltage of Q_2 is -12.7 V , $v_o = -12\text{ V}$, and the capacitor will hold its previous voltage at $V_C = -10.35\text{ V}$ and the voltage at the R_8 , and R_9 junction is,

$$V_x = v_o + V_{C3} = -12\text{ V} - 10.35\text{ V} = -22.35\text{ V}$$

So, the voltage across R_{10} will be,

$$V_{R10} = -12.7\text{ V} - V_x = -12.7\text{ V} + 22.35\text{ V} = 9.65\text{ V}$$

Once again, V_{R9} remains constant, however, note that the bootstrapping capacitor has actually driven the R_8 , R_9 junction to a level (-22.35 V) higher than the supply voltage ($V_{EE} = -20\text{ V}$). This allows the output transistors to be driven into saturation, and the voltage drop across ($R_8 + R_9$) is no longer involved in the supply voltage calculation.

The peak output voltage can also be limited by the output voltage range of the op-amp. For most op-amps the output voltage range is 1 V to 1.5 V less than the positive and negative supply levels. However, rail-to-rail op-amps are available with an output that ranges from $+V_{CC}$ to $-V_{EE}$.

15.21.1 Design Procedure

The BJT Power Amplifier with bootstrapping capacitor in Fig.15-61 is redrawn in Fig.15-63 using darlington power transistors. The design procedure of these two amplifiers will be same. The only difference is, the amplifier of Fig.15-63 uses four diodes to forward bias the base-emitter junctions of the Darlington output transistors.

As always, the peak output voltage and current are calculated from the specified output power and load resistance using Equ.(15-61) and Equ.(15-62). The supply voltage is determined using Equ.(15-96), and the emitter resistors for the output stage are typically selected as $0.1R_L$. The bias network current (I_1) should be larger than the peak base current for Q_1 and Q_2 . The resistance of R_6 (which equals $R_8 + R_9$) is calculated from I_1 and the circuit DC voltage drops as,

$$R_6 = R_8 + R_9 = \frac{V_{CC} - 2V_D}{I_1} \quad (15-97)$$

R_8 , and R_9 should typically be selected as $0.5R_6$, and then R_{10} , R_{11} will be equal to R_8 , and R_9 , respectively. R_1 and R_3 are equal-value resistors that bias the op-amp input terminals. R_2 is calculated from R_3 to give the required voltage gain as,

$$A_{vf} = 1 + \frac{R_3}{R_2} \quad (15-98)$$

C_1 is selected to have its impedance equal to R_2 at the desired lower cutoff frequency (f_L). The bootstrapping capacitors are calculated in terms of the resistance in series with them which is $R_8 \parallel R_9$ or $R_8 \parallel R_9$. The op-amp must have a suitable full power bandwidth to produce the peak output voltage at the desired upper cutoff frequency for the BJT power amplifier with op-amp driver.

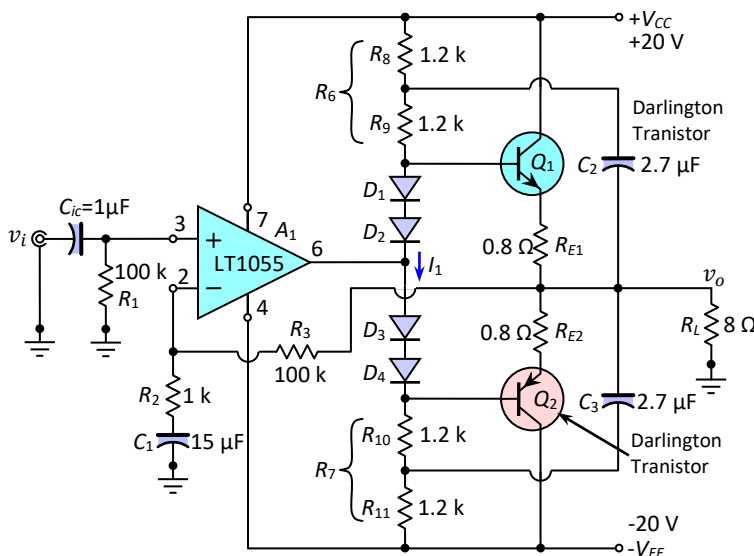


Fig.15-63: Power amplifier with bootstrap capacitors

Example 14-19

Design the amplifier of Fig.15-63 to deliver $P_o = 12\text{ W}$ to a load $R_L = 8\Omega$. The closed-loop voltage gain of the amplifier should be $A_{vf} = 100$. Assume that the Darlington transistors used in this circuit have: $V_{CE(sat)} = 3\text{ V}$, $h_{fe1} = h_{fe2} = 1200$. The lower cut-off frequency $f_L = 50\text{ Hz}$.

Solution:

Using Equ.(15-61),
$$V_p = \sqrt{2P_o R_L} = \sqrt{2(12\text{ W})(8\Omega)} \approx 14\text{ V}$$

Using Equ.(15-62),
$$I_p = \frac{V_p}{R_L} = \frac{14\text{ V}}{8\Omega} \approx 1.75\text{ A}$$

Let,
$$V_{RE1} = V_{RE2} = \frac{R_L}{10} = \frac{8\Omega}{10} = 0.8\Omega$$

Using Equ.(15-96),
$$V_{CC} = |V_{EE}| = 14\text{ V} + 1.75\text{ A} \times 0.8\Omega + 3\text{ V} = 18.4\text{ V}$$

Let us use, $V_{CC} = 20\text{ V}$, and $V_{EE} = -20\text{ V}$.

Let the quiescent current of Q_1 and Q_2 be $0.1I_p$. Therefore, the peak base current of these transistors will be,

$$I_{Bp1} = I_{Bp2} = \frac{1.1I_p}{h_{fe1}} = \frac{1.1 \times 1.75\text{ A}}{1200} \approx 1.6\text{ mA}$$

Let the biasing current $I_1 = 5 \times I_{Bp1} = 5 \times 1.6\text{ mA} = 8\text{ mA}$.

Using Equ.(15-97),
$$R_6 = R_8 + R_9 = \frac{V_{CC} - 2V_D}{I_1} = \frac{20\text{ V} - 2 \times 0.7\text{ V}}{8\text{ mA}} \approx 2.3\text{ k}$$

Let $R_8 = R_9 = 0.5R_6 \approx 1.2\text{ k}$. Similarly $R_{10} = R_{11} = 0.5R_6 \approx 1.2\text{ k}$.

Let $R_1 = R_3 = 100 \text{ k}$. Therefore, using Equ.(15-98),

$$A_{vf} = 1 + \frac{R_3}{R_2} \text{ or, } 100 = 1 + \frac{100}{R_2}$$

$$\therefore R_2 \approx 1 \text{ k}$$

Claculation of the capacitance of the capacitors:

In this circuit the lower cut-off frequency will be set by the bootstrapping capacitors. Therefore, at $f = f_L$,

$$\begin{aligned} X_{C2} &= R_8 \parallel R_9 \\ \frac{1}{2\pi f_L C_2} &= 1.2 \text{ k} \parallel 1.2 \text{ k} \\ C_2 &= \frac{1}{2\pi \times 100 \text{ Hz} \times 600 \Omega} \approx 2.6 \mu\text{F} \text{ [use } 2.7 \mu\text{F}] \end{aligned}$$

Also use $C_3 = 2.7 \mu\text{F}$. The other capacitances can be calculated by equating their reactance equal to one-tenth of their series resistance.

$$\begin{aligned} C_{ic} &= \frac{10}{2\pi f_L R_1} = \frac{10}{2\pi \times 100 \text{ Hz} \times 100 \text{ k}} \approx 0.6 \mu\text{F} \text{ [use } 1 \mu\text{F}] \\ C_1 &= \frac{10}{2\pi f_L R_2} = \frac{10}{2\pi \times 100 \text{ Hz} \times 1 \text{ k}} \approx 16 \mu\text{F} \text{ [use } 15 \mu\text{F}] \end{aligned}$$

Comments: The designed values are shown in Fig.15-63. The op-amp used in this circuit must have sufficient slew rate to amplify the maximum frequency signal without distortion.

15.22 Class-C Amplifier

A class-C amplifier is one whose output transistor conducts load current during *less* than one-half cycle of the input sine wave. Fig.15-64 shows a typical class-C current wave, and it is apparent that the total angle during which current flows is less than 180° . This angle is called the *conduction angle* and is denoted by θ_c . Class-C amplifiers are more efficient than either class-A or push-pull class-B and class-AB amplifiers. Thus, more output power can be obtained from class-C operation. As the output current does not flow for the entire cycle of the input, the output amplitude is a nonlinear function of the input. That is, the output of a class-C amplifier is highly distorted version of input. It could not be used in an application requiring high fidelity (*Hi-Fi*), such as audio amplifiers. Class-C amplifiers are primarily used in high-power, high-frequency applications, such as, radio frequency (RF)

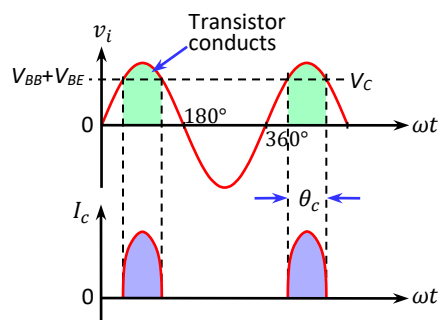


Fig.15-64: Input voltage and output current waveform

applications, including circuits, such as oscillators, that have a constant output amplitude, and modulators, where a high-frequency signal is controlled by a low-frequency signal. The principle advantage of class-C amplifier is that it has a very high efficiency.

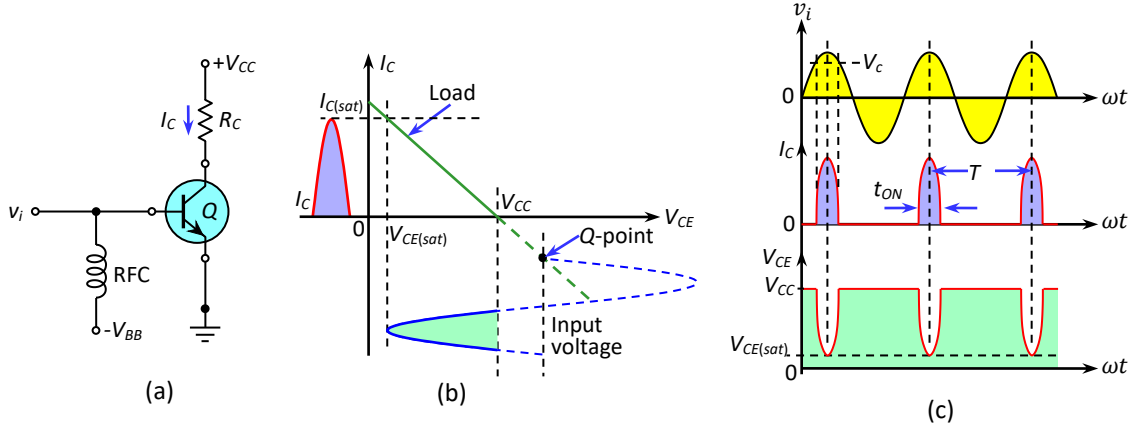


Fig.15-65: (a) Class-C amplifier with resistive load, (b) Position of Q-point, and (c) Waveforms

Fig.15-65 shows a simple class-C amplifier with a resistive load. Note that the base of the NPN transistor is biased by a negative voltage, $-V_{BB}$ connected through a coil labeled RFC.

The RFC is a *radio-frequency choke* whose inductance presents high impedance to the high-frequency input signals and thereby prevents the AC input being shorted by the DC source. In order for the transistor to begin conducting, the input signal must reach a level sufficient to overcome both the negative bias voltage and the V_{BE} drop of the transistor about 0.7 V. That is,

$$V_c = |V_{BB}| + V_{BE} = |V_{BB}| + 0.7 \text{ V} \quad (15-99)$$

where, V_c is the input voltage level at which the transistor begins to conduct. As shown in the Fig.15-64 and Fig.15-65(c), the transistor is cut-off until v_i reaches V_c , then it conducts, and then it cuts-off again when v_i falls below V_c . Clearly, the more negative the value of V_{BB} , the shorter the conduction interval [H_E_atten]. In most of the class-C applications, the amplifier is designed so that the peak value of the input, V_p , is just sufficient to drive the transistor into saturation, as shown in the Fig.15-65(b).

The conduction angle θ_c , can be calculated as,

$$\theta_c = 2 \cos^{-1} \left(\frac{V_c}{V_p} \right) \quad (15-100)$$

where V_p is the peak input voltage that drives the transistor to saturation. If the peak input only just reaches V_c , then $\theta_c = 2 \cos^{-1}(V_c/V_c) = 0^\circ$. At the other extreme, if $V_{BB} = 0$, then $V_c = 0 + 0.7 \text{ V} \approx 0$ and $\theta_c = 2 \cos^{-1}(0) = 180^\circ$, which corresponds to class-B operation.

Power Dissipation

The power dissipation of the transistor in a class-C amplifier is low because it is ON only for small time duration of the input cycle. Fig.15-65(c) shows the collector current pulses. The time between

the pulses is the period (T) of the AC signal. The collector current and the collector voltage during the ON time of the transistor are also shown in the same figure.

To avoid complex mathematics, we will assume ideal pulse approximations for the collector current. Using this simplification, if the output swings over the entire load line, the maximum current amplitude is $I_{C(sat)}$ and the minimum voltage amplitude is $V_{CE(sat)}$ during the time the transistor is ON [see Fig.15-65(b&c)]. The power dissipation during the ON time is, therefore,

$$P_{D(ON)} = I_{C(sat)} V_{CE(sat)} \quad (15-101)$$

The transistor is ON for a short time, t_{ON} , and OFF for the rest of the input cycle. The ON time of the transistor can be calculated from the time period of the input signal as,

$$t_{ON} = \frac{T}{360^\circ} \times \theta_c \quad (15-102)$$

Assuming the entire load line is used, the power dissipation averaged over the entire cycle is,

$$P_{D(avg)} = \left(\frac{t_{ON}}{T} \right) P_{D(ON)} = \left(\frac{t_{ON}}{T} \right) I_{C(sat)} V_{CE(sat)} \quad (15-103)$$

Example 14-20

A class-C amplifier is driven by a 200 kHz signal with $V_p = 6.2$ V. Assume that the amplifier is operating over 100% of its load line [as shown in Fig.15-65(b)] and $-V_{BB} = -5$ V. Calculate the conduction angle and conduction time of the transistor. If $I_{C(sat)} = 100$ mA, and $V_{CE(sat)} = 0.2$ V, also calculate the average power dissipation of the transistor.

Solution:

The period of the signal is,

$$T = \frac{1}{f} = \frac{1}{200 \text{ kHz}} = 5 \mu\text{s}$$

Using Equ.(15-99), $V_c = |V_{BB}| + V_{BE} = 5 \text{ V} + 0.7 \text{ V} = 5.7 \text{ V}$

Using Equ.(15-100), $\theta_c = 2 \cos^{-1} \left(\frac{V_c}{V_p} \right) = 2 \cos^{-1} \left(\frac{5.7 \text{ V}}{6.2 \text{ V}} \right) \approx 46^\circ$

The ON time the transistor will be,

Using Equ.(15-102), $t_{ON} = \frac{T}{360^\circ} \times \theta_c = \frac{5 \mu\text{s}}{360^\circ} \times 46^\circ = 0.64 \mu\text{s}$

Therefore, the average power dissipated by the transistor is,

Using Equ.(15-103), $P_{D(avg)} = \frac{t_{ON}}{T} I_{C(sat)} V_{CE(sat)}$

$$\therefore P_{D(avg)} = \frac{0.64 \mu\text{s}}{5 \mu\text{s}} \times 100 \text{ mA} \times 0.2 \text{ V} \approx 2.56 \text{ mW}$$

Comments: The low power dissipation of the transistor operated in class-C is important because, it leads to a very high efficiency.

15.22.1 Tuned Operation

Fig.15-66(a) shows the class-C amplifier as it is normally operated, with an LC tank circuit at the collector. Recall that the tank circuit is a resonant network (details description in Chapter 15). Neglecting small coil resistance, the centre (resonant) frequency is given by,

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

The purpose of the tank circuit is to produce the fundamental component of the pulsed class-C waveforms, which has the same frequency as v_i . This type of amplifier is called a *tuned amplifier*, because the center frequency of the tank circuit is set equal to (*tuned* to) the input frequency. The short pulse of collector current on each cycle of the input initiates and sustains the oscillation of the tank circuit so that an output sinusoidal voltage is produced, as illustrated in Fig.15-66(b). The tank circuit has high impedance only near the resonant frequency, so the gain is large only at this frequency. Since the impedance of the tank circuit is very large at its center frequency, the gain is correspondingly large at that frequency, while the impedance and the gain at harmonic frequencies are much smaller. For this reason the fundamental sinusoidal output is produced across the load.

The operation of this circuit can also be understood in a different way. We know that the tank circuit will generate a damped (decaying amplitudes due to loss) fundamental frequency, when power up. But current pulses produced at the collector will compensate the loss and a constant amplitude output is generated. As shown in Fig.15-66(b), at every cycle of input a collector current pulse is generated, which is synchronized with the output. **These pulses drive the output voltage (V_{CE}) to a little bit lower level than the value it would have been otherwise [H_E_atten].** The output waveform is 180° out of phase from the input as the amplifier is a CE amplifier.

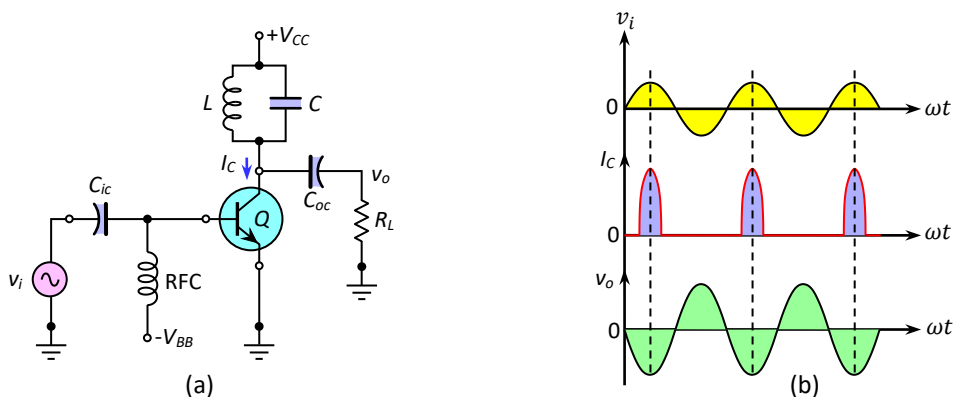


Fig.15-66: (a) Class-C tuned amplifier, and (b) Waveforms

15.22.2 The Maximum Output Power

As shown in Fig.15-65(b) and (c), the maximum possible output voltage swing $\Delta V_{CE(max)}$ is $V_{CC} - V_{CE(sat)} \approx V_{CC}$, that is the peak value of output voltage is $V_p \approx V_{CC}$. Hence, the maximum output power can be expressed as

$$P_{o(AC)} = \frac{V_{rms}^2}{R_C} = \frac{(V_p/\sqrt{2})^2}{R_C} = \frac{V_{CC}^2}{2R_C} \quad (15-104)$$

R_C is the simple collector resistance for the circuit of Fig.15-65(a) and for the circuit of Fig.15-66(a), R_C is the equivalent parallel resistance of the tank circuit at resonance and represents the parallel combination of the coil resistance and the load resistance (R_L). The total power that must be supplied to the amplifier from the DC source is,

$$P_{i(DC)} = P_{o(AC)} + P_{D(avg)} \quad (15-105)$$

Therefore, the efficiency is,

$$\eta = \frac{P_{o(AC)}}{P_{i(DC)}} \quad (15-106)$$

When $P_{o(AC)} \gg P_{D(avg)}$, the efficiency of class-C amplifier approaches to 1 (100%).

Example14-21

Suppose the class-C amplifier described in Example 14–20 has a V_{CC} equal to 24 V and the R_C is 50 Ω . Determine the efficiency of the class-C amplifier.

Solution:

From Example 14–20, $P_{D(avg)} = 2.56$ mW.

Using Equ.(15-99), $V_c = |V_{BB}| + V_{BE} = 5 \text{ V} + 0.7 \text{ V} = 5.7 \text{ V}$

Using Equ.(15-104), $P_{o(AC)} = \frac{V_{CC}^2}{2R_C} = \frac{(24 \text{ V})^2}{2 \times 50 \Omega} = 5.76 \text{ W}$

Using Equ.(15-105), $P_{i(DC)} = P_{o(AC)} + P_{D(avg)}$

Using Equ.(15-105), $P_{i(DC)} = 5.76 \text{ W} + 2.56 \text{ mW} \approx 5.763 \text{ W}$

$\therefore \eta = \frac{5.76 \text{ W}}{5.763 \text{ W}} \approx 99.95\%$

Comments: The efficiency of class-C amplifier is almost 100%.

15.22.3 Clamper Bias for a Class-C Amplifier

Fig.15-67 shows a class-C amplifier with a base bias clamping circuit. The base-emitter junction behaves as a diode. For the input signal, the capacitor, the parallel combination of R_B and the base-emitter junction diode form a parallel clamping circuit (as described in Chapter 4). When the input signal goes positive, capacitor C_{ic} is charged to $V_{ip} - 0.7$ V with the polarity shown in the figure. Where V_{ip} is the peak value of the input signal. For this positive signal, the voltage across the base-emitter junction will be a little bit higher than 0.7 V (due to the junction resistance). During the negative half-cycle, the voltage across this junction will be, $-(V_{ip} - 0.7 \text{ V} + V_{ip}) = -2V_{ip} + 0.7 \text{ V}$. That is the input signal will be clamped to negative direction as shown in the figure.

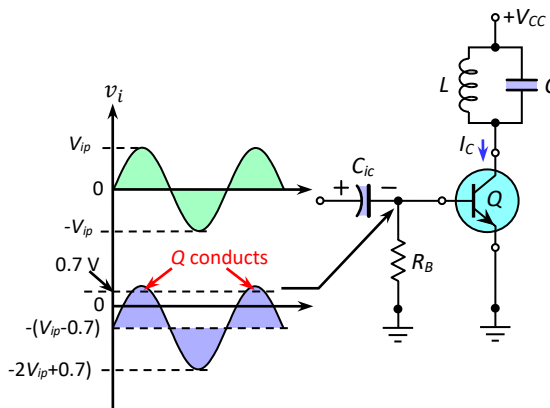


Fig.15-67: Clamper bias for class-C amplifier

This clamping action produces an average voltage at the base of approximately $-V_{ip}$ (exactly $-V_{ip} + 0.7 \text{ V}$). This places the transistor in cut-off except at the positive peaks, when the transistor conducts for a short interval. For good clamping action, the $R_B C_{ic}$ time constant of the clamping circuit must be much greater than the period of the input signal. Since the DC value of the input signal is zero (at positive side of C_1), the DC voltage at the base (negative side of C_1) is slightly more positive than $-(V_{ip} - 0.7 \text{ V})$, as indicated in the figure. Near the positive peaks of the input voltage, the base voltage goes slightly above 0.7 V and causes the transistor to conduct for a short time. The maximum peak negative voltage at the base will be, $-2V_{ip} + 0.7 \text{ V}$. The circuit's operation is exactly same as that using $-V_{BB}$.

Example 14-22

Determine the voltage at the base of the transistor, the resonant frequency, and the peak-to-peak value of the output signal voltage for the class-C amplifier in Fig.15-68.

Solution:

The peak value of the input signal will be,

$$V_{ip} = 1 \text{ V} \times \sqrt{2} \approx 1.4 \text{ V}$$

The base will be clamped at (the average DC voltage at the base),

$$-(V_{ip} - 0.7 \text{ V})$$

or,

$$-(1.4 \text{ V} - 0.7 \text{ V}) = -0.7 \text{ V}$$

The signal at the base has a positive peak of +0.7 V and a negative peak of,

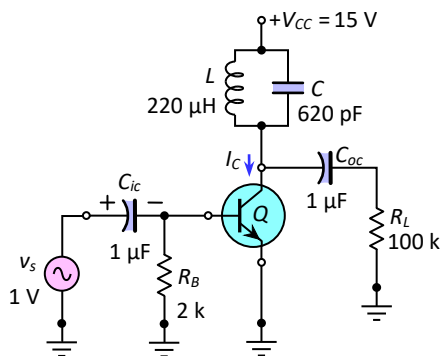


Fig.15-68: Class-C amplifier with clamper bias for Example 14-22

$$-2V_{ip} + 0.7 \text{ V}$$

or,
$$-2V_{ip} + 0.7 \text{ V} = -2 \times 1.4 \text{ V} + 0.7 \text{ V} = -2.1 \text{ V}$$

The resonant frequency is,

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{220 \mu\text{F} \times 620 \text{ pF}}} \approx 431 \text{ kHz}$$

The output signal will have peak-to-peak value of,

$$V_{pp} = 2 V_{CC} = 2 \times 15 \text{ V} = 30 \text{ V}$$

Comments: The waveform of the clamped input signal will be same as in Fig.15-67.

15.23 Class-D Amplifier

In a class-D amplifier, the output transistors (or any other power devices) are operated as switches instead of operating linearly as in class-A, -B, and -AB amplifiers. The main advantage is that a class-D amplifier can operate at a maximum theoretical efficiency of 100% compared to class-A at 25% and class-B/AB at 79%. In practice, efficiencies over 90% can be achieved with class-D, resulting in smaller heat sinks and overall cost savings. MOSFETs have superior switching characteristics and low ON-state resistance compared to BJTs and JFETs, particularly at the high frequencies required for class-D operation. This accounts for why they are used almost exclusively with class-D amplifiers. A typical class-D power amplifier consists of a triangular waveform generator, comparator (based on an OPAMP), switching circuit, and a low pass filter. The block diagram of a class-D amplifier is shown in the Fig.15-69. Most class-D amplifiers operate using dual-power supplies (split power supply). The MOSFETs are basically push-pull amplifiers that are operated as switching devices, rather than linear devices as in the case of other amplifier classes.

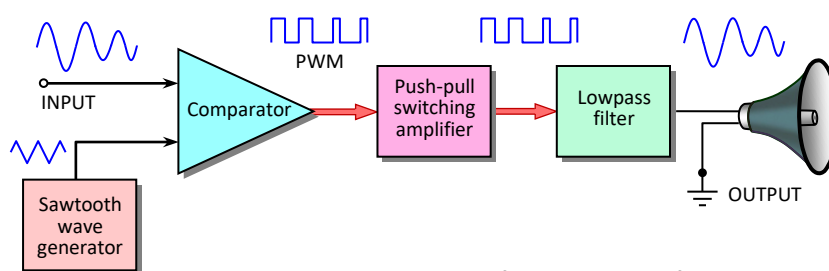


Fig.15-69: Block diagram of a class-D amplifier

15.23.1 Sawtooth waveform generator

The sawtooth waveform generator generates a high frequency sawtooth waveform for sampling the input audio signal. The frequency of the sawtooth waveform is usually selected 10 times the

maximum frequency in the input signal. In some circuit, instead of using sawtooth waveforms, triangular waveforms are used.

Pulse-width Modulation (PWM)

Pulse-width modulation is a process in which an input signal is converted to a series of pulses having widths proportional to the instantaneous value of the input signal. This is illustrated in Fig.15-70 for a sinusoidal input signal. Notice that the pulse-width is wider when the instantaneous value of the signal is high, and the widths are low when the input signal is low. The widths of the pulses, τ , are sometimes represented as the percentage of the period of the sawtooth wave $[(100 \times \tau/T)\%]$ which is called duty cycle. The output will be a square wave (50% duty cycle) if the input is zero. The PWM signal is typically produced using a comparator circuit. When the voltage on the inverting input exceeds the voltage on the noninverting input, the comparator switches to its negative saturated output state. When the voltage on the noninverting input exceeds the voltage on the inverting input, the comparator switches

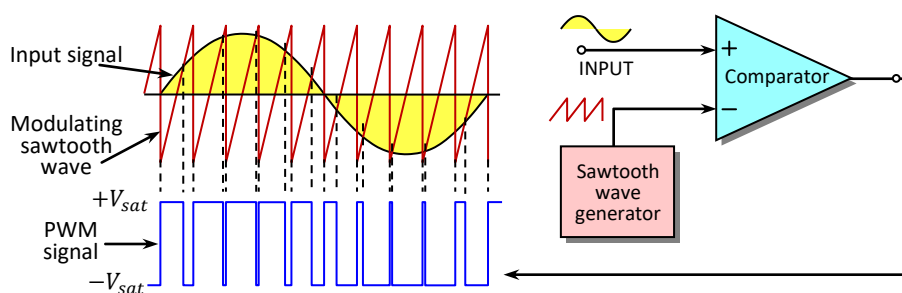


Fig.15-70: Pulse-width modulation (PWM)

to its positive saturated output state. This is illustrated in Fig.15-70 for one cycle of a sine wave voltage on the noninverting input and a higher frequency sawtooth wave voltage on the inverting input. When the value of the input signal is high, the comparator's output will be high ($+V_{sat}$), and when the value of input signal is low the comparator's output will be low ($-V_{sat}$).

15.23.2 The Complementary MOSFET Stage

The MOSFETs are arranged in a common-source complementary configuration to provide power gain. Each transistor switches between the ON state and the OFF state and when one transistor is ON, the other one is OFF, as shown in Fig.15-71. When a transistor is ON, there is very little voltage drop across it and, therefore, there is very little power dissipation even though it may have a high current through it. When a transistor is OFF, there is no current through it and, therefore, there is no power dissipation. The power is dissipated in the transistors

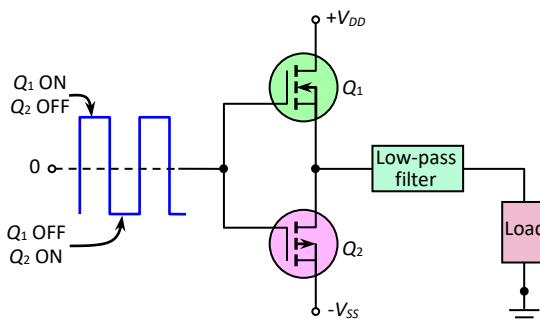


Fig.15-71: MOSFET power stage of class-D amplifier

during the very short switching time. Power delivered to a load can be very high because a load will have a voltage across it nearly equal to the supply voltages and a high current through it.

Efficiency:

When Q_1 is ON, it provides current to the load. However, ideally the voltage drop across it is zero so the internal power dissipation of Q_1 is,

$$P_{DQ1} = V_{Q1}I_{Q1} = V_{Q1}I_L = (0\text{ V})I_L = 0$$

At the same time, Q_2 is OFF and the current through this transistor is zero, so the internal power is,

$$P_{DQ2} = V_{Q2}I_{Q2} = V_{Q2}(0\text{ A}) = 0$$

Similarly, when Q_1 is OFF and Q_2 is ON, the internal power loss in transistors will be zero.

The output power delivered to the load will be the sum of power supplied to the load by the transistors. Ideally, the output power will be $V_{CC}I_L + V_{EE}I_L$. As, generally, $V_{CC} = V_{EE}$, therefore the output power will be, $2V_{CC}I_L$. The ideal maximum efficiency of class-D amplifier will be,

$$\eta_{max} = \frac{P_{out}}{P_{tot}} = \frac{P_{out}}{P_{out} + P_{DQ}} = \frac{P_{out}}{P_{out} + 0} = 1 = 100\%$$

In practical cases, each MOSFET would have a few tenths of a volt across it in the ON state. There is also a small internal power dissipation in the low-pass filter's resistance because it is in series with the power output and a small amount of power dissipated in other components of the circuit. Also, power is dissipated during the finite switching time, distortions in the switched waveform, ON-state resistance, timing errors due to finite switching time, and other subtle errors, so the ideal efficiency of 100% can never be reached in practice.

Low-Pass Filter:

Ideally, the low-pass filter removes the modulating frequency and harmonics and passes only the original signal to the output. The filter has a higher cut-off frequency (f_H) that is above the input signal frequency and below the modulating frequency and harmonics. Low-pass filters have been discussed in Chapter 13.

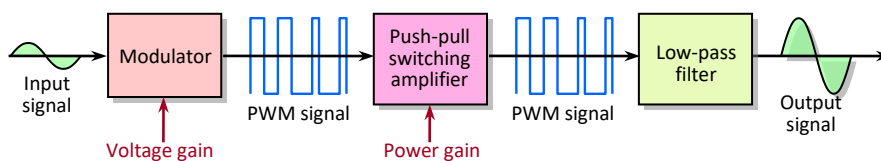


Fig.15-72: Signal flow block diagram of class-D amplifier

Signal Flow:

Fig.15-72 shows the signals at each point in a class-D amplifier. A small input signal is applied and pulse-width modulated to produce a PWM signal at the output of the modulator where voltage gain is achieved. **The required voltage gain can be achieved, by proper adjustment of the width of each pulse of the PWM signal.** The PWM drives the complementary MOSFET stage to achieve power amplification. The PWM signal is filtered and the amplified audio signal appears on the output with sufficient power to drive a speaker.

Example 14-23

A class-D amplifier is to be driven by a 10 kHz audio signal with peak value $V_{ip} = \pm 5$ V. Assuming $V_{DD} = 32$ V determine the minimum frequency of the sawtooth wave, the peak-to-peak value of the sawtooth wave, and the power delivered to a load of $16\ \Omega$ when the width of the driving PWM is 20%.

Solution:

Generally the frequency of the carrier sawtooth wave should be 10 times the signal frequency. Therefore, the frequency of the sawtooth wave will be,

$$f_{saw} = 10 \times 10\text{ kHz} = 100\text{ kHz}$$

The peak-to-peak value of the sawtooth wave should be greater than or equal to the peak-to-peak value of the signal. Thus,

$$V_{pp(saw)} = +10\text{ V} - (-10\text{ V}) = 20\text{ V [Ans.]}$$

20% duty-cycle means $\tau/T = 20/100 = 0.2$.

Therefore, the average voltage to the load will be,

$$V_{o(avg)} = \frac{\tau}{T} V_{DD} = 0.2 \times 32\text{ V} = 6.4\text{ V}$$

Therefore, the average output current will be,

$$I_{o(avg)} = \frac{V_{o(avg)}}{R_L} = \frac{6.4\text{ V}}{16\ \Omega} = 0.4\text{ A}$$

The power delivered to the load will be,

$$P_o = V_{o(avg)} I_{o(avg)} = 6.4\text{ V} \times 0.4\text{ A} = 2.56\text{ W [Ans.]}$$

Comments: According to Nyquist's theorem the frequency of the sawtooth signal can be calculated as twice the highest frequency component of the input signal.

15.24 Integrated Circuit Power Amplifiers

Nowadays a number of power amplifier ICs are available in the market. Different ICs have different power rating, bandwidth and power supply requirements. Some of the most common ICs are described in the following headings.

15.25 Power Amplifier Driver IC LM391

The LM391 is an integrated circuit audio power amplifier driver that contains amplification and driver stages for controlling an externally connected class-AB output stage delivering 10 W to 100 W. High power supply voltage operation and true high fidelity (Hi-Fi) performance distinguish this IC. The LM391 is internally protected for output faults and thermal overloads. The circuitry

providing output transistors protection is user programmable. This IC possesses many other important features such as: low distortion, dual slope SOA (safe operating area) protection, internally protected for output faults and thermal loads, low input noise, shutdown pin, high supply rejection, and selectable gain and bandwidth.

This IC comes in a 16-pin DIP (dual in-line package) package as shown in Fig.15-73. The equivalent internal schematic diagram is shown in Fig.15-74. Here, transistor Q_1 is used to shutdown the amplifier when Pin-14 is grounded. Q_2 generates the reference current for this circuit, and Q_3 provides biasing current for the differential input stage (Q_4 and Q_5) and Q_8 provides bias current for the next stage. Q_6 and Q_7 work as the active load for the input stage. Q_9 is used as V_{BE} multiplier to bias the power amplifier in class-AB mode. Q_{12} and Q_{13} are used to protect the output stage. Transistors Q_{10} , Q_{11} , and Q_{14} , Q_{15} constitute the final power stage of the IC. The diodes connected to the bases of Q_{12} and Q_{13} are used for dual slope SOA protection. The pin description of this circuit is given below,

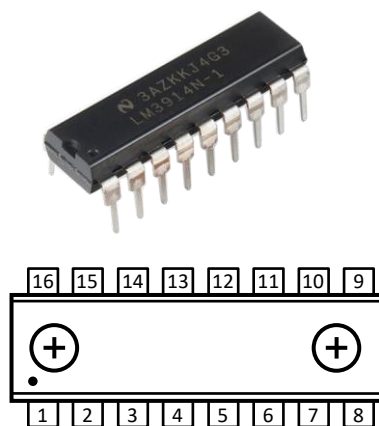


Fig.15-73: LM391 package and pin numbers

Pin no.	Pin Name	Comments
1	+Input	Audio input
2	-Input	Feedback input
3	Compensation	Sets the dominant pole
4	Ripple Filer	Improves negative supply rejection
5	Sink Output	Drives output devices and is emitter of AB bias V_{BE} multiplier
6	BIAS	Base of V_{BE} multiplier
7	BIAS	Collector of V_{BE} multiplier
8	Source Output	Drives output devices
9	Output Sense	Biases the IC and is used in protection circuits
10	+Current Limit	Base of positive side protection circuit transistor
11	+SOA Diode	Diode used for dual slope SOA protection
12	-SOA Diode	Diode used for dual slope SOA protection
13	-Current Limit	Base of negative side protection circuit transistor
14	Shutdown	Shuts off amplifier when current is pulled out of range

15	V+	Positive supply
16	V-	Negative supply

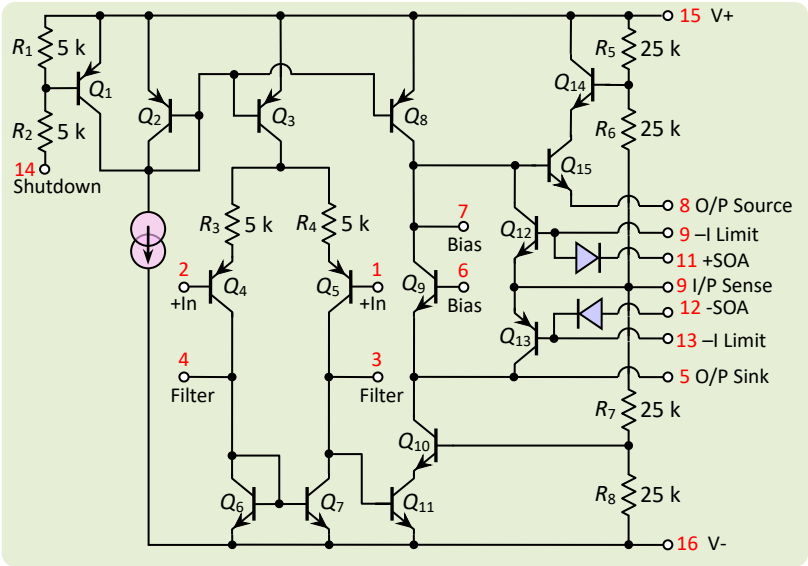


Fig.15-74: Internal schematic diagram of LM391

15.25.1 Audio Power Amplifier using LM391

LM391 IC can be used to drive class-AB power amplifier with a wide range of output power. Fig.15-75 illustrates a typical power amplifier using LM391.

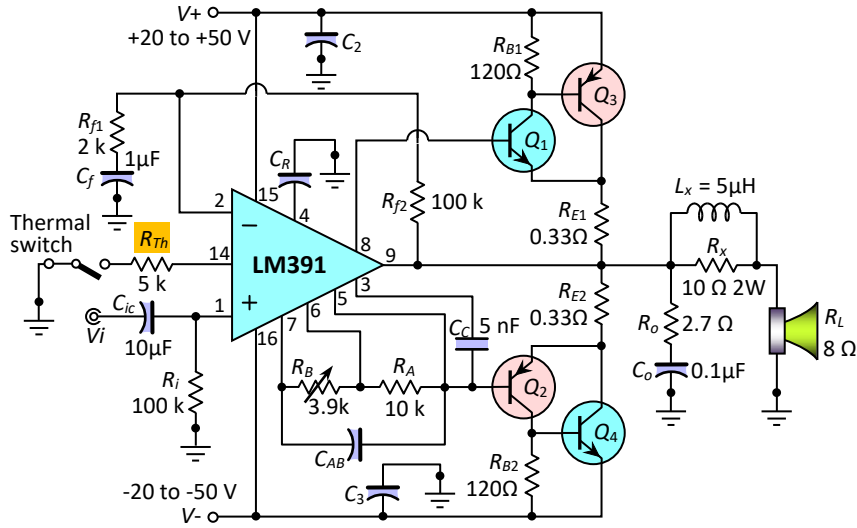


Fig.15-75: Power amplifier using LM391

At the output stage of this amplifier two complementary emitter followers with two low power transistors (Q_1 and Q_2) and two high power transistors (Q_3 and Q_4) connected in quasi-complementary form have been used to drive the load. The selection of these transistors will mainly depend on the output power and voltage. Some possible combinations of these output transistors are given in Table-1. The IC output at terminal 9 is connected to the amplifier

Table 1: Output transistors selection guide

Power Output	Driver Transistor		Output Transistor	
	PNP (Q_2)	NPN (Q_1)	PNP (Q_3)	NPN (Q_4)
20 W @ 8 Ω	MJE711	MJE721	TIP42A	TIP41A
30 W @ 4 Ω	MJE171	MJE181	2N6490	2N6487
	D43C8	D42C8		
40 W @ 8 Ω	MJE712	MJE722	2N5882	2N5880
60 W @ 4 Ω	MJE172	MJE182	MJ14002	MJ14003
	D43C11	D42C11		

output, and the output stage transistors are controlled from current source terminal 8 and current sink terminal 5. The circuit uses a split power supply (V_+ , V_-). The input signal is applied to the noninverting terminal of the IC (pin 1) via capacitor C_{ic} , and this input terminal is biased to ground through input resistor R_i . The inverting input terminal (pin 2) receives feedback from the output, so that the complete circuit operates as a noninverting amplifier with a closed-loop gain of $(1 + R_{f2}/R_{f1})$. Resistors R_A and R_B are connected to an internal transistor (via terminals 5, 6, and 7) to constitute a V_{BE} multiplier for controlling the bias voltage to the amplifier output stage. For better

understanding, the V_{BE} multiplier is redrawn in Fig.15-76(a). Capacitor C_{AB} by-passes the V_{BE} multiplier circuit to improve the amplifier high frequency response. Capacitor C_R helps to reject power supply ripples, and is a compensation capacitor for frequency stability. Components R_o , C_o , L_x and R_x are included for load compensation (see Section 18-5) [H_S_atten].

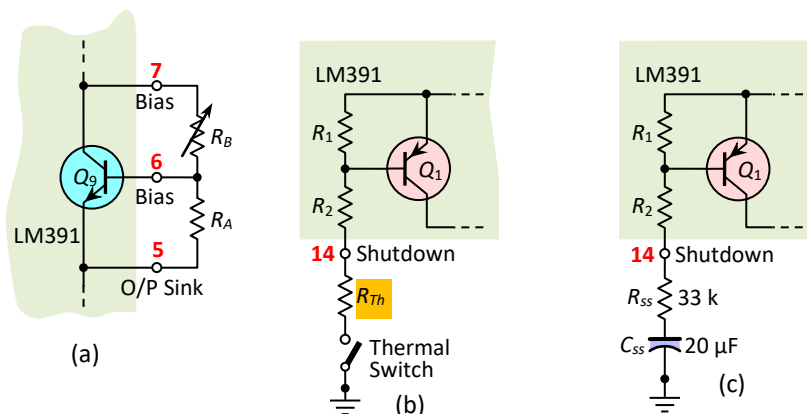


Fig.15-76: Bias control and shut-down circuits of LM391
(Shaded area shows the internal components of LM391)

The transistor Q_1 of LA4391, as shown in Fig.15-74 and redrawn in Fig.15-76(b), can shut the circuit down when turned on by a thermal switch. This allows the device to be protected from overheating that might occur with an excessive load current. This same transistor can be employed for soft *turn-ON* of the circuit as shown in Fig.15-76(c). If the amplifier supply voltage is switched ON, at that instant a peak input signal is applied, a high level output is passed to the speaker, causing a sharp unpleasant noise. Soft *turn-ON* causes the output to increase slowly, thus eliminating the speaker noise. The circuit in Fig.15-76(c) holds the amplifier in a *shut-down* condition until the capacitor charges.

Overload protection transistors are included in the LM391 (Q_{12} , and Q_{13} of Fig.15-74) which is redrawn in Fig.15-77. These transistors turn ON when excessive voltage drops occur across the emitter resistors (R_{E1} and R_{E2}) in the output stage, (see Section 18-5) [H_S_atten].

The output stage components in Fig.15-75 are selected in the same way as for other direct-coupled Class-AB amplifiers. The minimum levels of supply voltage are generally calculated by adding 5 V to the peak output voltage.

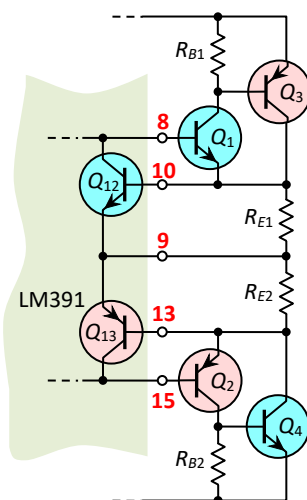


Fig.15-77: Overload protection circuit of LM391
(Internal components of LM391 are in shaded area)

$$V_+ = V_- = \pm(V_p + 5 \text{ V}) \quad (15-107)$$

The input resistance at terminal 1 of the LM391 is extremely high (see Fig.15-74), so the circuit input resistance is set by resistor R_i , which is typically selected to be 100 k Ω . Feedback resistors R_{f1} and R_{f2} set the amplifier closed-loop voltage gain. The feedback network components are determined in exactly the same way as for other feedback amplifiers. R_{f2} is made equal to R_i to minimize output offset, and R_{f1} is calculated from R_{f2} to give the desired voltage gain. Capacitor C_f is determined in terms of R_{f1} , to set lower cut-off frequency. Similarly, the high-frequency roll-off capacitor C_c determines the higher cut-off frequency of the amplifier circuit.

Example 14-24

For audio power amplifier of Fig.15-75, determine the maximum possible output power, the closed-loop voltage gain, lower cut-off frequency, and the gain bandwidth product. Assume the supply voltage is $\pm 30 \text{ V}$.

Solution:

The possible maximum output voltage swing can be calculated using Equ.(15-107) as,

$$\text{From Equ.(15-107),} \quad |V_p| = V_{CC} - 5 \text{ V} = 30 \text{ V} - 5 \text{ V} = 25 \text{ V}$$

Neglecting the power loss in the compensating impedance at the output stage, the maximum output power will be,

$$P_{o(max)} = \frac{V_p^2}{2R_L} = \frac{(25 \text{ V})^2}{2 \times (8 \Omega)} \approx 39 \text{ W [Ans.]}$$

The closed-loop voltage gain of the amplifier will be,

$$A_{vf} = \left(1 + \frac{R_{f2}}{R_{f1}}\right) = \left(1 + \frac{100 \text{ k}}{2 \text{ k}}\right) = 51 \text{ [Ans.]}$$

The lower cut-off frequency set by the input capacitor and input resistor will be,

$$f_{Li} = \frac{1}{2\pi R_i C_i} = \frac{1}{2\pi (100 \text{ k})(10 \mu\text{F})} \approx 0.16 \text{ Hz}$$

The lower cut-off frequency set by the feedback capacitor and resistor will be,

$$f_{Lf} = \frac{1}{2\pi R_{f1} C_f} = \frac{1}{2\pi (2 \text{ k})(1 \mu\text{F})} \approx 80 \text{ Hz}$$

So, the overall lower cut-off frequency of this amplifier will be,

$$f_L = 80 \text{ Hz}$$

The gain bandwidth product (GBW) will be set by the compensating capacitor C_c and is calculated as (given in datasheet),

$$\text{GBW} = \frac{1}{2\pi 5000 C_c} = \frac{1}{2\pi 5000 (5 \text{ nF})} \approx 6.34 \text{ kHz [Ans.]}$$

Comments: The components values of Fig.15-75 are taken from the datasheet of LM391 IC.

15.26 The TDA2030

The TDA2030 is a monolithic integrated circuit in Pentawatt package (zigzag package), intended for use as a low frequency class-AB amplifier. Typically it provides 14 W output power (with 0.5% distortion) at 14 V/4 Ω with ± 14 V or 28 V supply voltage. The guaranteed output power is 12 W on a 4 Ω load and 8 W on an 8 Ω . The TDA2030 provides high output current and has very low harmonic and cross-over distortion. Further, the device incorporates an original (and patented) short-circuit protection system comprising an arrangement for automatically limiting the dissipated power so as to keep the working point of the output transistors within their safe operating area. A conventional thermal shut-down system is also included. The TDA2030 is a 5-pin IC and available in TO220 package. A photograph and the pin-diagram of this IC is shown in Fig.15-78.

15.26.1 A 14 W Amplifier Circuit using TDA2030

The TDA2030 IC subwoofer (mid-range frequency) circuit using 28 V single power supply is shown in Fig.15-79. The input signal is applied to a 22 k potentiometer to control the volume of the amplifier. From this potentiometer the input signal is applied to the noninverting terminal (pin 1) through a 2.2 μ F capacitor (C_1). This input capacitor and the input resistor $R_1 = 100$ k set the lower cut-off frequency of the amplifier. The R_4 resistor which is connected in between pins 2 & 4 is known as feedback resistor. This resistor and R_3 determine the AC closed-loop gain of the amplifier. Capacitor C_4 is short for AC signal and open for DC signal. Thus, for DC signal the amplifier will work as buffer which reduces of off-set voltage at the output.

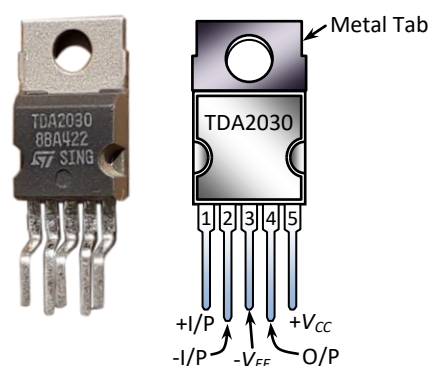


Fig.15-78: Photograph and pin diagram of TDA2030 IC

The output of the IC is connected to the loudspeaker through the 2000 μ F series capacitor which isolates the DC voltage from the load. This circuit uses 4 Ω to 6 Ω -ohm speakers to generate 12 W output. The high temperature developed in this IC should be removed by using a heat sink. The combination of the output resistor R_5 and C_5 are used to balance the load impedance (as described in

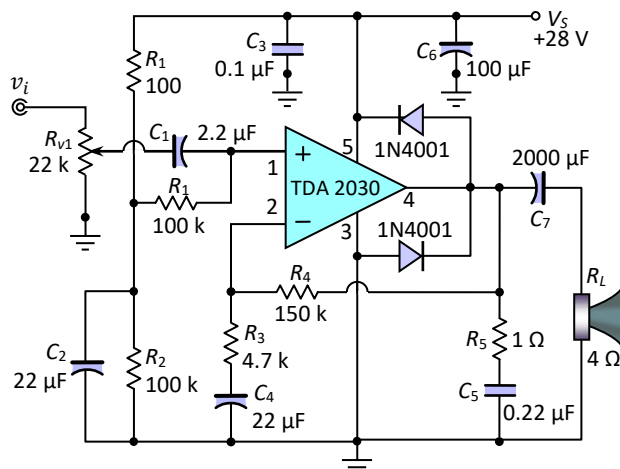


Fig.15-79: A 14 W power amplifier using TDA2030 IC

section) [H_S_atten]. They help in avoiding the surplus noises within the speaker. The diodes (1N4001) connected from the output terminal (pin 4) to the supply rails protect the internal power transistors from any high voltage generated by inductive load. Capacitors C_3 and C_6 are power supply decoupling capacitor that provides surge current to the load.

Although the device incorporates a short circuit protection system and a conventional thermal shut-down system, a heatsink must be used for the TDA2030 IC. If the junction temperature increases up to 150°C , the thermal shut-down mechanism simply reduces the power dissipation and the current consumption of the IC.

15.26.2 35 W Bridge Amplifier using TDA2030

Typically TDA2030 provides 14 W output power at $\pm 14\text{ V}$ or 28 V as shown in Fig.15-79. However, with a split supply voltage of $\pm 15\text{ V}$ it is possible to design 35 W amplifier using the TDA2030 in a bridge connection as shown in Fig.15-80. This type of amplifier is also called *Bridge Tied Load* (BTL) amplifier.

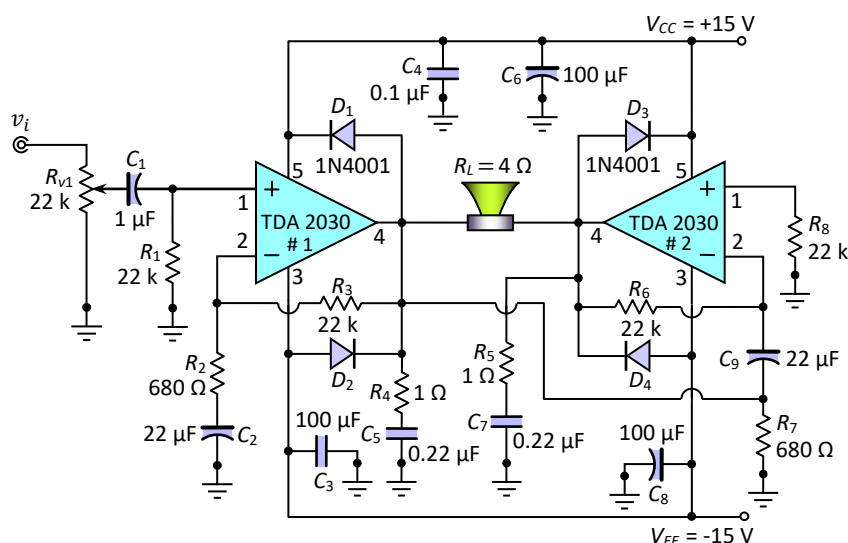


Fig.15-80: A 35 W bridge power amplifier using TDA2030 IC

In this circuit there are two TDA2030 ICs that are connected together. The components connection to each IC is almost similar to that of Fig.15-79. Only the IC #2 is connected in slightly different way. As shown in the Fig.15-80, the noninverting terminal of this IC is grounded through a 22 k resistor and the output from TDA2030 #1 is connected to the inverting input of TDA2030 #2 via C_9 . Thus, the output of this IC (IC #2) will be 180° out of phase from that of IC #1. Thus, both the output voltages will be added across the load (loud speaker).

15.27 DMOS Audio Amplifier TDA7294

The TDA7294 is a monolithic integrated circuit in Multiwatt 15 package, intended for use as audio class-AB amplifier in Hi-Fi field applications. The IC has the wide voltage range ($\pm 10\text{ V}$ to $\pm 40\text{ V}$) and high output current capability. It is able to supply the highest power into both 4 Ω and 8 Ω

loads even in presence of poor supply regulation, with high supply voltage rejection (SVR). The built-in muting function with turn-ON delay simplifies the remote operation avoiding switching ON-OFF noises. The TDA7294 is a monolithic MOS power amplifier which can be operated at 80 V supply voltage (100 V with no signal applied) while delivering output currents up to ± 10 A. The metal TAB of the TDA7294 is connected to the $-V_s$ pin internally. The IC has a DMOS (double diffused MOS) power stage, short-circuit protection, thermal shutdown and does not need boucherot cells (the RC circuit used across the loudspeaker). It can provide closed-loop voltage gain minimum 24 dB, and typically 30 dB. TDA7294 is mainly used in TV, radio receiver, car sound system, home sound systems, etc. The pin diagram of the TDA7294 IC is shown in Fig.15-

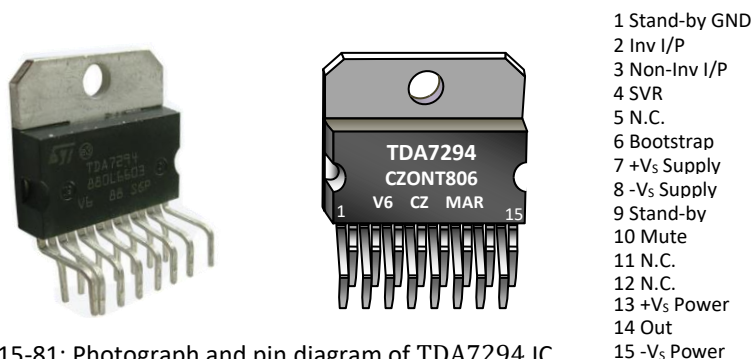


Fig.15-81: Photograph and pin diagram of TDA7294 IC

15.27.1 100 W Amplifier using TDA7294

Fig.15-82 shows a 100 W amplifier consisting of TDA7294. In this circuit, R_1 is the input resistance and its recommended value is 22 k and C_1 is the input coupling capacitor and its

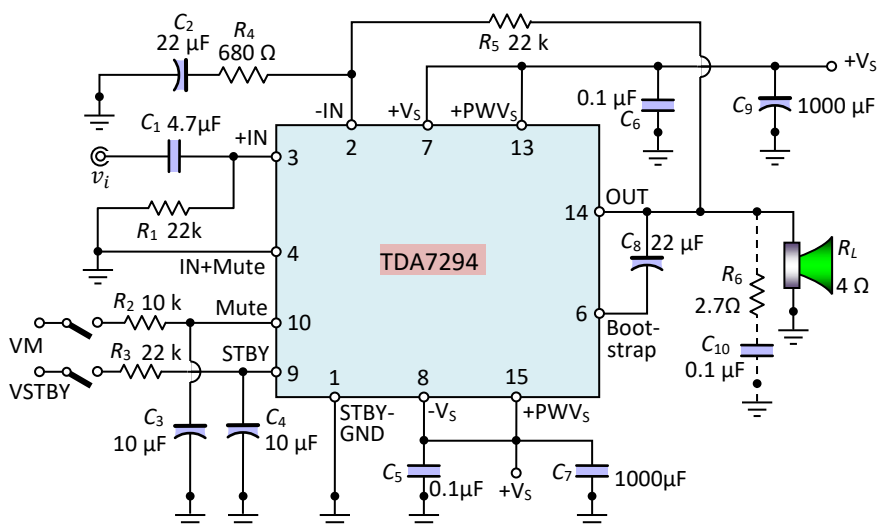


Fig.15-82: A 100 W power amplifier using TDA7294 IC

recommended value is 4.7 μ F. The $R_2 = 10$ k and $C_3 = 10$ μ F are used for deciding the mute time constant, and $R_3 = 22$ k and $C_4 = 10$ μ F determines the stand by ON/OFF time duration. $R_4 =$

$680\ \Omega$ and $R_5 = 22\ \text{k}$ decide the closed-loop gain of the amplifier. $C_2 = 22\ \mu\text{F}$ provides AC feedback and limits the closed-loop gain, but for DC signal this capacitor is open and provides maximum feedback. Capacitors C_5 , C_7 and C_6 , C_9 have been used for power supply decoupling purpose. $C_8 = 22\ \mu\text{F}$ is the bootstrapping capacitor. The RC circuit consisting of R_6 and C_{10} is called the *Boucherot cell* which is optional for this IC.

15.27.2 150 W Amplifier using TDA7294

This 150 watt audio amplifier circuit is made using TDA7294 integrated circuit in bridge mode configuration, where two TDA7294 ICs are used, as shown in the schematic diagram of Fig.15-83. The value of the load must not be less than $8\ \Omega$ to keep the power dissipation and output current within the safe range of the IC. The amplifier is capable to deliver 150 W power to an $8\ \Omega$ load and 170 W to a $16\ \Omega$ load with $\pm 35\ \text{V}$ supply voltage.

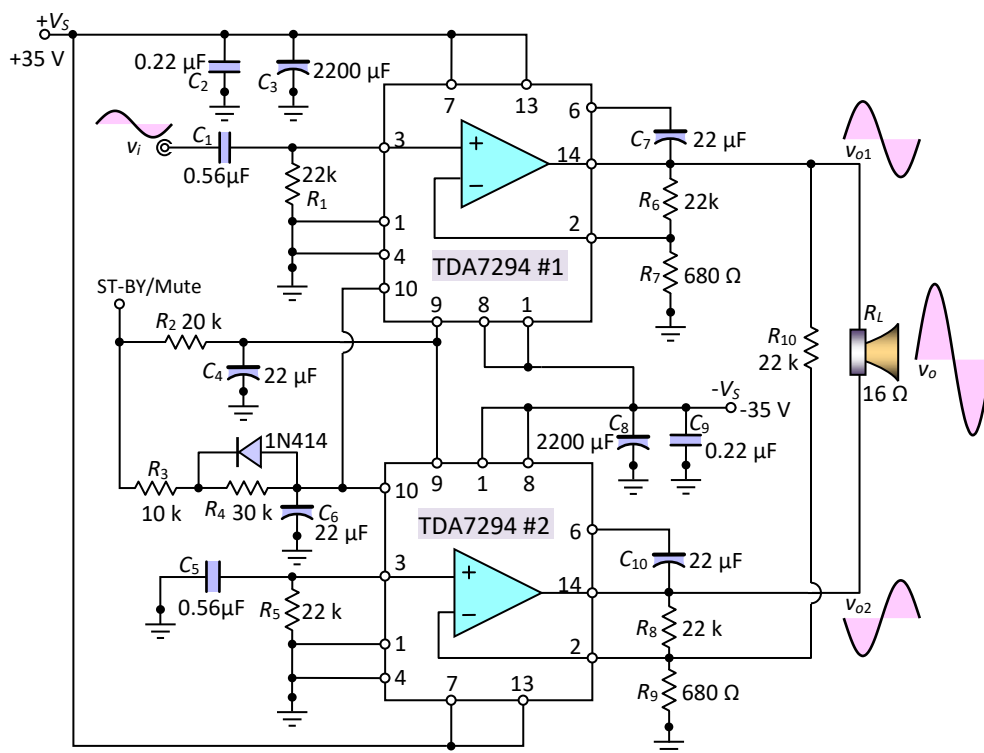


Fig.15-83: A 150 W power amplifier using TDA7294 ICs in bridge mode

The purpose of the components used in this circuit is same as the circuit of Fig.15-82. Here, C_1 and R_1 are the input capacitor and input resistor. Capacitors C_2 , C_3 , C_8 , and C_9 are the power supply decoupling capacitors and C_7 , C_{10} are the bootstrap capacitors. R_2 and C_4 determine the ST-BY time and R_3 , R_4 , C_6 set the time-constant for the mute signal. In both the ICs the closed-loop gain is determined by a $22\ \text{k}$ and a $680\ \Omega$ resistance (R_6 , R_7 , R_8 , R_9). The input signal is applied to the noninverting terminal of the first IC, and hence it works as a noninverting amplifier. The output of this amplifier is applied to the load as well as to the inverting terminal of the second IC, through R_{10} resistor. Therefore, the output of the first IC is in phase with the input signal, while the output

of the second IC is output phase from the input signal. As shown in Fig.15-83, these two output signals are added across the load (loudspeaker).

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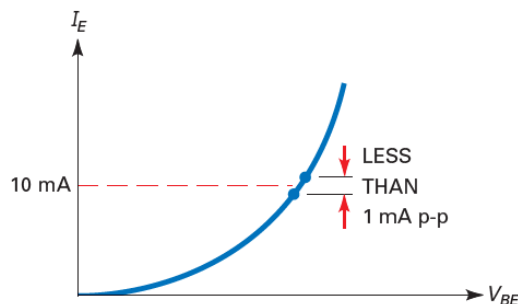
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Habib	9.5	
Beauty	29	
Habib,3m	20	
Land	37.5	
Agrani	13.5	
Prime	4.5	
IBBL	30	
Beauty,ac	2	
	151	

Distortion

The ac voltage on the base produces the ac emitter current shown in Fig. 8-10. This ac emitter current has the same frequency as the ac base voltage. For instance, if the ac generator driving the base has a frequency of 1 kHz, the ac emitter current has a frequency of 1 kHz. The ac emitter current also has approximately the same shape as the ac base voltage. If the ac base voltage is sinusoidal, the ac emitter current is approximately sinusoidal.

The ac emitter current is not a perfect replica of the ac base voltage because of the curvature of the graph. Since the graph is curved upward, the positive half-cycle of the ac emitter current is elongated (stretched) and the negative half-cycle is compressed. This stretching and compressing of alternate half-cycles is called **distortion**. It is undesirable in high-fidelity amplifiers because it changes the sound of voice and music.

Figure 8-11 Definition of small-signal operation.

Why is this important? Because there is negligible distortion for a small signal. When the signal is small, the changes in ac emitter current are almost directly proportional to the changes in ac base voltage because the graph is almost linear. In other words, if the ac base voltage is a small enough sine wave, the ac emitter current will also be a small sine wave with no noticeable stretching or compression of half-cycles.

The 10 Percent Rule

The total emitter current shown in Fig. 8-10 consists of a dc component and an ac component, which can be written as:

$$I_E = I_{EQ} + i_e$$

where I_E = the total emitter current

I_{EQ} = the dc emitter current

i_e = the ac emitter current

To minimize **distortion**, the peak-to-peak value of i_e must be small compared to I_{EQ} . Our definition of small-signal operation is:

$$\text{Small signal: } i_{e(p-p)} < 0.1 I_{EQ} \quad (8-6)$$

This says that the ac signal is small when the peak-to-peak ac emitter current is less than 10 percent of the dc emitter current. For instance, if the dc emitter current is 10 mA, as shown in Fig. 8-11, the peak-to-peak emitter current should be less than 1 mA in order to have small-signal operation.

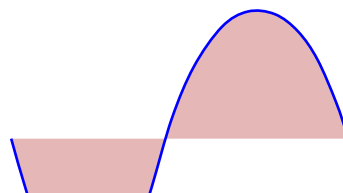
From now on, we will refer to amplifiers that satisfy the 10 percent rule as **small-signal amplifiers**. This type of amplifier is used at the front end of radio and television receivers because the signal coming in from the antenna is very weak. When coupled into a transistor amplifier, a weak signal produces very small variations in emitter current, much less than the 10 percent rule requires.

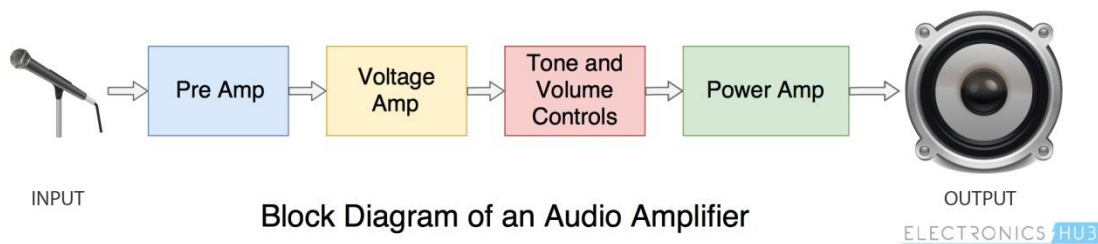
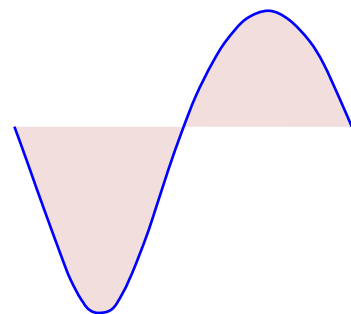
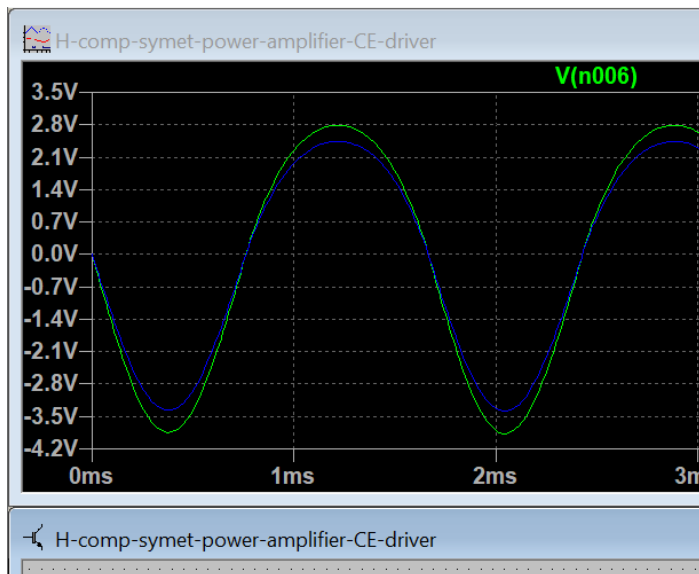
frequency distortion

frequency distortion [frē·kwēn·sē di'stór·shən]

(electronics)

Distortion in which the relative magnitudes of the different frequency components of a wave are changed during transmission or amplification. Also known as amplitude distortion; amplitude-frequency distortion; waveform-amplitude distortion.





Loudspeaker Concepts and Precautions

One of the most important components of a good audio system is the speaker and its enclosure. This section covers speaker operation, how they are connected, speaker enclosures, and tips on hooking up your speaker system.

HOW SPEAKERS ARE CONNECTED

Figure 2-14 shows connections for a woofer, mid-range horn, high-frequency tweeter, and a crossover network for a typical speaker system. In this set up, the crossover coil (1-mH choke) is installed in parallel with the 1-kHz exponential horn after the series crossover capacitor. In other systems, you might find the horn connected in series with the tweeter. The crossover point is usually at the 1-kHz frequency point.

The purpose of the crossover coil is to shunt the heavy bass frequencies around the 1-kHz horn and tweeter, thus affording added protection for the voice coils of the horn and tweeter. The crossover coil also serves to smooth the horn's acoustic crossover point and improves the speaker's sound reproduction.

A brief analysis of the speaker system (shown in Fig. 2-14) is as follows: Assume that the complete audio spectrum appears at the input of the 8-ohm speaker system. All frequencies will appear across the 12-inch woofer. The 12-inch woofer will reproduce audio frequencies from 30 Hz to 1 kHz. The crossover capacitor will block virtually all frequencies below 1 kHz and pass the audio frequencies above the 1 kHz (crossover) point. The 1-mH choke will act as a very low impedance to any frequencies below 1 kHz that might still be present after the blocking capacitor while acting as a high impedance to the frequencies above 1 kHz. Audio frequencies above 1 kHz will now be present across the 1-kHz horn. The acoustic audio output of the 1-kHz signal is essential flat, out to approximately 8 kHz. The capacitor blocks the frequencies below 8 kHz and passes the higher frequencies across to the 3-inch tweeter for a smooth acoustic output to approximately a frequency of 16 kHz.

If you are connecting new speaker to your audio system, be sure that they match for impedance (such as 8 or 16 ohms) and have proper power-handling capability for your power amplifier. It is possible to damage a speaker system—even if its power-handling capacity is the same as, or higher than, the power output rating of the amplifier to which it is connected. Damage can occur to the speaker system because almost all power amplifiers deliver more than their rated power output. This is especially true if the amplifier is operated at maximum, or very high volume settings while the tone controls are set at, or near, maximum boost. A safety margin should be allowed between the amplifier's rated output and

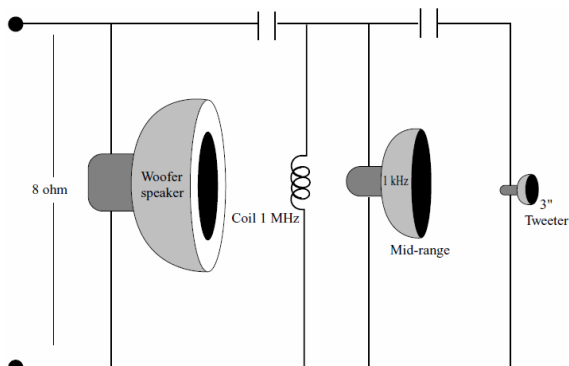


FIGURE 2-14 Crossover network circuit for a speaker and horn.

the speaker system's rated handling capability if the amplifier is to be operated at very high volume levels. If distortion is noticed at high volume levels, it is recommended that the volume level be reduced because you might be reaching the safe operating limits of the amplifier or the speaker system.

HOW TUNED-PORT SPEAKER SYSTEMS WORK

The speaker system shown in the (Fig. 2-15) drawing is of the tuned port type. The cross section view of this speaker enclosure is shown in (Fig. 2-16) and will be used for the following explanation. This enclosure has four openings in the front panel (one for each of the three speakers and one for the port), and the remaining panels are of solid construction.

A tuned-port speaker can be described as a tuned enclosure in which the air in the port will resonate with the air in the main area of the cabinet, at a given frequency. This frequency determines the effective low-frequency cutoff of the system (cabinet and enclosure combined). Below the selected frequency (30 Hz), the response drops very rapidly (approximately 24 db/octave).

This system could also be described as an acoustic phase inverter. That is, at some frequency, within its normal operating range, the air in the port is moving in an outward direction (to the front) while the speaker cone is also moving in an outward direction (to the front). These two movements would occur at the same time, and in phase.

Basic advantage of a tuned-port enclosure, as used in this speaker system, over a typical acoustic suspension (closed box) enclosure are:

- Reduced low frequency distortion.
- Increased efficiency. This requires less amplifier driving power for equal loudness level.